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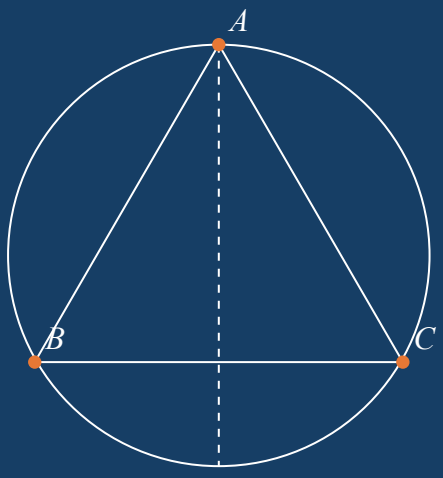
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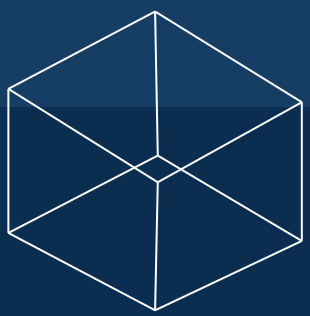
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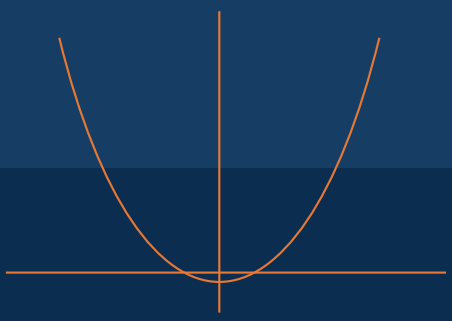
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2000-2026

810 problems / 44 sections

The Mathematics Atlas

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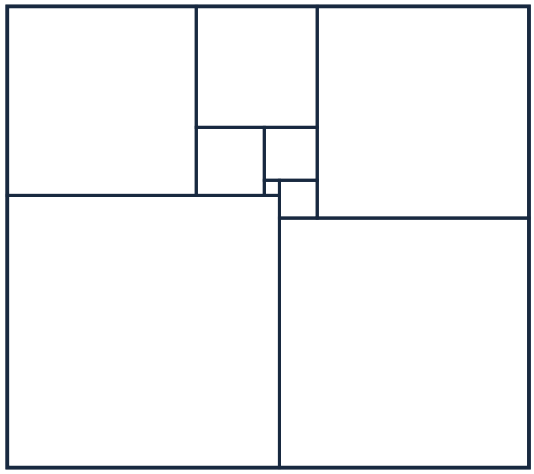
Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q04] | Part A (Q1-Q5)

A rectangle is tiled by nine squares in the arrangement shown. Its width and height are relatively prime positive integers. Find its perimeter.

一个矩形按图示方式分成九个正方形，宽和高为互质正整数。求矩形周长。



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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2004-II-Q05] | Part A (Q1-Q5)

A job is scheduled for 1000 equal-rate workers. After its first quarter, 100 leave; after the second quarter, another 100 leave. Once three quarters are finished, how many additional workers beyond the remaining 800 are minimally needed to meet the original deadline?

某工程原由一千名等效率工人按时完成。做完四分之一后离开一百人，再做完四分之一又离开一百人。完成四分之三时，除剩下的八百人外至少再招多少人，才能不超过原工期?

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2006-I-Q05] | Part A (Q1-Q5)

For positive integers a, b, c, $\sqrt{2006 + 104\sqrt{6} + 468\sqrt{10} + 144\sqrt{15}} = a\sqrt{2} + b\sqrt{3} + c\sqrt{5}$. Find abc.

正整数 a, b, c 满足 $\sqrt{2006 + 104\sqrt{6} + 468\sqrt{10} + 144\sqrt{15}} = a\sqrt{2} + b\sqrt{3} + c\sqrt{5}$，求 abc。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-I-Q02] | Part A (Q1-Q5)

A 100-foot walkway moves at 6 ft/s. Person A rides from time zero, B starts two seconds later walking at 4 ft/s relative to the walkway, and C starts two more seconds later beside it at 8 ft/s. After all start and before anyone finishes, one is midway between the others. Find that person’s distance from the start.

百英尺长的传送步道速度为六英尺每秒。甲在零时登上后站立，乙晚两秒登上并相对步道以四英尺每秒行走，丙再晚两秒沿旁边以八英尺每秒前行。三人都出发且尚未走完时，一人位于另两人正中，求此人的位置。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2007-II-Q04] | Part A (Q1-Q5)

Identical workers require fixed production times per item type. One hundred worker-hours produce 300 items of type X and 200 of Y; 120 worker-hours produce 240 X and 300 Y. With 150 worker-hours, producing 150 X leaves capacity for how many Y?

所有工人生产每种产品所需时间固定相同。一百工时可生产三百件甲及二百件乙；一百二十工时可生产二百四十件甲及三百件乙。一百五十工时生产一百五十件甲后，还能生产多少乙?

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-I-Q01] | Part A (Q1-Q5)

At a party, 60% are girls and 40% like dancing. Twenty additional boys, all dancers, arrive, reducing the girl fraction to 58%. How many attendees now like dancing?

聚会上女生占百分之六十，喜欢跳舞者占百分之四十。又来二十名男生且都喜欢跳舞，女生比例变为百分之五十八。求现有多少人喜欢跳舞。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2008-II-Q02] | Part A (Q1-Q5)

Cyclist R rests five minutes after every mile; cyclist J rides at three quarters R’s riding speed and rests five minutes after every two miles. They start together and first reach mile fifty simultaneously, before any final rest. Find elapsed minutes.

骑手甲每骑一英里休五分钟；乙骑行速度为甲的四分之三，每两英里休五分钟。两人同时出发且同时首次到达五十英里处，不计终点休息，求分钟数。

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方程与代数建模

[AIME-2009-II-Q01] | Part A (Q1-Q5)

Initially there are $130, 164, 188$ ounces of blue,red,white paint. Four equal-area stripes use equal paint amounts: one of each pure color and one pink mixture of red and white. All three original colors have equal amounts left. Find total paint left.

原有蓝、红、白漆分别一百三十、一百六十四、一百八十八盎司，用等量油漆刷四条同面积条纹：蓝、红、白及红白混合粉色。之后三种原漆余量相等，求总余量。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q01] | Part A (Q1-Q5)

Jar A has 4 L at 45% acid and jar B has 5 L at 48%. Split 1 L of $k\%$ acid between them so both become 50% acid. If the amount added to A is reduced m/n liters, find $k+m+n$.

A 瓶有四升 45% 酸液，B 瓶有五升 48% 酸液。将一升 $k\%$ 酸液分入两瓶，使两瓶均为 50%。加入 A 的升数为最简分数 m/n，求 $k+m+n$。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-II-Q01] | Part A (Q1-Q5)

A drinker consumes the fraction m/n of a purchase. Had the purchase been half as large and consumption twice as large, the leftover would have been $2/9$ of the actual leftover. Find $m+n$ for the reduced fraction.

某人喝掉所购饮料的最简分数 m/n。若购买量减半、喝掉量加倍，剩余量只有实际剩余的 $2/9$。求 $m+n$。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-I-Q04] | Part A (Q1-Q5)

Two travelers walk at 4 and 2.5 mph and share a horse traveling at 6 mph. The slower traveler initially rides, ties the horse at milepost one, and walks on. The other rides upon reaching the horse, passes the partner, and leaves the horse at the next milepost; they repeat this procedure. If their first meeting at a positive integer milepost is n miles from the start after t minutes, find $n + t$.

两人步行速度为每小时四英里、二点五英里，共用一匹每小时六英里的马。慢者先骑马，在一英里路桩拴马后步行；另一人到马处骑上，追过同伴后在下一路桩留下马，如此轮流。若首次在正整数英里路桩相遇时距起点 n 英里、历时 t 分钟，求 $n + t$。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2012-II-Q04] | Part A (Q1-Q5)

Three cyclists start at a rectangle northeast corner: one follows the boundary westward at 8.6 m/s, one southward at 6.2 m/s, and one rides straight at 5 m/s. Their first arrivals coincide at a point D on the south edge. The west-east length exceeds the north-south width. Write length:width:distance from D to the southeast corner as $p : q : r$, with coprime p, q, and find $p + q + r$.

三名骑车人从矩形东北角同时出发，沿边向西者速率 8.6 米/秒，沿边向南者 6.2，直线骑行者为五。三人首次同时到达南边点 D。东西长大于南北宽；将长、宽、D 到东南角距离之比写为 $p : q : r$，其中 p, q 互质，求三者和。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q01] | Part A (Q1-Q5)

An athlete swims $1/2$ mile, cycles thirty miles, and runs eight miles at constant speeds. Running is five times swimming speed, and cycling twice running speed. The total time is 4.25 hours. Find the cycling time in minutes.

运动员以各自恒定速度游半英里、骑车三十英里、跑八英里。跑速为游速五倍，骑速为跑速两倍，总时间四点二五小时。求骑车分钟数。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2013-I-Q03] | Part A (Q1-Q5)

Two lines parallel to adjacent sides divide a square into two smaller squares and two nonsquare rectangles. The small squares together occupy $9/10$ of the original area. If the two segments of one cut side have lengths x, y, find $x/y + y/x$.

两条平行于相邻边的直线将正方形分成两个小正方形及两个非正方形矩形。小正方形总面积为原面积的十分之九。一条被分边的两段长为 x, y，求 $x/y + y/x$。

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方程与代数建模

[AIME-2013-II-Q01] | Part A (Q1-Q5)

A decimal clock divides a day into ten hours of one hundred minutes each. Ordinary $6{:}36$ AM becomes $A{:}BC$. Find $100A + 10B + C$.

十进制时钟将一天分成十小时，每小时一百分。通常的早晨六点三十六分显示为 $A{:}BC$，求 $100A + 10B + C$。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q04] | Part A (Q1-Q5)

Two cyclists ride east and west at twenty mph. Two equal-length trains travel oppositely at different constant speeds, overtaking a cyclist when moving in the same direction. Each takes one minute to pass the eastbound cyclist; the westbound train takes ten times as long as the eastbound train to pass the westbound cyclist. Write the train length as reduced m/n miles and find $m+n$.

两名骑车人分别向东、向西，速度均为每小时二十英里。两列等长列车方向相反、速率不同，并会超越与其同向的骑车人。每列通过东行者均需一分钟；西行列车通过西行者的时间是东行列车的十倍。若列车长度为最简分数 m/n 英里，求 $m+n$。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-II-Q01] | Part A (Q1-Q5)

Worker A paints a room in fifteen hours; B works fifty percent faster and C twice as fast as A. First A works ninety minutes alone, then A, B work until half the room is done, then all three finish. Find the total minutes.

甲独刷一房需十五小时，乙效率比甲高百分之五十，丙为甲两倍。甲先单独工作九十分钟，再与乙一起刷到一半，最后三人共同完成。求总分钟数。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2017-II-Q05] | Part A (Q1-Q5)

Four distinct real numbers have six unordered pair sums, listed without order as $189, 320, 287, 234, x, y$. Find the largest possible $x + y$.

四个不同实数的六个无序两两和，未按顺序列为 $189, 320, 287, 234, x, y$。求 $x + y$ 的最大值。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2018-II-Q01] | Part A (Q1-Q5)

Points A, B, C lie in order on a straight path with $AC = 1800$ m. Runners start together: Ina at A toward C, Paul at B toward C, Eve at C toward A, with speed ratio Ina:Paul:Eve $= 2 : 4 : 1$. Paul turns back when he meets Eve and returns to B exactly as Ina first reaches it. Find AB.

直路上 A, B, C 依次排列，$AC = 1800$ 米。三人同时出发，伊娜从 A 向 C、保罗从 B 向 C、伊芙从 C 向 A，速度比为 $2 : 4 : 1$。保罗遇到伊芙后折返，恰与伊娜首次同时到达 B。求 AB。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2022-I-Q05] | Part A (Q1-Q5)

A river of width 264 m flows east at 14 m/min. Two swimmers start on the south bank, the faster one D m east of the slower. Their speeds relative to water are 80 and 60 m/min. Both swim straight to the same north-bank point, equidistant from their starts, and arrive together. Find D.

河宽 264 米，水流向东，速度 14 米/分。两人从南岸出发，较快者在较慢者以东 D 米；静水速度分别为 80、60 米/分。两人沿直线游向北岸同一点，该点到两出发点等距，且同时到达。求 D。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2023-II-Q04] | Part A (Q1-Q5)

Real numbers satisfy $xy + 4z = yz + 4x = zx + 4y = 60$. Let S contain every possible value of x. Compute $\sum_{s \in S} s^2$.

实数 x, y, z 满足 $xy + 4z = yz + 4x = zx + 4y = 60$。令 S 为 x 的所有可能值组成的集合，求各元素平方和。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2024-I-Q01] | Part A (Q1-Q5)

A daily outing consists of a 9 km walk and a fixed stop of t minutes. At walking speed s km/h the total is 4 h; at $s+2$ km/h it is 144 min. Find the total minutes at speed $s+1/2$ km/h.

一次日常外出包括步行 9 千米及固定停留 t 分钟。步速为 s 千米/时时总用时 4 小时；步速为 $s+2$ 时总用时 144 分钟。求步速为 $s+1/2$ 时的总分钟数。

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Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2026-I-Q01] | Part A (Q1-Q5)

Three travelers follow the same route and arrive together. The second starts one hour after the first at 2 mph faster; the third starts one hour after the second at 7 mph faster than the second. Write the route length as a reduced fraction m/n and find $m+n$.

三人走同一路线并同时到达。第二人比第一人晚一小时出发，速度快 2 英里/时；第三人又晚一小时出发，速度比第二人快 7 英里/时。路程为最简分数 m/n 英里，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q07] | Part B (Q6-Q10)

Positive real numbers satisfy $xyz = 1$, $x + 1/z = 5$, and $y + 1/x = 29$. Write $z + 1/y = m/n$ reduced and find $m + n$.

正实数满足 $xyz = 1, x + 1/z = 5, y + 1/x = 29$。将 $z + 1/y$ 约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2000-I-Q10] | Part B (Q6-Q10)

For $k = 1, \ldots, 100$, a list $x_1, \ldots, x_{100}$ satisfies $x_k = \sum_{j \neq k} x_j - k$. If $x_{50} = m/n$ reduced with $m, n > 0$, find $m + n$.

一列数满足：对 $k = 1$ 至 100，$x_k = \sum_{j \neq k} x_j - k$。若 x_{50} 约分为正整数之比 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2010-I-Q09] | Part B (Q6-Q10)

Real x, y, z satisfy $x^3 - xyz = 2, y^3 - xyz = 6, z^3 - xyz = 20$. The maximum of $x^3 + y^3 + z^3$ is the reduced fraction m/n. Find $m + n$.

实数满足 $x^3 - xyz = 2, y^3 - xyz = 6, z^3 - xyz = 20$。$x^3 + y^3 + z^3$ 的最大值为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2011-I-Q06] | Part B (Q6-Q10)

A parabola $y = ax^2 + bx + c$ has vertex $(1/4, -9/8)$, $a > 0$, and integer $a + b + c$. If the least possible a is reduced p/q, find $p + q$.

抛物线 $y = ax^2 + bx + c$ 的顶点为 $(1/4, -9/8)$，$a > 0$，且 $a + b + c$ 为整数。若 a 的最小值为最简分数 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2021-II-Q07] | Part B (Q6-Q10)

Real numbers a, b, c, d satisfy $a+b=-3$, $ab+bc+ca=-4$, $abc+bcd+cda+dab=14$, and $abcd=30$. If $a^2+b^2+c^2+d^2=m/n$ reduced with $m, n>0$, find $m+n$.

实数 a, b, c, d 满足 $a+b=-3$、$ab+bc+ca=-4$、$abc+bcd+cda+dab=14$、$abcd=30$。若平方和约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2001-I-Q11] | Part C (Q11-Q15)

A $5 \times N$ point array is numbered first by rows, then renumbered by columns. Choose P_i in row i, with old number x_i and new number y_i. Require $x_1 = y_2, x_2 = y_1, x_3 = y_4, x_4 = y_5, x_5 = y_3$. Find the least N.

一个五行 N 列点阵先逐行编号，再逐列重新编号。在第 i 行选点 P_i，原编号为 x_i、新编号为 y_i。要求 $x_1 = y_2, x_2 = y_1, x_3 = y_4, x_4 = y_5, x_5 = y_3$，求最小 N。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2014-I-Q14] | Part C (Q11-Q15)

Let m be the greatest real solution of $3/(x-3)+5/(x-5)+17/(x-17)+19/(x-19)=x^2-11x-4$. If $m=a+\sqrt{b+\sqrt{c}}$ for positive integers, find $a+b+c$.

令 m 为方程 $3/(x-3)+5/(x-5)+17/(x-17)+19/(x-19)=x^2-11x-4$ 的最大实根。若 $m=a+\sqrt{b+\sqrt{c}}$，求正整数 $a+b+c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.1 Equations & algebraic modeling

方程与代数建模

[AIME-2015-II-Q14] | Part C (Q11-Q15)

Real numbers x, y satisfy $x^4y^5 + y^4x^5 = 810$ and $x^3y^6 + y^3x^6 = 945$. Evaluate $2x^3 + (xy)^3 + 2y^3$.

实数 x, y 满足 $x^4y^5 + y^4x^5 = 810$、$x^3y^6 + y^3x^6 = 945$。求 $2x^3 + (xy)^3 + 2y^3$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2001-I-Q03] | Part A (Q1-Q5)

Find the sum of all complex roots of $x^{2001} + (1/2 - x)^{2001} = 0$, given that all roots are simple.

方程 $x^{2001} + (1/2 - x)^{2001} = 0$ 的根均为单根，求全部复根之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2013-I-Q05] | Part A (Q1-Q5)

The real solution of $8x^3 - 3x^2 - 3x - 1 = 0$ is $(\sqrt[3]{a} + \sqrt[3]{b} + 1)/c$ with positive integers. Find $a + b + c$.

方程 $8x^3 - 3x^2 - 3x - 1 = 0$ 的实根为 $(\sqrt[3]{a} + \sqrt[3]{b} + 1)/c$，各参数为正整数，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-II-Q05] | Part A (Q1-Q5)

Real r, s are roots of $p(x) = x^3 + ax + b$, while $r + 4, s - 3$ are roots of $p(x) + 240$. Find the sum of distinct possible $|b|$.

实数 r, s 为 $p(x) = x^3 + ax + b$ 的根，而 $r + 4, s - 3$ 为 $p(x) + 240$ 的根。求所有不同可能 $|b|$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2018-I-Q01] | Part A (Q1-Q5)

Count integer pairs (a, b) with $1 \leq a \leq 100, b \geq 0$ for which $x^2 + ax + b$ factors into integer linear factors, allowing repeated factors. Report the count modulo 1000.

求整数对 (a, b) 数，满足 $1 \leq a \leq 100, b \geq 0$，且 $x^2 + ax + b$ 可分解为整系数一次因子，允许重因子，取总数模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2021-II-Q04] | Part A (Q1-Q5)

Two real monic cubics are $x^3 + ax + b$ and $x^3 + cx^2 + d$. They have real roots $-20, -21$ respectively and share a root $m + i\sqrt{n}$, where m, n are positive integers. Find $m + n$.

两个实系数首一三次式为 $x^3 + ax + b$ 和 $x^3 + cx^2 + d$，分别有实根 $-20, -21$，并有公共根 $m + i\sqrt{n}$，其中 m, n 为正整数。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2022-I-Q01] | Part A (Q1-Q5)

Two quadratic functions have leading coefficients 2 and -2. Each takes the values 54 at $x = 16$ and 53 at $x = 20$. Determine the sum of their constant terms.

两个二次函数的二次项系数分别为 2 和 -2；它们在 $x = 16$ 时均取值 54，在 $x = 20$ 时均取值 53。求两个常数项之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2003-II-Q09] | Part B (Q6-Q10)

Let $P = x^6 - x^5 - x^3 - x^2 - x$ and $Q = x^4 - x^3 - x^2 - 1$. For the four roots z_i of Q, counted with multiplicity, find $\sum_i P(z_i)$.

令 $P = x^6 - x^5 - x^3 - x^2 - x, Q = x^4 - x^3 - x^2 - 1$。设 Q 的四根为 z_i（按重数计），求 $\sum_i P(z_i)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2004-I-Q07] | Part B (Q6-Q10)

Let C be the x^2 coefficient of $\prod_{j=1}^{15}(1+(-1)^j jx)$. Find $|C|$.

令 C 为 $\prod_{j=1}^{15}(1+(-1)^j jx)$ 中 x^2 系数，求 $|C|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2005-II-Q07] | Part B (Q6-Q10)

Let $x = 4/[(\sqrt{5}+1)(5^{1/4}+1)(5^{1/8}+1)(5^{1/16}+1)]$. Find $(x+1)^{48}$.

令 $x = 4/[(\sqrt{5}+1)(5^{1/4}+1)(5^{1/8}+1)(5^{1/16}+1)]$，求 $(x+1)^{48}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2007-I-Q08] | Part B (Q6-Q10)

A cubic is divisible by both $x^2 + (k-29)x - k$ and $2x^2 + (2k-43)x + k$. Find the largest real k possible.

一个三次多项式同时被 $x^2 + (k-29)x - k$ 与 $2x^2 + (2k-43)x + k$ 整除，求可能的最大实数 k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2008-II-Q07] | Part B (Q6-Q10)

If r, s, t are the roots of $8x^3 + 1001x + 2008 = 0$, find $(r+s)^3 + (s+t)^3 + (t+r)^3$.

若 r, s, t 是 $8x^3 + 1001x + 2008 = 0$ 的根，求 $(r+s)^3 + (s+t)^3 + (t+r)^3$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2010-I-Q06] | Part B (Q6-Q10)

A real quadratic P satisfies $x^2 - 2x + 2 \le P(x) \le 2x^2 - 4x + 3$ for every real x. Given $P(11) = 181$, find $P(16)$.

实系数二次多项式 P 对一切实数 x 满足 $x^2 - 2x + 2 \le P(x) \le 2x^2 - 4x + 3$。已知 $P(11) = 181$，求 $P(16)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2010-II-Q06] | Part B (Q6-Q10)

Find the least positive integer n for which $x^4 - nx + 63$ factors into two nonconstant integer-coefficient polynomials.

求最小正整数 n，使 $x^4 - nx + 63$ 能分解为两个非常数整系数多项式的乘积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-I-Q06] | Part B (Q6-Q10)

The parabolas $y = 3(x-h)^2 + j$ and $y = 2(x-h)^2 + k$ have vertical intercepts $2013, 2014$ respectively, and each has two positive integer horizontal intercepts. Find h.

抛物线 $y = 3(x-h)^2 + j$ 与 $y = 2(x-h)^2 + k$ 的纵截距分别为 $2013, 2014$，各自均有两个正整数横截距，求 h。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2014-I-Q09] | Part B (Q6-Q10)

The cubic $\sqrt{2014}\,x^3 - 4029x^2 + 2 = 0$ has real roots $x_1 < x_2 < x_3$. Find $x_2(x_1 + x_3)$.

方程 $\sqrt{2014}\,x^3 - 4029x^2 + 2 = 0$ 的实根依次为 $x_1 < x_2 < x_3$，求 $x_2(x_1 + x_3)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2015-I-Q10] | Part B (Q6-Q10)

A real cubic polynomial f has $|f(t)| = 12$ at $t = 1, 2, 3, 5, 6, 7$. Find $|f(0)|$.

实系数三次多项式 f 在 $t = 1, 2, 3, 5, 6, 7$ 处均满足 $|f(t)| = 12$。求 $|f(0)|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2015-II-Q06] | Part B (Q6-Q10)

A polynomial $2x^3 - 2ax^2 + (a^2 - 81)x - c$ has three positive integral roots, counting multiplicity, with positive integers a, c. Among all such polynomials, knowing a still leaves exactly two possible values of c. Find their sum.

多项式 $2x^3 - 2ax^2 + (a^2 - 81)x - c$ 的三个根（按重数计）均为正整数，且 a, c 为正整数。在所有符合条件的多项式中，已知 a 后仍恰有两个可能的 c。求这两个 c 的和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2016-II-Q06] | Part B (Q6-Q10)

Let $P(x) = 1 - x/3 + x^2/6$ and $Q(x) = P(x)P(x^3)P(x^5)P(x^7)P(x^9) = \sum_{j=0}^{50} a_j x^j$. Express $\sum |a_j|$ as reduced m/n and find $m+n$.

令 $P(x) = 1 - x/3 + x^2/6$，$Q(x) = P(x)P(x^3)P(x^5)P(x^7)P(x^9) = \sum_{j=0}^{50} a_j x^j$。将 $\sum |a_j|$ 约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2018-II-Q06] | Part B (Q6-Q10)

Choose real a uniformly from $[-20, 18]$. The probability that all four roots of $x^4 + 2ax^3 + (2a-2)x^2 + (-4a+3)x - 2$ are real, counting multiplicity, is reduced m/n. Find $m+n$.

在 $[-20, 18]$ 均匀选实数 a，四次式 $x^4 + 2ax^3 + (2a-2)x^2 + (-4a+3)x - 2$ 的四根全部为实数（计重数）的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2019-I-Q08] | Part B (Q6-Q10)

For real x, $\sin^{10} x + \cos^{10} x = 11/36$. Write $\sin^{12} x + \cos^{12} x = m/n$ reduced and find $m + n$.

实数 x 满足 $\sin^{10} x + \cos^{10} x = 11/36$。将 $\sin^{12} x + \cos^{12} x = m/n$ 约简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2019-I-Q10] | Part B (Q6-Q10)

Distinct complex numbers $z_1, \ldots, z_{673}$ satisfy $\prod_j (x - z_j)^3 = x^{2019} + 20x^{2018} + 19x^{2017} + g(x)$, with $\deg g \le 2016$. If $\left|\sum_{j<k} z_j z_k\right| = m/n$ reduced, find $m + n$.

不同复数 $z_1, \ldots, z_{673}$ 满足 $\prod_j (x - z_j)^3 = x^{2019} + 20x^{2018} + 19x^{2017} + g(x)$，其中 g 次数至多 2016。若 $\left|\sum_{j<k} z_j z_k\right| = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2023-I-Q09] | Part B (Q6-Q10)

For $p(x) = x^3 + ax^2 + bx + c$, all three coefficients are integers from -20 through 20. Count the polynomials for which exactly one integer $m \neq 2$ satisfies $p(m) = p(2)$.

三次多项式 $p(x) = x^3 + ax^2 + bx + c$ 的系数 a, b, c 均为 -20 至 20 的整数。求恰有一个整数 $m \neq 2$ 满足 $p(m) = p(2)$ 的多项式个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2000-II-Q13] | Part C (Q11-Q15)

The equation $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ has exactly two real roots, one of which is $(m + \sqrt{n})/r$ with integers m, n, r, $r > 0$, and $\gcd(m, r) = 1$. Find $m + n + r$.

方程 $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ 恰有两个实根，其中一个为 $(m + \sqrt{n})/r$，其中 m, n, r 为整数，$r > 0$，m, r 互质。求 $m + n + r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2008-I-Q13] | Part C (Q11-Q15)

A real polynomial $p(x, y)$ of total degree at most three vanishes at $(0,0), (1,0), (-1,0), (0,1), (0,-1), (1,1), (1,-1), (2,2)$. It has another positive common zero $(a/c, b/c)$ for every such polynomial, with $c > 1$ and $\gcd(a, c) = 1$. Find $a + b + c$.

实系数总次数至多三的二元多项式在 $(0,0), (1,0), (-1,0), (0,1), (0,-1), (1,1), (1,-1), (2,2)$ 全为零。所有这种多项式还共有正点零点 $(a/c, b/c)$，$c > 1, \gcd(a, c) = 1$，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2013-II-Q12] | Part C (Q11-Q15)

Count monic integer-coefficient cubics all of whose complex roots have modulus either 13 or 20, counting multiplicity.

求首一整系数三次多项式的个数，使每个复根（计重数）的模均为十三或二十。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2016-I-Q11] | Part C (Q11-Q15)

A nonzero polynomial satisfies $(x-1)P(x+1)=(x+2)P(x)$ identically and $P(2)^2=P(3)$. Reduce $P(7/2)$ to m/n and find $m+n$.

非零多项式恒满足 $(x-1)P(x+1)=(x+2)P(x)$，且 $P(2)^2=P(3)$。将 $P(7/2)$ 约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-I-Q11] | Part C (Q11-Q15)

Let $f(x) = x^2 + ax + b$ and $g(x) = x^2 + cx + d$ have integer coefficients. Count ordered triples $(a, b, c) \in [-10, 10]^3$ for which some integer d makes $g(f(2)) = g(f(4)) = 0$.

整数系数多项式 $f(x) = x^2 + ax + b$、$g(x) = x^2 + cx + d$。统计满足 $-10 \le a, b, c \le 10$ 且存在整数 d 使 $g(f(2)) = g(f(4)) = 0$ 的有序三元组 (a, b, c)。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-I-Q14] | Part C (Q11-Q15)

A monic quadratic P over $\mathbb{C}$ has the property that $P(P(x)) = 0$ has four distinct roots $3, 4, a, b$. Add all distinct possible values of $(a+b)^2$.

复系数首一二次多项式 P 满足 $P(P(x)) = 0$ 有四个不同根 $3, 4, a, b$。求 $(a+b)^2$ 所有不同可能值之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2020-II-Q11] | Part C (Q11-Q15)

Let $P = x^2 - 3x - 7$, and let Q, R be monic quadratics. Each pair among $P + Q, P + R, Q + R$ shares a root, and the three shared roots are distinct. If $Q(0) = 2$ and $R(0) = m/n$ reduced with $m, n > 0$, find $m + n$.

设 $P = x^2 - 3x - 7$，Q, R 为首一二次多项式。$P + Q, P + R, Q + R$ 两两有公共根，且三个公共根互异。已知 $Q(0) = 2$，$R(0) = m/n$ 已约分且为正，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

[AIME-2024-II-Q11] | Part C (Q11-Q15)

Count ordered nonnegative integer triples (a, b, c) with $a + b + c = 300$ and $a^2b + a^2c + b^2a + b^2c + c^2a + c^2b = 6000000$.

求非负整数有序三元组 (a, b, c) 的个数，满足 $a + b + c = 300$ 及 $a^2b + a^2c + b^2a + b^2c + c^2a + c^2b = 6000000$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2007-I-Q03] | Part A (Q1-Q5)

For $z = 9 + bi$ with $b > 0$, the imaginary parts of z^2, z^3 are equal. Find b.

$z = 9 + bi, b > 0$，若 z^2, z^3 虚部相等，求 b。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-I-Q02] | Part A (Q1-Q5)

A complex z has imaginary part 164 and satisfies $z/(z+n) = 4i$ for a positive integer n. Find n.

复数 z 的虚部为 164，且正整数 n 满足 $z/(z+n) = 4i$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-II-Q05] | Part A (Q1-Q5)

Complex numbers satisfy $xy = -80 - 320i, yz = 60, zx = -96 + 24i$. If $x + y + z = a + bi$ with real a, b, find $a^2 + b^2$.

复数满足 $xy = -80 - 320i, yz = 60, zx = -96 + 24i$。若 $x + y + z = a + bi$，a, b 为实数，求 $a^2 + b^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2022-I-Q04] | Part A (Q1-Q5)

Set $w = (\sqrt{3} + i)/2$ and $z = (-1 + i\sqrt{3})/2$. Count integer pairs (r, s) with $1 \le r, s \le 100$ for which $iw^r = z^s$.

令 $w = (\sqrt{3} + i)/2, z = (-1 + i\sqrt{3})/2$。求满足 $iw^r = z^s$ 且 $1 \le r, s \le 100$ 的整数有序对数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-I-Q05] | Part A (Q1-Q5)

For $AB = 1$, rotate A counterclockwise through acute θ about B to $A\prime$, then rotate B clockwise through θ about $A\prime$ to $B\prime$. Given $AB\prime = 4/3$, write $\cos\theta = m/n$ in lowest terms and find $m + n$.

$AB = 1$，将 A 绕 B 逆时针转锐角 θ 得 $A\prime$，再将 B 绕 $A\prime$ 顺时针转 θ 得 $B\prime$。若 $AB\prime = 4/3$，将 $\cos\theta$ 化为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2000-II-Q09] | Part B (Q6-Q10)

A complex number satisfies $z + z^{-1} = 2\cos 3^\circ$. Find the smallest integer strictly greater than $z^{2000} + z^{-2000}$.

复数满足 $z + z^{-1} = 2\cos 3^\circ$。求严格大于 $z^{2000} + z^{-2000}$ 的最小整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-I-Q06] | Part B (Q6-Q10)

Let P be the product of the nonreal roots of $x^4 - 4x^3 + 6x^2 - 4x = 2005$. Find $\lfloor P \rfloor$.

令 P 为方程 $x^4 - 4x^3 + 6x^2 - 4x = 2005$ 的所有非实根之积，求 $\lfloor P \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-II-Q09] | Part B (Q6-Q10)

Count $1 \leq n \leq 1000$ for which $(\sin t + i\cos t)^n = \sin(nt) + i\cos(nt)$ holds for every real t.

求 $1 \leq n \leq 1000$ 中使 $(\sin t + i\cos t)^n = \sin(nt) + i\cos(nt)$ 对所有实数 t 成立的整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q09] | Part B (Q6-Q10)

Start at $(5,0)$ and repeat 150 times: rotate counterclockwise 45° about the origin, then translate ten units right. If the endpoint is (p,q), find $\lfloor |p|+|q| \rfloor$.

从 $(5,0)$ 出发，重复一百五十次“绕原点逆时针转四十五度，再向右平移十”，终点为 (p,q)，求 $\lfloor |p|+|q| \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2010-II-Q07] | Part B (Q6-Q10)

A monic cubic with real coefficients has roots $w+3i, w+9i, 2w-4$ for some complex w. If it is z^3+az^2+bz+c, find $|a+b+c|$.

某首一实系数三次多项式的根为 $w+3i, w+9i, 2w-4$。若其为 z^3+az^2+bz+c，求 $|a+b+c|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2011-II-Q08] | Part B (Q6-Q10)

Let $z_1, \ldots, z_{12}$ be all roots of $z^{12} = 2^{36}$. Independently choose $w_j = z_j$ or iz_j. The largest real part of $\sum w_j$ is $m + \sqrt{n}$. Find $m + n$.

令 $z_1, \ldots, z_{12}$ 为 $z^{12} = 2^{36}$ 的全部根。每项独立选择 $w_j = z_j$ 或 iz_j。若 $\sum w_j$ 实部的最大值为 $m + \sqrt{n}$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q06] | Part B (Q6-Q10)

Complex numbers obey $z^{13} = w, w^{11} = z$, and $\operatorname{Im} z = \sin(m\pi/n)$ for coprime integers $0 < m < n$. Find n.

复数满足 $z^{13} = w, w^{11} = z$，且 $\operatorname{Im} z = \sin(m\pi/n)$，其中整数 $0 < m < n$ 互质。求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q06] | Part B (Q6-Q10)

Among complex $z = a + bi$ with $|z| = 5, b > 0$, choose one maximizing the distance from $(1 + 2i)z^3$ to z^5. If $z^4 = c + di$, find $c + d$.

复数 $z = a + bi$ 满足 $|z| = 5, b > 0$，取使 $(1 + 2i)z^3$ 与 z^5 的距离最大的 z。若 $z^4 = c + di$，求 $c + d$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q08] | Part B (Q6-Q10)

Nonzero complex z, w satisfy $z + 20i/w = 5 + i$ and $w + 12i/z = -4 + 10i$. Find the least $|zw|^2$.

非零复数满足 $z + 20i/w = 5 + i, w + 12i/z = -4 + 10i$，求 $|zw|^2$ 的最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q10] | Part B (Q6-Q10)

For nonzero integers a, b, r, s, the cubic $x^3 - ax^2 + bx - 65$ has root $r + si$. For each distinct possible coefficient pair (a, b), take the sum of its three roots. Find the sum of these values over all pairs.

非零整数 a, b, r, s 使 $x^3 - ax^2 + bx - 65$ 有根 $r + si$。对每个不同的可行系数对 (a, b)，取其三根之和，再将所有这些值相加，求总和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-I-Q07] | Part B (Q6-Q10)

Complex w, z satisfy $|w| = 1, |z| = 10$. Let $\theta = \arg((w - z)/z)$. If the greatest possible $\tan^2 \theta$ is reduced p/q, find $p + q$.

复数满足 $|w| = 1, |z| = 10$，令 $\theta = \arg((w - z)/z)$。若 $\tan^2 \theta$ 的最大值为最简分数 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-II-Q10] | Part B (Q6-Q10)

Fix complex z with $|z| = 2014$. A polygon has vertices z and every complex w satisfying $1/(z+w) = 1/z + 1/w$. Its area is $n\sqrt{3}$ for an integer n. Find $n \bmod 1000$.

固定模为二千零一十四的复数 z，以 z 及所有满足 $1/(z+w) = 1/z + 1/w$ 的复数 w 为顶点构成多边形。面积为整数 n 乘 $\sqrt{3}$，求 n 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2016-I-Q07] | Part B (Q6-Q10)

For integers a, b, consider $z = [\sqrt{ab + 2016} - i\sqrt{|a + b|}]/(ab + 100)$, with the principal complex square root. Count ordered pairs for which the expression is defined and real.

对整数 a, b，考虑 $z = [\sqrt{ab + 2016} - i\sqrt{|a + b|}]/(ab + 100)$，根号取复数主平方根。求使表达式有定义且为实数的有序对数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-I-Q10] | Part B (Q6-Q10)

Let $z_1 = 18 + 83i, z_2 = 18 + 39i, z_3 = 78 + 99i$. Among complex $z \neq z_3$ for which $[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ is real, choose the one with largest imaginary part. Find its real part.

令 $z_1 = 18 + 83i, z_2 = 18 + 39i, z_3 = 78 + 99i$。在使 $[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ 为实数的 $z \neq z_3$ 中，选择虚部最大者。求其实部。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-I-Q06] | Part B (Q6-Q10)

Count unit complex numbers z for which $z^{720} - z^{120}$ is real. Find the count modulo 1000.

求单位复数 z 的个数，使 $z^{720} - z^{120}$ 为实数，取个数模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-II-Q08] | Part B (Q6-Q10)

Real coefficients $a, b, c \leq 2019$ define $f(z) = az^{2018} + bz^{2017} + cz^{2016}$ with $f((1+i\sqrt{3})/2) = 2015 + 2019i\sqrt{3}$. Find $f(1) \bmod 1000$.

实系数 $a, b, c \leq 2019$ 定义 $f(z) = az^{2018} + bz^{2017} + cz^{2016}$，且 $f((1+i\sqrt{3})/2) = 2015 + 2019i\sqrt{3}$。求 $f(1)$ 除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2020-I-Q08] | Part B (Q6-Q10)

Starting at O, a walker moves 5 units east. Each subsequent displacement is half the preceding length and is rotated 60° counterclockwise. The positions converge to P. If $OP^2 = m/n$ in lowest terms, find $m+n$.

从 O 出发，首段向东走 5 单位；以后每段长度为前段一半，方向相对前段逆时针转 60°。位置趋于 P。若 $OP^2 = m/n$ 已约分，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-II-Q08] | Part B (Q6-Q10)

Let $\omega = e^{2\pi i/7}$. Evaluate $\prod_{k=0}^{6}(1+\omega^k+\omega^{3k})$.

令 $\omega = e^{2\pi i/7}$，求 $\prod_{k=0}^{6}(1+\omega^k+\omega^{3k})$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-I-Q07] | Part B (Q6-Q10)

For complex z with $|z| = 4$, find the maximum real part of $(75 + 117i)z + (96 + 144i)/z$.

复数 z 满足 $|z| = 4$，求 $(75 + 117i)z + (96 + 144i)/z$ 实部的最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2001-II-Q14] | Part C (Q11-Q15)

List the unit-modulus roots of $z^{28} - z^8 - 1 = 0$ by increasing arguments $0 \le \theta_1 < \cdots < \theta_{2n} < 360^\circ$. Find $\theta_2 + \theta_4 + \cdots + \theta_{2n}$.

将 $z^{28} - z^8 - 1 = 0$ 的单位模根按辐角 $0 \le \theta_1 < \cdots < \theta_{2n} < 360^\circ$ 排序，求偶下标辐角之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2002-I-Q12] | Part C (Q11-Q15)

Let $F(z) = (z+i)/(z-i)$ and $z_n = F(z_{n-1})$, starting from $z_0 = 1/137 + i$. If $z_{2002} = a + bi$, find $a + b$.

令 $F(z) = (z+i)/(z-i)$，$z_n = F(z_{n-1})$，初值 $z_0 = 1/137 + i$。若 $z_{2002} = a + bi$，求 $a + b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2003-II-Q15] | Part C (Q11-Q15)

Let $P(x) = 24x^{24} + \sum_{j=1}^{23}(24-j)(x^{24-j} + x^{24+j})$. For each distinct zero z_k, write $z_k^2 = a_k + ib_k$. If $\sum_k |b_k| = m + n\sqrt{p}$ with squarefree p, find $m+n+p$.

令 $P(x) = 24x^{24} + \sum_{j=1}^{23}(24-j)(x^{24-j} + x^{24+j})$。对每个不同零点写 $z_k^2 = a_k + ib_k$。若 $\sum_k |b_k| = m + n\sqrt{p}$，$p$ 无平方因子，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2004-I-Q13] | Part C (Q11-Q15)

The 34 roots of $P(x) = (1 + x + \cdots + x^{17})^2 - x^{17}$ have arguments $2\pi\alpha_k$ ordered by $0 < \alpha_1 \le \cdots \le \alpha_{34} < 1$. If $\alpha_1 + \cdots + \alpha_5 = m/n$ reduced, find $m + n$.

多项式 $P(x) = (1 + x + \cdots + x^{17})^2 - x^{17}$ 的三十四根辐角为 $2\pi\alpha_k$，按 $0 < \alpha_1 \le \cdots \le \alpha_{34} < 1$ 排序。若前五个 α 和约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q13] | Part C (Q11-Q15)

A regular hexagon centered at zero in the complex plane has opposite sides distance one apart, with one pair vertical. Let R be its exterior and $S = \{1/z : z \in R\}$. If S has area $a\pi + \sqrt{b}$ for positive integers, find $a + b$.

复平面正六边形中心为零，对边距离为一，其中一对竖直。令 R 为其外部，$S = \{1/z : z \in R\}$。若面积为 $a\pi + \sqrt{b}$，求 $a + b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-II-Q13] | Part C (Q11-Q15)

Divide a radius-two semicircle from endpoint A to endpoint B into seven equal arcs by points $C_1, \ldots, C_6$. Multiply all twelve lengths AC_i, BC_i to obtain N. Find $N \bmod 1000$.

半径二的半圆弧端点为 A, B，用六点 C_i 分成七等弧。十二条弦 AC_i, BC_i 长度之积为 N，求其除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q14] | Part C (Q11-Q15)

The roots a, b, c of $z^3 + qz + r$ are vertices of a right triangle in the complex plane. Given $|a|^2 + |b|^2 + |c|^2 = 250$, find the squared hypotenuse length.

多项式 $z^3 + qz + r$ 的三个复根在复平面构成直角三角形，且模平方和为 250。求斜边长的平方。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q14] | Part C (Q11-Q15)

For $\pi \le \theta < 2\pi$, define $P = \sum_{k\ge 1} 2^{-k} \sin(k(\theta + \pi/2))$ and $Q = 1 + \sum_{k\ge 1} 2^{-k} \cos(k(\theta + \pi/2))$. Given $P/Q = 2\sqrt{2}/7$, write $\sin\theta = -m/n$ in lowest terms and find $m + n$.

$\pi \le \theta < 2\pi$，定义 $P = \sum_{k\ge 1} 2^{-k} \sin(k(\theta + \pi/2))$，$Q = 1 + \sum_{k\ge 1} 2^{-k} \cos(k(\theta + \pi/2))$。若 $P/Q = 2\sqrt{2}/7$，将 $\sin\theta = -m/n$ 约为最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-II-Q12] | Part C (Q11-Q15)

Circle C_0 has radius1 and marked point A_0. For $0 < r < 1$, circle C_n has radius r^n and is internally tangent to C_{n-1} at A_{n-1}. Mark A_n by rotating the radius from its center to A_{n-1} counterclockwise by90 degrees. All these disks have one common point B. When $r = 11/60$, express the distance from the center of C_0 to B as reduced m/n and find $m + n$.

圆 C_0 半径为1，标记点为 A_0。对 $0 < r < 1$，圆 C_n 半径为 r^n，并在 A_{n-1} 与 C_{n-1} 内切。将从其圆心指向 A_{n-1} 的半径逆时针旋转90度得到 A_n。所有闭圆盘共有一点 B。当 $r = 11/60$，将 C_0 圆心到 B 的距离写成最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-I-Q12] | Part C (Q11-Q15)

Let $f(z) = z^2 - 19z$. A value $z = m + \sqrt{n} + 11i$ with positive integers m, n makes $z, f(z), f(f(z))$ a nondegenerate right triangle with right angle at $f(z)$. Find $m + n$.

令 $f(z) = z^2 - 19z$。某 $z = m + \sqrt{n} + 11i$（m, n 为正整数）使 $z, f(z), f(f(z))$ 成非退化直角三角形，直角在 $f(z)$。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-I-Q15] | Part C (Q11-Q15)

Find the largest prime $p < 1000$ for which a Gaussian integer z has $|z| = \sqrt{p}$ and the three positive numbers $p, \mathrm{Re}(z^3), \mathrm{Im}(z^3)$ are triangle side lengths.

求小于 1000 的最大素数 p，使存在实部、虚部均为整数的复数 z，满足 $|z| = \sqrt{p}$，且 $p, \mathrm{Re}(z^3), \mathrm{Im}(z^3)$ 均为正并可作三角形三边。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-II-Q13] | Part C (Q11-Q15)

For a nontrivial thirteenth root of unity ω, find $\prod_{k=0}^{12}(2-2\omega^k+\omega^{2k})$ modulo 1000.

设 $\omega \neq 1$ 为十三次单位根，求 $\prod_{k=0}^{12}(2-2\omega^k+\omega^{2k})$ 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-II-Q11] | Part C (Q11-Q15)

Find the greatest integer n for which $x^3 - (n/6)x^2 + (n-11)x - 400$ has roots $\alpha^2, \beta^2, \gamma^2$ allowing exactly seven distinct sums $\alpha + \beta + \gamma$.

求最大整数 n，使三次多项式 $x^3 - (n/6)x^2 + (n-11)x - 400$ 的根为 $\alpha^2, \beta^2, \gamma^2$ 时，$\alpha + \beta + \gamma$ 恰有七个不同可能值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2000-II-Q01] | Part A (Q1-Q5)

Write $2/\log_4(2000^6) + 3/\log_5(2000^6) = m/n$ in lowest positive terms. Find $m+n$.

将 $2/\log_4(2000^6) + 3/\log_5(2000^6)$ 约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2003-I-Q04] | Part A (Q1-Q5)

Given $\log_{10}(\sin x)+\log_{10}(\cos x)=-1$ and $\log_{10}(\sin x+\cos x)=(\log_{10} n-1)/2$, find n.

已知 $\log_{10}(\sin x)+\log_{10}(\cos x)=-1$，且 $\log_{10}(\sin x+\cos x)=(\log_{10} n-1)/2$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2005-II-Q05] | Part A (Q1-Q5)

Count integer pairs $2 \le a, b \le 2005$ satisfying $\log_a b + 6 \log_b a = 5$.

求整数对 $2 \le a, b \le 2005$ 满足 $\log_a b + 6 \log_b a = 5$ 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2006-II-Q02] | Part A (Q1-Q5)

For how many positive integers n do $\log_{10} 12, \log_{10} 75, \log_{10} n$ form the sides of a positive-area triangle?

有多少个正整数 n 使 $\log_{10} 12, \log_{10} 75, \log_{10} n$ 构成非退化三角形的三边?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2009-II-Q02] | Part A (Q1-Q5)

Positive a, b, c satisfy $a^{\log_3 7} = 27, b^{\log_7 11} = 49, c^{\log_{11} 25} = \sqrt{11}$. Find $a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}$.

正数 a, b, c 满足 $a^{\log_3 7} = 27, b^{\log_7 11} = 49, c^{\log_{11} 25} = \sqrt{11}$，求 $a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2010-I-Q03] | Part A (Q1-Q5)

Positive reals satisfy $y = 3x/4$ and $x^y = y^x$. Write $x + y = r/s$ in lowest terms and find $r + s$.

正实数满足 $y = 3x/4$ 及 $x^y = y^x$。将 $x + y$ 约为 r/s，求 $r + s$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2010-II-Q05] | Part A (Q1-Q5)

Positive x, y, z obey $xyz = 10^{81}$ and $(\log_{10} x)(\log_{10}(yz)) + (\log_{10} y)(\log_{10} z) = 468$. Find $\sqrt{(\log_{10} x)^2 + (\log_{10} y)^2 + (\log_{10} z)^2}$.

正数满足 $xyz = 10^{81}$ 及 $(\log_{10} x)(\log_{10}(yz)) + (\log_{10} y)(\log_{10} z) = 468$。求三者十进对数平方和的平方根。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2013-II-Q02] | Part A (Q1-Q5)

Positive integers a, b satisfy $\log_2(\log_{2^a}(\log_{2^b}(2^{1000}))) = 0$. Find the sum of the distinct possible values of $a + b$.

正整数满足 $\log_2(\log_{2^a}(\log_{2^b}(2^{1000}))) = 0$。求所有不同可能 $a + b$ 值之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2016-II-Q03] | Part A (Q1-Q5)

Real numbers x, y, z satisfy $\log_2(xyz - 3 + \log_5 x) = 5$, $\log_3(xyz - 3 + \log_5 y) = 4$, and $\log_4(xyz - 3 + \log_5 z) = 4$. Find $|\log_5 x| + |\log_5 y| + |\log_5 z|$.

实数 x, y, z 满足 $\log_2(xyz - 3 + \log_5 x) = 5$、$\log_3(xyz - 3 + \log_5 y) = 4$、$\log_4(xyz - 3 + \log_5 z) = 4$。求 $|\log_5 x| + |\log_5 y| + |\log_5 z|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2018-I-Q05] | Part A (Q1-Q5)

For pairs (x, y) satisfying $\log_2(2x + y) = \log_4(x^2 + xy + 7y^2)$, consider every real K making $\log_3(3x + y) = \log_9(3x^2 + 4xy + Ky^2)$ valid for that pair. Find the product of the distinct possible K.

实数对满足 $\log_2(2x + y) = \log_4(x^2 + xy + 7y^2)$。考虑使该数对还满足有意义的 $\log_3(3x + y) = \log_9(3x^2 + 4xy + Ky^2)$ 的实数 K，求所有不同可能 K 的乘积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2020-I-Q02] | Part A (Q1-Q5)

For $x > 0$, the ordered list $\log_8(2x), \log_4 x, \log_2 x$ is geometric with a positive ratio. If $x = m/n$ is reduced, find $m + n$.

当 $x > 0$ 时，$\log_8(2x), \log_4 x, \log_2 x$ 依次构成公比为正的等比数列。将 x 约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2020-II-Q03] | Part A (Q1-Q5)

Solve $\log_{2^x}(3^{20}) = \log_{2^{x+3}}(3^{2020})$. If the solution is m/n reduced with positive integers m, n, find $m+n$.

解 $\log_{2^x}(3^{20}) = \log_{2^{x+3}}(3^{2020})$。若解约分为正整数之比 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2022-II-Q04] | Part A (Q1-Q5)

A positive real $x \notin \{1/20, 1/2\}$ satisfies $\log_{20x}(22x) = \log_{2x}(202x)$. Write their common value as $\log_{10}(m/n)$ with coprime positive m, n. Find $m + n$.

正实数 $x \notin \{1/20, 1/2\}$ 满足 $\log_{20x}(22x) = \log_{2x}(202x)$。将共同值表示为 $\log_{10}(m/n)$，其中 m, n 为互质正整数。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2023-I-Q02] | Part A (Q1-Q5)

Positive $b \neq 1, n$ obey $\sqrt{\log_b n} = \log_b \sqrt{n}$ and $b \log_b n = \log_b(bn)$. Express $n = j/k$ in lowest terms and compute $j + k$.

正数 $b \neq 1, n$ 满足 $\sqrt{\log_b n} = \log_b \sqrt{n}$ 和 $b \log_b n = \log_b(bn)$。将 n 写成最简分数 j/k，求 $j + k$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2024-I-Q02] | Part A (Q1-Q5)

For real $x, y > 1$, both $\log_x(y^x)$ and $\log_y(x^{4y})$ equal 10. Determine xy.

实数 $x, y > 1$ 满足 $\log_x(y^x) = \log_y(x^{4y}) = 10$，求 xy。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2024-II-Q04] | Part A (Q1-Q5)

Positive x, y, z satisfy $\log_2(x/(yz)) = 1/2$, $\log_2(y/(xz)) = 1/3$, and $\log_2(z/(xy)) = 1/4$. If $|\log_2(x^4y^3z^2)| = m/n$ in lowest terms, find $m+n$.

正数 x, y, z 满足 $\log_2(x/(yz)) = 1/2, \log_2(y/(xz)) = 1/3, \log_2(z/(xy)) = 1/4$。若 $|\log_2(x^4y^3z^2)| = m/n$ 为最简分数，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2025-II-Q04] | Part A (Q1-Q5)

Write $\prod_{k=4}^{63} \frac{\log_k(5^{k^2-1})}{\log_{k+1}(5^{k^2-4})} = m/n$ in lowest terms and find $m+n$.

将 $\prod_{k=4}^{63} \frac{\log_k(5^{k^2-1})}{\log_{k+1}(5^{k^2-4})}$ 化为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2000-I-Q09] | Part B (Q6-Q10)

The positive-real system is $\log_{10}(2000xy) - (\log_{10} x)(\log_{10} y) = 4$, $\log_{10}(2yz) - (\log_{10} y)(\log_{10} z) = 1$, $\log_{10}(zx) - (\log_{10} z)(\log_{10} x) = 0$. Its two solutions have second coordinates y_1, y_2. Find $y_1 + y_2$.

正实数方程组为 $\log_{10}(2000xy) - (\log_{10} x)(\log_{10} y) = 4$、$\log_{10}(2yz) - (\log_{10} y)(\log_{10} z) = 1$、$\log_{10}(zx) - (\log_{10} z)(\log_{10} x) = 0$。两组解的第二坐标为 y_1, y_2，求其和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2002-I-Q06] | Part B (Q6-Q10)

The two solutions of $\log_{225} x + \log_{64} y = 4$ and $\log_x 225 - \log_y 64 = 1$ are $(x_1, y_1), (x_2, y_2)$. Find $\log_{30}(x_1 y_1 x_2 y_2)$.

方程组 $\log_{225} x + \log_{64} y = 4, \log_x 225 - \log_y 64 = 1$ 有两组解 $(x_1, y_1), (x_2, y_2)$。求 $\log_{30}(x_1 y_1 x_2 y_2)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2005-I-Q08] | Part B (Q6-Q10)

The equation $2^{333x-2} + 2^{111x+2} = 2^{222x+1} + 1$ has three real roots with reduced sum m/n. Find $m+n$.

方程 $2^{333x-2} + 2^{111x+2} = 2^{222x+1} + 1$ 有三个实根，其和为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2011-I-Q09] | Part B (Q6-Q10)

For $0 \le x \le \pi/2$, the logarithm $\log_{24\sin x}(24\cos x)$ exists and equals $3/2$. Find $24\cot^2 x$.

$x \in [0, \pi/2]$，且有意义的对数 $\log_{24\sin x}(24\cos x) = 3/2$，求 $24\cot^2 x$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2012-I-Q09] | Part B (Q6-Q10)

Positive x, y, z have valid logarithms satisfying $2\log_x(2y) = 2\log_{2x}(4z) = \log_{2x^4}(8yz) \neq 0$. If $xy^5z = 2^{-p/q}$ with coprime positive p, q, find $p + q$.

正数 x, y, z 的各对数有意义，且 $2\log_x(2y) = 2\log_{2x}(4z) = \log_{2x^4}(8yz) \neq 0$。若 $xy^5z = 2^{-p/q}$ 且 p, q 为互质正整数，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2013-I-Q08] | Part B (Q6-Q10)

For positive integers $m > 1, n$, the real domain of $\arcsin(\log_m(nx))$ has length $1/2013$. Find the least possible $m + n$ modulo 1000.

正整数 $m > 1, n$ 使 $\arcsin(\log_m(nx))$ 的实数定义域长度为 $1/2013$。求 $m + n$ 最小值模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2017-II-Q07] | Part B (Q6-Q10)

For how many integers $k \in [-500, 500]$ does $\log(kx) = 2\log(x+2)$ have exactly one real solution, using common logarithms?

在整数 $k \in [-500, 500]$ 中，有多少个使常用对数方程 $\log(kx) = 2\log(x+2)$ 恰有一个实数解?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2019-II-Q06] | Part B (Q6-Q10)

All unspecified logarithms use one base $b \geq 2$. The equations $3\log_b(\sqrt{x}\log_b x) = 56$ and $\log_{\log_b x} x = 54$ have a common real solution $x > 1$. Find b.

所有未注明的对数均以同一 $b \geq 2$ 为底。方程 $3\log_b(\sqrt{x}\log_b x) = 56$、$\log_{\log_b x} x = 54$ 有实数公共解 $x > 1$，求 b。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2026-I-Q06] | Part B (Q6-Q10)

The product of all positive solutions of $\sqrt[20]{x^{\log_{2026} x}} = 26x$ is an integer P. Find its number of positive divisors.

方程 $\sqrt[20]{x^{\log_{2026} x}} = 26x$ 的所有正实根之积为整数 P，求 P 的正约数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.4 Logarithms & exponentials

对数与指数

[AIME-2017-I-Q14] | Part C (Q11-Q15)

Numbers $a, x > 1$ satisfy $\log_a(\log_a(\log_a 2) + \log_a 24 - 128) = 128$ and $\log_a(\log_a x) = 256$. Find the remainder of x modulo1000.

$a, x > 1$ 满足 $\log_a(\log_a(\log_a 2) + \log_a 24 - 128) = 128$、$\log_a(\log_a x) = 256$。求 x 模1000的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2001-II-Q03] | Part A (Q1-Q5)

Let $x_1 = 211, x_2 = 375, x_3 = 420, x_4 = 523$ and $x_n = x_{n-1} - x_{n-2} + x_{n-3} - x_{n-4}$ for $n \geq 5$. Find $x_{531} + x_{753} + x_{975}$.

数列初值为 $211, 375, 420, 523$，且 $n \geq 5$ 时 $x_n = x_{n-1} - x_{n-2} + x_{n-3} - x_{n-4}$。求 $x_{531} + x_{753} + x_{975}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2002-I-Q04] | Part A (Q1-Q5)

For $a_k = 1/(k^2+k)$, positive integers $m < n$ satisfy $\sum_{k=m}^{n-1} a_k = 1/29$. Find $m+n$.

令 $a_k = 1/(k^2+k)$，正整数 $m < n$ 满足 $\sum_{k=m}^{n-1} a_k = 1/29$，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2004-I-Q02] | Part A (Q1-Q5)

Set A contains m consecutive integers with sum $2m$. Set B contains $2m$ consecutive integers with sum m. Their largest elements differ in absolute value by 99. Find m.

集合 A 有 m 个连续整数，总和 $2m$；集合 B 有 $2m$ 个连续整数，总和 m。两集合最大元之差绝对值为 99，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q03] | Part A (Q1-Q5)

An infinite geometric series sums to 2005. Squaring each term gives a series with sum 20050. If the original ratio is reduced $m/n > 0$, find $m + n$.

无穷等比级数和为 2005，每项平方后的级数和为 20050。若原公比为最简正分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q01] | Part A (Q1-Q5)

Let $N = 100^2 + 99^2 - 98^2 - 97^2 + 96^2 + 95^2 - \cdots + 4^2 + 3^2 - 2^2 - 1^2$. Find $N \bmod 1000$.

令 $N = 100^2 + 99^2 - 98^2 - 97^2 + 96^2 + 95^2 - \cdots + 4^2 + 3^2 - 2^2 - 1^2$，求其除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q03] | Part A (Q1-Q5)

The interior angles of a convex eighteen-gon are distinct increasing integers in arithmetic progression, measured in degrees. Find the least angle.

凸十八边形的内角度数为严格递增的整数等差数列。求最小内角。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q05] | Part A (Q1-Q5)

A geometric series has sums $S_{2011} = 200$ and $S_{4022} = 380$. Find S_{6033}.

某等比数列前 2011 项和为 200，前 4022 项和为 380，求前 6033 项和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2012-I-Q02] | Part A (Q1-Q5)

An arithmetic sequence sums to 715. Adding $2k-1$ to its kth term changes the total to 836. Find the sum of its original first, middle, and last terms.

一等差数列各项和为 715。将第 k 项增加 $2k-1$ 后，总和为 836。求原数列首项、中间项及末项之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2012-II-Q02] | Part A (Q1-Q5)

Two geometric sequences have the same nonzero common ratio, first terms $27, 99$, and $a_{15} = b_{11}$. Find a_9.

两个等比数列有相同的非零公比，首项分别为 $27, 99$，且 $a_{15} = b_{11}$。求 a_9。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2013-II-Q03] | Part A (Q1-Q5)

A 119-cm candle takes $10k$ seconds to burn its kth centimeter from the top. If the full burn lasts T, its remaining height at time $T/2$ is h. Find $10h$.

一根一百一十九厘米的蜡烛，从顶端起第 k 厘米需燃烧 $10k$ 秒。完全燃尽时间为 T，在 $T/2$ 时剩余高度为 h，求 $10h$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q01] | Part A (Q1-Q5)

Let $A = \sum_{k=1}^{19}(2k-1)(2k) + 39$ and $B = 1 + \sum_{k=1}^{19}(2k)(2k+1)$. Calculate $|A - B|$.

令 $A = \sum_{k=1}^{19}(2k-1)(2k) + 39$，$B = 1 + \sum_{k=1}^{19}(2k)(2k+1)$。求 $|A - B|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q01] | Part A (Q1-Q5)

For $|r| < 1$, define $S(r) = 12 \sum_{k=0}^{\infty} r^k$. If $|a| < 1$ and $S(a)S(-a) = 2016$, find $S(a) + S(-a)$.

对 $|r| < 1$，定义 $S(r) = 12 \sum_{k=0}^{\infty} r^k$。若 $|a| < 1$ 且 $S(a)S(-a) = 2016$，求 $S(a) + S(-a)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q05] | Part A (Q1-Q5)

A reader finishes a 374-page book over at least two days. On day j, she reads $n+j-1$ pages in $t+j-1$ minutes, with positive integers n, t. Her total reading time is 319 minutes. Find $n+t$.

某人用至少两天读完 374 页书。第 j 天读 $n+j-1$ 页，用时 $t+j-1$ 分钟，n, t 为正整数。总用时 319 分钟。求 $n+t$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q01] | Part A (Q1-Q5)

Three people initially hold increasing peanut counts $A < B < C$, in geometric progression, totaling 444. After they eat respectively $5, 9, 25$ peanuts, the counts in the same order form an arithmetic progression. Find A.

三人起初的花生数 $A < B < C$ 成等比数列，总计 444。他们依次吃掉 $5, 9, 25$ 颗后，按同一顺序的剩余数量成等差数列。求 A。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2023-II-Q01] | Part A (Q1-Q5)

Six tree harvests form an arithmetic progression with sum 990, and the largest harvest is twice the smallest. Find the largest harvest.

六棵树的果实数成等差数列，总数 990，最大项为最小项的两倍。求最大项。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2026-II-Q01] | Part A (Q1-Q5)

Sum the tenth terms of all integer arithmetic sequences whose first term is 4 and which contain both 24 and 34.

求所有首项为 4 且包含 $24, 34$ 的整数等差数列的第十项之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q06] | Part B (Q6-Q10)

Find the nearest integer to $1000\sum_{n=3}^{10000} 1/(n^2-4)$.

求最接近 $1000\sum_{n=3}^{10000} 1/(n^2-4)$ 的整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2003-II-Q08] | Part B (Q6-Q10)

A sequence begins $1440, 1716, 1848$ and is the termwise product of two arithmetic sequences. Find its eighth term.

数列首三项为 $1440, 1716, 1848$，它由两个等差数列对应项相乘得到，求第八项。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2004-II-Q09] | Part B (Q6-Q10)

Positive integers a_j begin with $a_1 = 1$. Every triple at odd starting index is geometric; every triple at even starting index is arithmetic. Given $a_9 + a_{10} = 646$, let a_n be the largest sequence term below 1000. Find $n + a_n$.

正整数数列首项为一，奇数起始下标的每组三项等比，偶数起始下标的每组三项等差。已知 $a_9 + a_{10} = 646$，设 a_n 为小于一千的最大项，求 $n + a_n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2008-I-Q06] | Part B (Q6-Q10)

The top row of a triangular sum array is $1, 3, 5, \ldots, 99$. Each lower cell is the sum of the two just above it, until one bottom cell remains. How many array entries are divisible by 67?

三角求和阵首行为 $1, 3, 5, \ldots, 99$，每下一行少一项，每项为上方相邻两项之和，直至底部一项。求全阵中被六十七整除的项数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.5 Sequences & summation

数列与求和

[AIME-2008-II-Q06] | Part B (Q6-Q10)

Sequences a_n, b_n both obey $u_n = u_{n-1} + u_{n-1}^2/u_{n-2}$ for $n \geq 2$. Initial terms are $(a_0, a_1) = (1, 1)$ and $(b_0, b_1) = (1, 3)$. Find b_{32}/a_{32}.

两数列都满足 $u_n = u_{n-1} + u_{n-1}^2/u_{n-2}$，初值分别为 $(1, 1)$ 与 $(1, 3)$，求 b_{32}/a_{32}。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q07] | Part B (Q6-Q10)

A real sequence has $a_1 = 1$ and $5^{a_{n+1}-a_n} - 1 = 1/(n + 2/3)$. Find the smallest integer $k > 1$ for which a_k is an integer.

实数列满足 $a_1 = 1, 5^{a_{n+1}-a_n} - 1 = 1/(n + 2/3)$，求使 a_k 为整数的最小整数 $k > 1$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2009-I-Q08] | Part B (Q6-Q10)

Take all positive differences between elements of $\{2^0, 2^1, \ldots, 2^{10}\}$. If their sum is N, find $N \bmod 1000$.

取集合 $\{2^0, 2^1, \ldots, 2^{10}\}$ 中元素两两的所有正差，和为 N，求除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.5 Sequences & summation

数列与求和

[AIME-2014-II-Q07] | Part B (Q6-Q10)

Let $f(x) = (x^2 + 3x + 2)^{\cos \pi x}$. Find the sum of positive integers n such that $|\sum_{k=1}^{n} \log_{10} f(k)| = 1$.

令 $f(x) = (x^2 + 3x + 2)^{\cos \pi x}$，求使 $|\sum_{k=1}^{n} \log_{10} f(k)| = 1$ 的所有正整数 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2015-I-Q09] | Part B (Q6-Q10)

Each ordered starting triple $(a_1, a_2, a_3) \in \{1, \ldots, 10\}^3$ generates $a_n = a_{n-1}|a_{n-2} - a_{n-3}|$ for $n \geq 4$. Count the starting triples for which some term is zero.

每个起始有序三元组 $(a_1, a_2, a_3) \in \{1, \ldots, 10\}^3$ 按 $a_n = a_{n-1}|a_{n-2} - a_{n-3}|$ （$n \geq 4$）生成数列。求最终出现零项的起始三元组数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-I-Q10] | Part B (Q6-Q10)

An infinite strictly increasing sequence of positive integers has each triple $(a_{2k-1}, a_{2k}, a_{2k+1})$ geometric and each $(a_{2k}, a_{2k+1}, a_{2k+2})$ arithmetic. Given $a_{13} = 2016$, find a_1.

一个无限严格递增正整数数列中，每个 $(a_{2k-1}, a_{2k}, a_{2k+1})$ 为等比数列，每个 $(a_{2k}, a_{2k+1}, a_{2k+2})$ 为等差数列。已知 $a_{13} = 2016$，求 a_1。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2016-II-Q09] | Part B (Q6-Q10)

An increasing positive-integer arithmetic sequence a_n and an increasing positive-integer geometric sequence b_n both start at1. Put $c_n = a_n + b_n$. For some integer k, $c_{k-1} = 100, c_{k+1} = 1000$. Find c_k.

递增正整数等差数列 a_n、递增正整数等比数列 b_n 均从1开始，令 $c_n = a_n + b_n$。某整数 k 满足 $c_{k-1} = 100, c_{k+1} = 1000$。求 c_k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2020-II-Q06] | Part B (Q6-Q10)

Define $t_1 = 20, t_2 = 21$ and $t_n = (5t_{n-1} + 1)/(25t_{n-2})$ for $n \geq 3$. Write $t_{2020} = p/q$ reduced and find $p + q$.

数列满足 $t_1 = 20, t_2 = 21$，且 $n \geq 3$ 时 $t_n = (5t_{n-1} + 1)/(25t_{n-2})$。将 t_{2020} 约分为 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 1.5 Sequences & summation

数列与求和

[AIME-2002-II-Q11] | Part C (Q11-Q15)

Two distinct convergent real geometric series each sum to 1 and have the same second term. One has third term $1/8$. Their common second term is $(\sqrt{m}-n)/p$ with squarefree m. Find $100m+10n+p$.

两个不同的收敛实等比级数和都为 1，且第二项相同，其中一个第三项为 $1/8$。共同第二项为 $(\sqrt{m}-n)/p$，m 无平方因子，求 $100m+10n+p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2005-II-Q11] | Part C (Q11-Q15)

A real sequence has $a_0 = 37, a_1 = 72$, $a_m = 0$, and $a_{k+1} = a_{k-1} - 3/a_k$ for $1 \leq k < m$. Find the positive integer m.

实数列满足 $a_0 = 37, a_1 = 72, a_m = 0$，且 $1 \leq k < m$ 时 $a_{k+1} = a_{k-1} - 3/a_k$，求正整数 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2006-I-Q15] | Part C (Q11-Q15)

A real sequence satisfies $x_0 = 0$ and $|x_k| = |x_{k-1} + 3|$ for $k \geq 1$. Minimize $|\sum_{k=1}^{2006} x_k|$.

实数列满足 $x_0 = 0, |x_k| = |x_{k-1} + 3|$，$k \geq 1$。求 $|\sum_{k=1}^{2006} x_k|$ 的最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2006-II-Q11] | Part C (Q11-Q15)

The tribonacci sequence has $a_1 = a_2 = a_3 = 1$ and $a_{j+3} = a_{j+2} + a_{j+1} + a_j$. Given $a_{28} = 6090307, a_{29} = 11201821, a_{30} = 20603361$, find $\left(\sum_{j=1}^{28} a_j\right) \bmod 1000$.

数列前三项为一，且 $a_{j+3} = a_{j+2} + a_{j+1} + a_j$。已知 $a_{28} = 6090307, a_{29} = 11201821, a_{30} = 20603361$，求前二十八项和除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2007-I-Q14] | Part C (Q11-Q15)

Let $a_1 = a_2 = 3$ and $a_{n+1}a_{n-1} = a_n^2 + 2007$ for $n \geq 2$. Find $\left\lfloor (a_{2007}^2 + a_{2006}^2)/(a_{2007}a_{2006}) \right\rfloor$.

令 $a_1 = a_2 = 3$，$n \geq 2$ 时 $a_{n+1}a_{n-1} = a_n^2 + 2007$。求 $\left\lfloor (a_{2007}^2 + a_{2006}^2)/(a_{2007}a_{2006}) \right\rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2007-II-Q12] | Part C (Q11-Q15)

An increasing geometric sequence consists of integer powers of three, satisfies $\sum_{j=0}^{7} \log_3 x_j = 308$ and $56 \leq \log_3(\sum_{j=0}^{7} x_j) \leq 57$. Find $\log_3 x_{14}$.

递增等比数列各项均为三的整数次幂，满足前八项对数和为 308，且 $56 \leq \log_3(\sum_{j=0}^{7} x_j) \leq 57$。求 $\log_3 x_{14}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.5 Sequences & summation

数列与求和

[AIME-2011-II-Q11] | Part C (Q11-Q15)

Let D_n be the determinant of the $n \times n$ tridiagonal matrix with diagonal entries 10, entries immediately above and below the diagonal 3, and all others zero. Reduce $\sum_{n=1}^{\infty}(8D_n+1)^{-1}=p/q$ and find $p+q$.

令 D_n 为 n 阶三对角矩阵的行列式：主对角线均为十，上下紧邻对角线均为三，其余为零。将 $\sum_{n=1}^{\infty}1/(8D_n+1)$ 约为 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q05] | Part A (Q1-Q5)

An integer Fahrenheit temperature $32 \leq T \leq 1000$ is converted by $C = 5(T - 32)/9$, rounded to nearest integer, converted back and rounded again. How many T return unchanged?

将整数华氏温度 $32 \leq T \leq 1000$ 用 $C = 5(T - 32)/9$ 转成摄氏并四舍五入，再转回华氏并四舍五入。求最终与原数相同的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2002-II-Q08] | Part B (Q6-Q10)

Find the least positive integer k that is not $\lfloor 2002/n \rfloor$ for any nonzero integer n.

求最小正整数 k，使不存在非零整数 n 满足 $\lfloor 2002/n \rfloor = k$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q07] | Part B (Q6-Q10)

Let $N = \sum_{k=1}^{1000} k(\lceil \log_{\sqrt{2}} k \rceil - \lfloor \log_{\sqrt{2}} k \rfloor)$. Find $N \bmod 1000$.

令 $N = \sum_{k=1}^{1000} k(\lceil \log_{\sqrt{2}} k \rceil - \lfloor \log_{\sqrt{2}} k \rfloor)$，求除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-II-Q07] | Part B (Q6-Q10)

For an integer k, exactly seventy positive multiples n_i of k satisfy $\lfloor \sqrt[3]{n_i} \rfloor = k$. Find the largest n_i/k.

某整数 k 恰有七十个正倍数 n_i 满足 $\lfloor \sqrt[3]{n_i} \rfloor = k$，求最大的 n_i/k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2008-I-Q07] | Part B (Q6-Q10)

For $i = 0, \ldots, 999$, let S_i consist of integers from $100i$ through $100i + 99$. How many S_i contain no perfect square?

对 $i = 0$ 至 999，令 $S_i = \{100i, \ldots, 100i + 99\}$，求不含平方数的集合个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2009-I-Q06] | Part B (Q6-Q10)

Count positive integers $N < 1000$ for which $x^{\lfloor x \rfloor} = N$ has a real solution where the expression is defined.

求正整数 $N < 1000$ 的个数，使 $x^{\lfloor x \rfloor} = N$ 在有定义的实数范围内有解。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2010-I-Q08] | Part B (Q6-Q10)

Let $R = \{(x, y) : \lfloor x \rfloor^2 + \lfloor y \rfloor^2 = 25\}$. The smallest disk enclosing R has radius $\sqrt{m}/n$, with squarefree m. Find $m + n$.

设 $R = \{(x, y) : \lfloor x \rfloor^2 + \lfloor y \rfloor^2 = 25\}$。包含 R 的最小圆盘半径为 $\sqrt{m}/n$，其中 m 无平方因子。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2012-II-Q10] | Part B (Q6-Q10)

How many positive integers $n < 1000$ can be written $n = x\lfloor x \rfloor$ for some positive real x?

求小于一千的正整数 n 的个数，使存在正实数 x 满足 $n = x\lfloor x \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2021-I-Q08] | Part B (Q6-Q10)

For how many integers c does $\big||20|x| - x^2| - c\big| = 21$ have exactly twelve distinct real solutions?

有多少个整数 c，使 $\big||20|x| - x^2| - c\big| = 21$ 恰有十二个不同实数解?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2023-I-Q10] | Part B (Q6-Q10)

For the unique positive integer a making $-1000 < U < 1000$, where $U = \sum_{n=1}^{2023} \lfloor (n^2 - na)/5 \rfloor$, determine $a + U$.

正整数 a 中，恰有一个使 $U = \sum_{n=1}^{2023} \lfloor (n^2 - na)/5 \rfloor$ 严格介于 -1000 与 1000 之间。求此时 $a + U$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2004-I-Q12] | Part C (Q11-Q15)

In $(0,1]\times(0,1]$, take points where both $\lfloor \log_2(1/x)\rfloor$ and $\lfloor \log_5(1/y)\rfloor$ are even. If this set has area m/n reduced, find $m+n$.

在 $(0,1]\times(0,1]$ 中，取使 $\lfloor \log_2(1/x)\rfloor$、$\lfloor \log_5(1/y)\rfloor$ 均为偶数的点集。若面积约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2007-I-Q11] | Part C (Q11-Q15)

Let $b(p)$ be the nearest positive integer to $\sqrt{p}$. Find $\left(\sum_{p=1}^{2007} b(p)\right) \bmod 1000$.

令 $b(p)$ 为最接近 $\sqrt{p}$ 的正整数，求 $\sum_{p=1}^{2007} b(p)$ 除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2008-I-Q12] | Part C (Q11-Q15)

Cars of length four meters move at common speed v km/h. Their bumper-to-bumper gap is exactly one car length per fifteen km/h or fraction thereof. Let M be the maximum whole number of cars passing a fixed point in an hour. Find the integer quotient of M by ten.

长四米的汽车同速 v 千米每小时行驶，前后车间净距恰为每十五千米每小时（不足也计一份）一个车长。设一小时通过某点的最大整数车数为 M，求 M 除以十的整数商。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2009-II-Q11] | Part C (Q11-Q15)

For positive integers $m \geq n$, exactly fifty positive integers k satisfy $|\log m - \log k| < \log n$. Sum all possible products mn.

正整数 $m \geq n$ 使恰有五十个正整数 k 满足 $|\log m - \log k| < \log n$，求所有可能乘积 mn 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2010-I-Q14] | Part C (Q11-Q15)

For positive integers n, put $f(n) = \sum_{k=1}^{100} \lfloor \log_{10}(kn) \rfloor$. Find the greatest n with $f(n) \leq 300$.

对正整数 n，定义 $f(n) = \sum_{k=1}^{100} \lfloor \log_{10}(kn) \rfloor$。求满足 $f(n) \leq 300$ 的最大 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2011-II-Q15] | Part C (Q11-Q15)

Let $P(x) = x^2 - 3x - 9$ and choose x uniformly in $[5, 15]$. The probability that $\lfloor \sqrt{P(x)} \rfloor = \sqrt{P(\lfloor x \rfloor)}$ is $(\sqrt{a} + \sqrt{b} + \sqrt{c} - d)/e$ for positive integers. Find $a + b + c + d + e$.

令 $P(x) = x^2 - 3x - 9$，在 $[5, 15]$ 内均匀选 x。若 $\lfloor \sqrt{P(x)} \rfloor = \sqrt{P(\lfloor x \rfloor)}$ 的概率为 $(\sqrt{a} + \sqrt{b} + \sqrt{c} - d)/e$，求五个正整数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2015-I-Q14] | Part C (Q11-Q15)

For an integer $n \geq 2$, let $A(n)$ be the area defined by $1 \leq x < n$ and $0 \leq y \leq x\lfloor\sqrt{x}\rfloor$. How many integers $2 \leq n \leq 1000$ make $A(n)$ an integer?

对整数 $n \geq 2$，令 $A(n)$ 为区域 $1 \leq x < n, 0 \leq y \leq x\lfloor\sqrt{x}\rfloor$ 的面积。求 $2 \leq n \leq 1000$ 中使 $A(n)$ 为整数的 n 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2020-II-Q14] | Part C (Q11-Q15)

For real x, let $\{x\} = x - \lfloor x \rfloor$ and $f(x) = x\{x\}$. Let N count real $x \in [0, 2020]$ satisfying $f(f(f(x))) = 17$. Find $N \bmod 1000$.

定义小数部分 $\{x\} = x - \lfloor x \rfloor$，$f(x) = x\{x\}$。设 N 为区间 $[0, 2020]$ 内满足 $f(f(f(x))) = 17$ 的实数解个数，求 N 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2024-I-Q12] | Part C (Q11-Q15)

Let $f(t)=||t|-1/2|$ and $g(t)=||t|-1/4|$. How many points satisfy both $y=4g(f(\sin(2\pi x)))$ and $x=4g(f(\cos(3\pi y)))$?

令 $f(t)=||t|-1/2|, g(t)=||t|-1/4|$，求方程组 $y=4g(f(\sin(2\pi x))), x=4g(f(\cos(3\pi y)))$ 的解点个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

[AIME-2025-I-Q11] | Part C (Q11-Q15)

Extend $f(x) = x$ for $-1 \le x < 1$ and $f(x) = 2 - x$ for $1 \le x < 3$ periodically with period 4. Sum the y-coordinates of intersections with $x = 34y^2$. If the sum is $(a + b\sqrt{c})/d$ in the specified reduced squarefree form, find $a + b + c + d$.

将 $f(x) = x$（$-1 \le x < 1$）及 $f(x) = 2 - x$（$1 \le x < 3$）以周期 4 延拓。求其与 $x = 34y^2$ 所有交点的纵坐标和，写成约简且 c 无平方因子的 $(a + b\sqrt{c})/d$，求 $a + b + c + d$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-II-Q03] | Part A (Q1-Q5)

A $10 \times 13 \times 14$ rectangular block loses ten slices, each one unit thick and parallel to one of its faces. Maximize the volume left after all slices.

尺寸 $10 \times 13 \times 14$ 的长方体，依次切去十片，每片厚一且平行于某个面。求剩余体积最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2011-II-Q09] | Part B (Q6-Q10)

Nonnegative $x_1, \ldots, x_6$ sum to one and satisfy $x_1x_3x_5 + x_2x_4x_6 \geq 1/540$. Let p/q be the greatest possible cyclic sum $\sum_{i=1}^{6} x_i x_{i+1} x_{i+2}$, with indices modulo six. Find $p + q$ in lowest terms.

非负数 $x_1, \ldots, x_6$ 的和为一，且 $x_1x_3x_5 + x_2x_4x_6 \geq 1/540$。设循环三项积和 $\sum_{i=1}^{6} x_i x_{i+1} x_{i+2}$ 的最大值为最简分数 p/q，下标模六，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-I-Q09] | Part B (Q6-Q10)

Triangle ABC has $AB = 40, AC = 31$ and $\sin A = 1/5$. It lies in rectangle $AQRS$ with B on side segment QR and C on side segment RS. Determine the maximum rectangle area. (A source-feasibility note accompanies the solution.)

三角形 ABC 满足 $AB = 40, AC = 31, \sin A = 1/5$，内接于矩形 $AQRS$，其中 B 在边线段 QR 上、C 在边线段 RS 上。求矩形的最大面积。（解析附来源可达性核查说明。）

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2022-II-Q06] | Part B (Q6-Q10)

Reals $x_1 \le \cdots \le x_{100}$ obey $\sum x_i = 0$ and $\sum |x_i| = 1$. If the maximum of $x_{76} - x_{16}$ is reduced m/n, find $m+n$.

实数 $x_1 \le \cdots \le x_{100}$ 满足 $\sum x_i = 0, \sum |x_i| = 1$。若 $x_{76} - x_{16}$ 的最大值为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2005-I-Q11] | Part C (Q11-Q15)

Find the largest possible diameter d of a semicircle contained in a side-8 square. If $d = m - \sqrt{n}$ with integers m, n, find $m + n$.

边长八的正方形内能容纳的半圆，其最大直径为 $d = m - \sqrt{n}$，其中 m, n 为整数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2008-I-Q14] | Part C (Q11-Q15)

Circle diameter AB has length eighteen. Extend AB past A to C, draw a tangent CT, and drop perpendicular AP to CT. If m is the greatest possible BP, find m^2.

圆的直径 AB 长十八，延长 AB 过 A 至 C，作切线 CT，从 A 向 CT 作垂足 P。若 BP 最大为 m，求 m^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2009-II-Q15] | Part C (Q11-Q15)

A circle has diameter $MN = 1$. On one semicircle, A is its arc midpoint and $MB = 3/5$. For C on the opposite semicircle, chords AC, BC meet MN at two points distance d apart. Its greatest value is $r - s\sqrt{t}$ in squarefree form. Find $r + s + t$.

圆的直径 $MN = 1$，一侧半圆弧中点为 A，该侧另有点 B 满足 $MB = 3/5$。C 在另一半圆弧上，AC, BC 与直径交点间距为 d，其最大值为 $r - s\sqrt{t}$，求 $r + s + t$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2016-II-Q15] | Part C (Q11-Q15)

For $1 \le i \le 215$, set $a_i = 2^{-i}$, and set $a_{216} = 2^{-215}$. Positive reals x_i satisfy $\sum x_i = 1$ and $\sum_{i<j} x_i x_j = 107/215 + \sum_i a_i x_i^2/[2(1-a_i)]$. If the largest possible x_2 is reduced m/n, find $m+n$.

对 $1 \le i \le 215$，令 $a_i = 2^{-i}$，并令 $a_{216} = 2^{-215}$。正实数 x_i 满足 $\sum x_i = 1$ 及 $\sum_{i<j} x_i x_j = 107/215 + \sum_i a_i x_i^2/[2(1-a_i)]$。若 x_2 的最大值为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2024-I-Q15] | Part C (Q11-Q15)

Consider all rectangular boxes with surface area 54 and volume 23. Let r be the smallest radius such that every box can fit inside some sphere of radius r. Write $r^2 = p/q$ in lowest terms and find $p+q$.

考虑所有表面积为 54、体积为 23 的长方体。若半径为 r 的球能够分别容纳其中每个长方体，求最小的 r。将其平方写成最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-I-Q14] | Part C (Q11-Q15)

A convex pentagon $ABCDE$ has side lengths $14, 7, 24, 13, 26$ in order and interior angles $B = E = 60^\circ$. For arbitrary plane points X, the minimum of $AX + BX + CX + DX + EX$ is $m + n\sqrt{p}$ with squarefree p. Find $m + n + p$.

凸五边形 $ABCDE$ 的连续边长为 $14, 7, 24, 13, 26$，内角 $B = E = 60^\circ$。平面点 X 到五顶点距离之和的最小值为 $m + n\sqrt{p}$，p 无平方素因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

[AIME-2025-II-Q15] | Part C (Q11-Q15)

For exactly three positive parameters k, the function $f(x) = (x-18)(x-72)(x-98)(x-k)/x$ on $x > 0$ has its global minimum at exactly two distinct points. Find the sum of these k.

对恰好三个正参数 k，函数 $f(x) = (x-18)(x-72)(x-98)(x-k)/x$ 在 $x > 0$ 上的全局最小值恰在两个不同点取到。求这三个 k 的和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2002-II-Q10] | Part B (Q6-Q10)

The two least positive real solutions of $\sin(x^\circ) = \sin(x \text{ radians})$ have forms $m\pi/(n-\pi)$ and $p\pi/(q+\pi)$ with positive integers m, n, p, q. Find their parameter sum.

方程 $\sin(x^\circ) = \sin(x \text{ 弧度})$ 的两个最小正实数解分别可写成 $m\pi/(n-\pi)$ 和 $p\pi/(q+\pi)$，参数为正整数。求四参数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2008-I-Q08] | Part B (Q6-Q10)

Find positive integer n satisfying $\arctan(1/3) + \arctan(1/4) + \arctan(1/5) + \arctan(1/n) = \pi/4$.

求满足 $\arctan(1/3) + \arctan(1/4) + \arctan(1/5) + \arctan(1/n) = \pi/4$ 的正整数 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2008-II-Q08] | Part B (Q6-Q10)

For $a = \pi/2008$, find the smallest positive n such that $2\sum_{k=1}^{n} \cos(k^2 a)\sin(ka)$ is an integer.

令 $a = \pi/2008$，求使 $2\sum_{k=1}^{n} \cos(k^2 a)\sin(ka)$ 为整数的最小正整数 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2012-II-Q09] | Part B (Q6-Q10)

Real angles satisfy $\sin x/\sin y = 3$ and $\cos x/\cos y = 1/2$. Reduce $\sin(2x)/\sin(2y) + \cos(2x)/\cos(2y) = p/q$ and find $p+q$.

实角满足 $\sin x/\sin y = 3, \cos x/\cos y = 1/2$。将 $\sin(2x)/\sin(2y) + \cos(2x)/\cos(2y)$ 约为 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2021-I-Q07] | Part B (Q6-Q10)

Count integer pairs $1 \le m < n \le 30$ for which $\sin(mx) + \sin(nx) = 2$ has at least one real solution x.

统计 $1 \le m < n \le 30$ 的整数对，使方程 $\sin(mx) + \sin(nx) = 2$ 有实数解。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2025-II-Q09] | Part B (Q6-Q10)

In $0 < x < 2\pi$, let n be the number of zeros of $f(x) = \sin(7\pi \sin 5x)$ and t the number at which the graph is tangent to the horizontal axis. Find $n + t$.

在 $0 < x < 2\pi$ 内，令 n 为 $f(x) = \sin(7\pi \sin 5x)$ 的零点数，t 为其中与横轴相切的零点数，求 $n + t$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2000-II-Q15] | Part C (Q11-Q15)

Find the least positive integer n satisfying $\sum_{k=45,47,\ldots,133}[\sin k^\circ \sin(k+1)^\circ]^{-1} = 1/\sin n^\circ$.

求最小正整数 n，使 $\sum_{k=45,47,\ldots,133}[\sin k^\circ \sin(k+1)^\circ]^{-1} = 1/\sin n^\circ$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2006-I-Q12] | Part C (Q11-Q15)

Sum degree values $100 < x < 200$ satisfying $\cos^3(3x) + \cos^3(5x) = 8\cos^3(4x)\cos^3 x$.

角度以度计，求 $100 < x < 200$ 内满足 $\cos^3(3x) + \cos^3(5x) = 8\cos^3(4x)\cos^3 x$ 的所有 x 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2013-II-Q15] | Part C (Q11-Q15)

Triangle angles A, C are acute and B obtuse. They satisfy $\cos^2 A + \cos^2 B + 2\sin A \sin B \cos C = 15/8$ and $\cos^2 B + \cos^2 C + 2\sin B \sin C \cos A = 14/9$. Write $\cos^2 C + \cos^2 A + 2\sin C \sin A \cos B = (p - q\sqrt{r})/s$, with $p + q$ coprime to s and squarefree r. Find $p + q + r + s$.

三角形锐角 A, C、钝角 B 满足 $\cos^2 A + \cos^2 B + 2\sin A \sin B \cos C = 15/8$，$\cos^2 B + \cos^2 C + 2\sin B \sin C \cos A = 14/9$。若 $\cos^2 C + \cos^2 A + 2\sin C \sin A \cos B = (p - q\sqrt{r})/s$，其中 $p + q$ 与 s 互质且 r 无平方因子，求四者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2014-II-Q12] | Part C (Q11-Q15)

Triangle angles satisfy $\cos 3A + \cos 3B + \cos 3C = 1$. Two side lengths are ten and thirteen. If the greatest third side is $\sqrt{m}$, find m.

三角形角满足 $\cos 3A + \cos 3B + \cos 3C = 1$，其中两边长为十、十三。若第三边最大值为 $\sqrt{m}$，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2015-I-Q13] | Part C (Q11-Q15)

With angles in degrees, $\prod_{k=1}^{45} \csc^2((2k-1)^\circ) = m^n$ for integers $m, n > 1$. Find $m+n$.

角度以度计，$\prod_{k=1}^{45} \csc^2((2k-1)^\circ) = m^n$，其中 $m, n > 1$ 为整数。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2015-II-Q13] | Part C (Q11-Q15)

Define $a_n = \sum_{k=1}^{n} \sin k$, with angles measured in radians. Find the index of the 100th negative term of this sequence.

令 $a_n = \sum_{k=1}^{n} \sin k$，角度用弧度。求该数列第 100 个负项的下标。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2022-I-Q15] | Part C (Q11-Q15)

Positive reals x, y, z satisfy $\sqrt{x(2-y)}+\sqrt{y(2-x)}=1$, $\sqrt{y(2-z)}+\sqrt{z(2-y)}=\sqrt{2}$, and $\sqrt{z(2-x)}+\sqrt{x(2-z)}=\sqrt{3}$. If $[(1-x)(1-y)(1-z)]^2=m/n$ in lowest terms, determine $m+n$.

正实数 x, y, z 满足 $\sqrt{x(2-y)}+\sqrt{y(2-x)}=1$，$\sqrt{y(2-z)}+\sqrt{z(2-y)}=\sqrt{2}$，$\sqrt{z(2-x)}+\sqrt{x(2-z)}=\sqrt{3}$。若 $[(1-x)(1-y)(1-z)]^2=m/n$ 为最简分数，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

[AIME-2023-II-Q13] | Part C (Q11-Q15)

An acute angle A satisfies $\tan A = 2\cos A$. Count the integers $1 \le n \le 1000$ for which $\sec^n A + \tan^n A$ is a positive integer ending in 9.

锐角 A 满足 $\tan A = 2\cos A$。求 $1 \le n \le 1000$ 中使 $\sec^n A + \tan^n A$ 为个位是 9 的正整数的 n 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2001-II-Q08] | Part B (Q6-Q10)

For positive x, $f(3x) = 3f(x)$; on $1 \le x \le 3$, $f(x) = 1 - |x - 2|$. Find the least positive x with $f(x) = f(2001)$.

对正数 x 有 $f(3x) = 3f(x)$，且 $1 \le x \le 3$ 时 $f(x) = 1 - |x - 2|$。求满足 $f(x) = f(2001)$ 的最小正数 x。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2018-II-Q10] | Part B (Q6-Q10)

Count functions on a labeled five-element set satisfying $f(f(x)) = f(f(f(x)))$ for every element x.

求固定标号五元集合到自身的函数数，使对每个 x 均有 $f(f(x)) = f(f(f(x)))$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2019-II-Q10] | Part B (Q6-Q10)

A unique $0 < \theta < 90^\circ$ has $\tan(2^n\theta) > 0$ exactly when $3 \mid n$, and negative for all other nonnegative integers n. Write $\theta = p/q$ degrees reduced and find $p + q$.

唯一角 $0 < \theta < 90^\circ$ 满足对所有非负整数 n，$\tan(2^n\theta)$ 恰在三整除 n 时为正，其余为负。若度数为最简分数 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2020-II-Q08] | Part B (Q6-Q10)

Let $f_1(x) = |x - 1|$ and $f_n(x) = f_{n-1}(|x - n|)$ for $n > 1$. Find the first positive index n for which the sum of all distinct real zeros of f_n is greater than 500000.

定义 $f_1(x) = |x - 1|$，$n > 1$ 时 $f_n(x) = f_{n-1}(|x - n|)$。求使 f_n 全部不同实零点之和首次超过 500000 的正整数下标 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2000-I-Q12] | Part C (Q11-Q15)

A real-valued function obeys $f(x) = f(398 - x) = f(2158 - x) = f(3214 - x)$ for every real x. What is the maximum number of distinct values among $f(0), \ldots, f(999)$?

实函数对每个实数 x 满足 $f(x) = f(398 - x) = f(2158 - x) = f(3214 - x)$。求 $f(0)$ 至 $f(999)$ 中不同函数值的最大个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2007-II-Q14] | Part C (Q11-Q15)

A real-coefficient polynomial obeys $f(0)=1$, $f(2)+f(3)=125$, and $f(x)f(2x^2)=f(2x^3+x)$ identically. Find $f(5)$.

实系数多项式满足 $f(0)=1, f(2)+f(3)=125$，且恒有 $f(x)f(2x^2)=f(2x^3+x)$，求 $f(5)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2009-I-Q13] | Part C (Q11-Q15)

An infinite sequence of positive integers satisfies $a_{n+2} = (a_n + 2009)/(1 + a_{n+1})$. Minimize $a_1 + a_2$.

无限正整数列满足 $a_{n+2} = (a_n + 2009)/(1 + a_{n+1})$，求 $a_1 + a_2$ 最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2009-II-Q14] | Part C (Q11-Q15)

Let $a_0 = 0$ and $a_{n+1} = 8a_n/5 + (6/5)\sqrt{4^n - a_n^2}$. Find $\lfloor a_{10} \rfloor$.

令 $a_0 = 0$，$a_{n+1} = 8a_n/5 + (6/5)\sqrt{4^n - a_n^2}$，求 $\lfloor a_{10} \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2012-II-Q11] | Part C (Q11-Q15)

Define $f(x) = 2/3 - 3/(3x+1)$ and let $f^{\circ 1001}$ denote its 1001-fold composition. The real solution of $f^{\circ 1001}(x) = x - 3$ is reduced m/n. Find $m+n$.

定义 $f(x) = 2/3 - 3/(3x+1)$，以 $f^{\circ 1001}$ 表示复合一千零一次。将方程 $f^{\circ 1001}(x) = x - 3$ 的实数解约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2013-II-Q11] | Part C (Q11-Q15)

On a labeled seven-element set, let N count functions f for which $f \circ f$ is constant. Find $N \bmod 1000$.

在有固定标号的七元集合上，设 N 为使 $f \circ f$ 是常函数的函数 f 的个数，求 N 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2021-II-Q15] | Part C (Q11-Q15)

For positive integers, define $f(n)=\sqrt{n}$ at squares and $f(n)=1+f(n+1)$ otherwise. Define $g(n)=\sqrt{n}$ at squares and $g(n)=2+g(n+2)$ otherwise. Find the least n with $f(n)/g(n)=4/7$.

对正整数，若 n 是平方数，令 $f(n)=g(n)=\sqrt{n}$；否则 $f(n)=1+f(n+1)$、$g(n)=2+g(n+2)$。求使 $f(n)/g(n)=4/7$ 的最小 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 1.9 Functional constraints & iteration

函数约束与迭代

[AIME-2025-II-Q13] | Part C (Q11-Q15)

Let $x_1 = 25/11$ and $x_{j+1} = (x_j + 1/x_j - 1)/3$. Express $x_{2025} = m/n$ in lowest positive terms and find $(m+n) \bmod 1000$.

令 $x_1 = 25/11$，$x_{j+1} = (x_j + 1/x_j - 1)/3$。将 $x_{2025} = m/n$ 约为最简正分数，求 $m+n$ 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

[AIME-2000-I-Q03] | Part A (Q1-Q5)

In $(ax+b)^{2000}$, positive coprime integers a, b make the coefficients of x^2 and x^3 equal. Find $a+b$.

在 $(ax+b)^{2000}$ 中，正整数 a, b 互质，且 x^2, x^3 系数相等。求 $a+b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

[AIME-2002-I-Q07] | Part B (Q6-Q10)

Find the first three digits after the decimal point of $(10^{2002}+1)^{10/7}$. You may use the binomial expansion for noninteger exponents.

求 $(10^{2002}+1)^{10/7}$ 小数点后前三位数字。 可使用非整数指数的二项式展开。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2008-II-Q05] | Part A (Q1-Q5)

Trapezoid $ABCD$ has $BC \parallel AD$, $BC = 1000$, $AD = 2008$, and base angles $A = 37^\circ$, $D = 53^\circ$. If M, N are the base midpoints, find MN.

梯形 $ABCD$ 中 $BC \parallel AD, BC = 1000, AD = 2008$，底角 $A = 37^\circ, D = 53^\circ$。$M, N$ 为两底中点，求 MN。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2009-I-Q05] | Part A (Q1-Q5)

Triangle ABC has $AC = 450, BC = 300$. Let K be the midpoint of AC, CL the internal angle bisector meeting AB, and $P = BK \cap CL$. Reflect P across K to point M. Given $AM = 180$, find LP.

三角形 ABC 中 $AC = 450, BC = 300$，K 为 AC 中点，内角平分线 CL 交 AB 于 L，$P = BK \cap CL$。将 P 关于 K 对称得 M，若 $AM = 180$，求 LP。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2016-II-Q05] | Part A (Q1-Q5)

Right triangle ABC_0 has right angle at C_0 and pairwise coprime positive integral sides, with perimeter p. Let C_1 be the perpendicular foot from C_0 to AB. Recursively, C_n is the foot from C_{n-1} to $C_{n-2}B$ for $n \geq 2$. If $\sum_{n\geq 1} C_{n-1}C_n = 6p$, find p.

直角三角形 ABC_0 在 C_0 处为直角，三边为两两互质的正整数，周长为 p。C_1 为 C_0 到 AB 的垂足；$n \geq 2$ 时，C_n 为 C_{n-1} 到 $C_{n-2}B$ 的垂足。若 $\sum_{n\geq 1} C_{n-1}C_n = 6p$，求 p。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2024-II-Q05] | Part A (Q1-Q5)

A convex hexagon has six equal sides and parallel opposite sides. The lines containing alternating sides AB, CD, EF form a triangle with sides $200, 240, 300$. Determine the hexagon side length.

凸六边形六边等长，三对对边分别平行。交替三边 AB, CD, EF 的所在直线围成边长为 $200, 240, 300$ 的三角形。求六边形边长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-I-Q07] | Part B (Q6-Q10)

In ABC with $AB = 21, AC = 22, BC = 20$, the line through the incenter parallel to BC cuts AB, AC at D, E. If $DE = m/n$ reduced, find $m + n$.

三角形 ABC 中 $AB = 21, AC = 22, BC = 20$。过内心且平行 BC 的直线交 AB, AC 于 D, E。若 $DE = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-I-Q09] | Part B (Q6-Q10)

A side-$3, 4, 5$ triangle is cut into a triangle U_1 and trapezoid V_1. A 6×7 rectangle is likewise cut into a triangle U_2 and trapezoid V_2, with $U_1 \sim U_2$ and $V_1 \sim V_2$. If the minimum possible $[U_1] = m/n$ reduced, find $m + n$.

边长 $3, 4, 5$ 的三角形被一线段切为三角形 U_1 与梯形 V_1，6×7 矩形也被切为 U_2, V_2，并要求两三角形相似、两梯形相似。若 U_1 最小面积约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2015-II-Q07] | Part B (Q6-Q10)

Triangle ABC has $AB = 12, AC = 17, BC = 25$. A rectangle $PQRS$ has P on AB, Q on AC, and R, S on BC, with $PQ = w$. Its area is $\alpha w - \beta w^2$. If $\beta = m/n$ in lowest terms, find $m + n$.

三角形 ABC 中 $AB = 12, AC = 17, BC = 25$。内接矩形 $PQRS$ 的 P, Q 分别在 AB, AC 上，R, S 在 BC 上，$PQ = w$。面积为 $\alpha w - \beta w^2$。若 $\beta = m/n$ 为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2018-II-Q07] | Part B (Q6-Q10)

Triangle ABC has $AB = 9, AC = 12, BC = 5\sqrt{3}$. Parallel cuts $P_kQ_k \parallel BC$ divide it into 2450 equal-area bands, with $P_0 = Q_0 = A, P_{2450} = B, Q_{2450} = C$. Count rational lengths among P_kQ_k, $1 \le k \le 2450$.

三角形 ABC 中 $AB = 9, AC = 12, BC = 5\sqrt{3}$，用平行于 BC 的 P_kQ_k 将其分成二千四百五十块等面积区域，$P_0 = Q_0 = A, P_{2450} = B, Q_{2450} = C$。求 $1 \le k \le 2450$ 的截线中长度为有理数的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-I-Q06] | Part B (Q6-Q10)

In convex quadrilateral $KLMN$, $MN \perp KM$, $KL \perp LN$, $MN = 65, KL = 28$. The line from L perpendicular to KN meets diagonal KM at O, with $KO = 8$. Find MO.

凸四边形 $KLMN$ 中 $MN \perp KM, KL \perp LN, MN = 65, KL = 28$。从 L 作 KN 垂线交对角线 KM 于 O，$KO = 8$，求 MO。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2019-II-Q07] | Part B (Q6-Q10)

Triangle ABC has $AB = 120, BC = 220, CA = 180$. Three lines parallel respectively to BC, CA, AB cut interior segments of lengths $55, 45, 15$. Find the perimeter of the triangle formed by these lines.

三角形 ABC 边长为 $AB = 120, BC = 220, CA = 180$。三线分别平行于 BC, CA, AB，在原三角形内截段长为五十五、四十五、十五。求三线围成三角形的周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2001-II-Q13] | Part C (Q11-Q15)

Quadrilateral $ABCD$ satisfies $\angle BAD = \angle ADC$ and $\angle ABD = \angle BCD$, with $AB = 8, BD = 10, BC = 6$. If $CD = m/n$ reduced, find $m + n$.

四边形 $ABCD$ 满足 $\angle BAD = \angle ADC, \angle ABD = \angle BCD$，且 $AB = 8, BD = 10, BC = 6$。若 $CD = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2002-II-Q13] | Part C (Q11-Q15)

In ABC, $AB = 8$, $CD = 2, DB = 5$ with $D \in BC$, and $CE = 1, EA = 3$ with $E \in AC$. Put $P = AD \cap BE$. Lines through P parallel to CA, CB meet AB at Q, R. Write $[PQR]/[ABC] = m/n$ reduced and find $m + n$.

三角形 ABC 中 $AB = 8$，$D \in BC$ 且 $CD = 2, DB = 5$，$E \in AC$ 且 $CE = 1, EA = 3$。令 $P = AD \cap BE$，过 P 分别作 CA, CB 平行线交 AB 于 Q, R。若面积比约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2004-II-Q13] | Part C (Q11-Q15)

Convex pentagon $ABCDE$ satisfies $AB \parallel CE, BC \parallel AD, AC \parallel DE$, with $\angle ABC = 120^\circ$, $AB = 3, BC = 5, DE = 15$. If $[ABC]/[EBD] = m/n$ reduced, find $m + n$.

凸五边形满足 $AB \parallel CE, BC \parallel AD, AC \parallel DE$，且 $\angle ABC = 120^\circ, AB = 3, BC = 5, DE = 15$。若面积比 $[ABC]/[EBD] = m/n$ 已约分，求其和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2006-II-Q12] | Part C (Q11-Q15)

Equilateral ABC has circumradius two. Extend AB, AC past B, C to D, E with $AD = 13, AE = 11$. Complete parallelogram $ADFE$. Line AF meets the circle again at G. If $[CBG] = p\sqrt{q}/r$ is reduced with squarefree q, find $p + q + r$.

等边三角形 ABC 外接圆半径为二，延长 AB, AC 至 D, E，使 $AD = 13, AE = 11$，补成平行四边形 $ADFE$。AF 与圆再交于 G。若 CBG 面积为约简的 $p\sqrt{q}/r$，求 $p + q + r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2007-I-Q15] | Part C (Q11-Q15)

In equilateral ABC, $F \in AB, D \in BC$ satisfy $AF = 5, CD = 2$. A point $E \in CA$ has $\angle DEF = 60^\circ$ and $[DEF] = 14\sqrt{3}$. The two possible side lengths are $p \pm q\sqrt{r}$ with rational p, q and squarefree integer r. Find r.

等边三角形 ABC 内，$F \in AB, D \in BC$ 满足 $AF = 5, CD = 2$。$E \in CA$ 使 $\angle DEF = 60^\circ$，面积 $[DEF] = 14\sqrt{3}$。原三角形边长两个可能值为有理数 p, q 构成的 $p \pm q\sqrt{r}$，r 无平方素因子，求 r。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

[AIME-2013-I-Q13] | Part C (Q11-Q15)

Begin with triangle AB_0C_0 having sides $12, 17, 25$ in that order. For each $n \geq 1$, take B_n on AB_{n-1} and C_n on AC_{n-1} so that the ordered triangles AB_nC_n, $B_{n-1}C_nC_{n-1}$, $AB_{n-1}C_{n-1}$ are similar. The union of triangles $B_{n-1}C_nB_n$ has reduced area p/q. Find q.

从三角形 AB_0C_0 开始，$AB_0 = 12, B_0C_0 = 17, C_0A = 25$。对每个 n，在 AB_{n-1}, AC_{n-1} 上分别取 B_n, C_n，使有序三角形 $AB_nC_n, B_{n-1}C_nC_{n-1}, AB_{n-1}C_{n-1}$ 相似。所有 $B_{n-1}C_nB_n$ 的并集面积为最简分数 p/q，求 q。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2002-II-Q04] | Part A (Q1-Q5)

In a regular hexagonal cluster of side-1 regular hexagon cells, there are n cells along each of the six outer sides. Only its outermost cell ring is paved; the enclosed interior is a garden. At $n = 202$, write garden area as $m\sqrt{3}/2$. Find $m \bmod 1000$.

用边长一的正六边形单元形成规则六边形簇，每条外侧边上有 n 个单元。仅最外一圈铺成小路，内部未铺部分为花园。取 $n = 202$，若花园面积为 $m\sqrt{3}/2$，求 m 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-I-Q02] | Part A (Q1-Q5)

Concentric circles have radii $1, 2, \ldots, 100$. Starting with a red center disk, color successive regions alternately red and green. Write the green area divided by the radius-100 disk area as m/n reduced. Find $m+n$.

同心圆半径依次为 1 至 100。中心圆盘染红，向外各环红绿交替。绿色总面积占最大圆盘面积的比约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-I-Q04] | Part A (Q1-Q5)

In a side-2 square, consider length-2 segments with endpoints on adjacent sides. Their midpoint arcs enclose a central region. If its area rounded to the nearest hundredth is k, find $100k$.

边长为二的正方形中，取端点位于相邻两边且长为二的线段。它们的中点弧围出中央区域，面积四舍五入到百分位为 k，求 $100k$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-II-Q01] | Part A (Q1-Q5)

A chord is perpendicular to a radius at its midpoint. The larger-to-smaller disk-region ratio is $(a\pi + b\sqrt{c})/(d\pi - e\sqrt{f})$, with positive integers, $\gcd(a, e) = 1$, and squarefree c, f. Find $abcdef \bmod 1000$.

一条弦在半径中点处与半径垂直。它分圆盘所得大、小区域面积比写为 $(a\pi + b\sqrt{c})/(d\pi - e\sqrt{f})$，参数为正整数，$a, e$ 互质，c, f 无平方因子。求乘积 $abcdef$ 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-II-Q01] | Part A (Q1-Q5)

A convex equilateral hexagon $ABCDEF$ has right angles at A, D and four other equal angles. Its area is $2116(1+\sqrt{2})$. Find its side length.

凸等边六边形 $ABCDEF$ 在 A, D 为直角，其余四角相等，面积为 $2116(1+\sqrt{2})$。求边长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2008-I-Q02] | Part A (Q1-Q5)

An isosceles triangle has as its base one side of a side-ten square, with apex on the same side as the square interior. Its overlap with the square has area eighty. Find its altitude to that base.

等腰三角形以边长十的正方形一边为底，顶点在正方形内部所在一侧，两图重叠面积八十。求三角形底边上的高。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2014-II-Q03] | Part A (Q1-Q5)

A hinged rectangle has side lengths $a, 36$, with hinges at its corners and at the midpoints of its length-36 sides. Press the a-sides symmetrically inward, keeping them parallel and their midpoints perpendicularly aligned, to form a convex hexagon. When the separation is 24, its area equals the rectangle area. Find a^2.

边长为 $a, 36$ 的矩形在各顶点及三十六长边中点设铰链。将两条 a 边对称向内压，保持平行且两中点连线垂直于它们，形成凸六边形。两边相距二十四时，面积与原矩形相同。求 a^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2019-I-Q03] | Part A (Q1-Q5)

Triangle PQR has $PR = 15, QR = 20, PQ = 25$. On PQ mark A, B, on QR mark C, D, and on PR mark E, F, with $PA = QB = QC = RD = RE = PF = 5$. Find $[ABCDEF]$.

三角形 PQR 边长为十五、二十、二十五。A, B 在 PQ，C, D 在 QR，E, F 在 PR，满足 $PA = QB = QC = RD = RE = PF = 5$。求六边形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2021-II-Q02] | Part A (Q1-Q5)

An equilateral triangle ABC has side 840. On the perpendicular ray from B toward the side containing A, choose D. The line through D parallel to BC meets AB, AC at E, F. Choose G on the same line beyond F from E, so AFG is isosceles and $[AFG] : [BED] = 8 : 9$. Find AF.

等边三角形 ABC 边长为 840。由 B 向含 A 一侧作 BC 的垂线并取点 D。过 D 作 BC 平行线交 AB, AC 于 E, F。在该线上取 G，使 F 位于 E, G 之间，AFG 为等腰三角形，且 $[AFG] : [BED] = 8 : 9$。求 AF。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-I-Q02] | Part A (Q1-Q5)

On triangle sides, A, D, E, B occur in order with successive lengths $4, 16, 8$, and A, F, G, C occur in order with lengths $13, 52, 26$. Reflect D across point F to obtain M, and G across point E to obtain N. If $[DEGF] = 288$, find the area of polygon $AFNBCEM$.

三角形两边上，A, D, E, B 依次分段为 $4, 16, 8$，A, F, G, C 依次分段为 $13, 52, 26$。将 D 关于点 F 对称得 M，将 G 关于点 E 对称得 N。若 $DEGF$ 面积为 288，求多边形 $AFNBCEM$ 的面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2000-II-Q06] | Part B (Q6-Q10)

A trapezoid has bases differing by 100. Its leg-midpoint segment divides the area in ratio $2:3$. A parallel section of length x divides the area equally. Find $\lfloor x^2/100 \rfloor$.

梯形两底差为 100。两腰中点连线将面积分成 $2:3$。另有一条平行两底、长为 x 的截线将面积平分，求 $\lfloor x^2/100 \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2001-I-Q09] | Part B (Q6-Q10)

In a triangle with $AB = 13, BC = 15, CA = 17$, take $D \in AB, E \in BC, F \in CA$ with $AD/AB = p, BE/BC = q, CF/CA = r > 0$. Given $p + q + r = 2/3$ and $p^2 + q^2 + r^2 = 2/5$, write $[DEF]/[ABC] = m/n$ reduced and find $m + n$.

三角形边长 $AB = 13, BC = 15, CA = 17$，边上点满足 $AD/AB = p, BE/BC = q, CF/CA = r > 0$。已知 $p + q + r = 2/3, p^2 + q^2 + r^2 = 2/5$。若面积比 $[DEF]/[ABC] = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-II-Q06] | Part B (Q6-Q10)

Triangle ABC has sides $13, 14, 15$. Rotate it 180° about its centroid to obtain a second triangle. Find the area of the union of the two filled triangles.

边长为 $13, 14, 15$ 的三角形绕重心旋转 180° 得第二个三角形，求两个实心三角形并集面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q07] | Part B (Q6-Q10)

An angle is cut by seven equally spaced parallel cross-lines numbered outward from its vertex. Region A is the initial triangle; B, C, D are strips between lines 2–3, 4–5, and 6–7. Given $[C]/[B] = 11/5$, find $[D]/[A]$.

一个角被七条等距平行截线依次截开。A 为顶点到第一线的三角形；B, C, D 分别为第二至三、第四至五、第六至七线间的带状区。若面积比 $[C]/[B] = 11/5$，求 $[D]/[A]$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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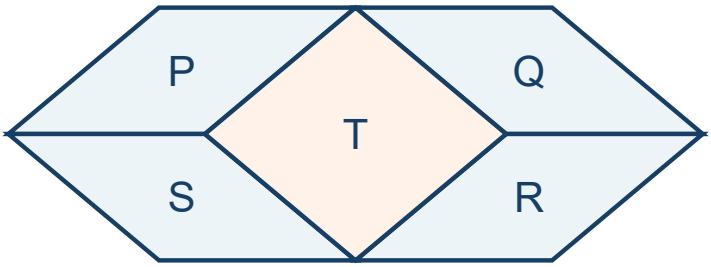
Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q08] | Part B (Q6-Q10)

A hexagon is tiled by five equal-side rhombi as in the companion diagram: a central rhombus T and four congruent outer rhombi P, Q, R, S. At the top vertex of T, the incident angles of P, T, Q form a straight angle; likewise at the bottom. Each outer rhombus has area $\sqrt{2006}$. How many positive integer areas can T have?

如配图，六边形被五个等边长菱形铺满：中央为 T，四周 P, Q, R, S 全等。T 顶点处 P, T, Q 的角拼成平角，底点处同理。每个外围菱形面积为 $\sqrt{2006}$。中央面积有多少个正整数可能值?



Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2011-I-Q08] | Part B (Q6-Q10)

A triangular sheet has sides $23, 27, 30$. At each vertex cut off a corner by a fold line parallel to the opposite side, with the three corner flaps nonoverlapping along the original edges. Fold all three corners down by 90° to form a table with horizontal top and all three vertices on the floor. Its greatest height is $k\sqrt{m}/n$, with coprime k, n and squarefree m. Find $k + m + n$.

一张三角形纸的边长为 $23, 27, 30$。在三顶点各作平行于对边的折线，三个角片在原各边上不重叠。将三角片向下折九十度，使桌面水平、三个原顶点落地。最大高度为 $k\sqrt{m}/n$，k, n 互质、m 无平方因子，求 $k + m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2013-II-Q10] | Part B (Q6-Q10)

For a circle of radius $\sqrt{13}$, an exterior point A is $4+\sqrt{13}$ from its center. Let B be its nearest circle point. A secant through A meets the circle at K, L. The greatest area of BKL is $(a-b\sqrt{c})/d$, with coprime a, d and squarefree c. Find $a+b+c+d$.

半径 $\sqrt{13}$ 的圆外有点 A，距圆心 $4+\sqrt{13}$。B 为圆上离 A 最近点。过 A 的割线交圆于 K, L。若 BKL 最大面积为 $(a-b\sqrt{c})/d$，a, d 互质、c 无平方因子，求四者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2007-I-Q12] | Part C (Q11-Q15)

Isosceles triangle ABC has $A=(0,0), B=(20,0)$, $AC=BC$, C in the first quadrant and $\angle BAC=75^\circ$. Rotate about A until C reaches the positive y axis. The overlap area is $p\sqrt{2}+q\sqrt{3}+r\sqrt{6}+s$ with integer coefficients. Find $(p-q+r-s)/2$.

等腰三角形 ABC 中 $A=(0,0), B=(20,0), AC=BC$，C 在第一象限，$\angle BAC=75^\circ$。绕 A 逆时针旋转至 C 落在正 y 轴，重叠面积为 $p\sqrt{2}+q\sqrt{3}+r\sqrt{6}+s$，求 $(p-q+r-s)/2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2009-I-Q11] | Part C (Q11-Q15)

Choose two distinct nonnegative lattice points P, Q on $41x + y = 2009$ and form triangle OPQ with the origin. Count distinct triangles of positive integer area.

从直线 $41x + y = 2009$ 上非负整数格点中选不同两点 P, Q，与原点组成三角形，求正整数面积的不同三角形数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2010-I-Q13] | Part C (Q11-Q15)

A rectangle $ABCD$ and a semicircle on diameter AB have disjoint interiors. A line meets the semicircle at N, AB at U, and CD at T, all distinct, dividing their combined region in area ratio $1:2$. Given $AU=84, UB=168, AN=126$, write $DA=m\sqrt{n}$ with squarefree n and find $m+n$.

矩形 $ABCD$ 与以 AB 为直径的半圆内部不重叠。一条直线依次交半圆弧、AB, CD 于不同点 N, U, T，将合并区域面积分成 $1:2$。已知 $AU=84, UB=168, AN=126$，若 $DA=m\sqrt{n}$ 且 n 无平方因子，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-II-Q14] | Part C (Q11-Q15)

Right triangle ABC has right angle A and hypotenuse $BC = 38$. Interior points K, L satisfy $AK = AL = BK = CL = KL = 14$. If quadrilateral $BKLC$ has area $n\sqrt{3}$, find n.

直角三角形 ABC 在 A 处为直角，斜边 $BC = 38$。内部点 K, L 满足 $AK = AL = BK = CL = KL = 14$。若四边形 $BKLC$ 面积为 $n\sqrt{3}$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2001-I-Q04] | Part A (Q1-Q5)

In ABC, $A = 60^\circ, B = 45^\circ$. The internal bisector AT meets BC and has length 24. Write the area as $a + b\sqrt{c}$ with positive integers and squarefree c. Find $a + b + c$.

三角形 ABC 中 $A = 60^\circ, B = 45^\circ$，内角平分线 AT 交 BC，且 $AT = 24$。面积写为 $a + b\sqrt{c}$，其中 c 无平方因子，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-I-Q01] | Part A (Q1-Q5)

Quadrilateral $ABCD$ has $B = 90^\circ$, $AC \perp CD$, and $AB = 18, BC = 21, CD = 14$. Find its perimeter.

四边形 $ABCD$ 中 $B = 90^\circ, AC \perp CD, AB = 18, BC = 21, CD = 14$，求周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q03] | Part A (Q1-Q5)

Square $ABCD$ has side thirteen. Exterior points E, F satisfy $AE = CF = 12$ and $BE = DF = 5$, with E outside across AB and F outside across CD. Find EF^2.

边长十三的正方形 $ABCD$ 外，点 E 在 AB 外侧、点 F 在 CD 外侧，满足 $AE = CF = 12, BE = DF = 5$，求 EF^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-II-Q05] | Part A (Q1-Q5)

Points D, E trisect one side of an equilateral triangle ABC. Express $\sin\angle DAE = a\sqrt{b}/c$, with coprime positive a, c and squarefree b, and find $a+b+c$.

D, E 三等分等边三角形 ABC 的边 BC。将 $\sin\angle DAE$ 写为 $a\sqrt{b}/c$，a, c 为互质正整数、b 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2014-I-Q01] | Part A (Q1-Q5)

A shoelace joins eight eyelets on a 50-by-80 mm rectangle, four equally spaced on each long side, including the corners. It crosses the bottom width once, then its two strands each cross diagonally between successive eyelet levels three times. Each free end must extend another 200 mm. Find the shortest lace length.

八个鞋带孔位于五十乘八十毫米矩形两长边上，各边四孔等距且含顶点。鞋带先横跨底边一次，再由两股各斜跨至下一层孔，连续三层；两端另各留二百毫米。求最短鞋带长度。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2015-II-Q04] | Part A (Q1-Q5)

An isosceles trapezoid has parallel bases $\log 3, \log 192$ and altitude $\log 16$, using common logarithms. Its perimeter is $\log(2^p 3^q)$ for positive integers p, q. Find $p+q$.

等腰梯形的两底为常用对数 $\log 3, \log 192$，高为 $\log 16$。周长为 $\log(2^p 3^q)$，其中 p, q 为正整数。求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-I-Q04] | Part A (Q1-Q5)

In triangle ABC, $AB = AC = 10, BC = 12$. Interior side points $D \in AB, E \in AC$ satisfy $AD = DE = EC$. Reduce $AD = p/q$ and find $p + q$.

三角形 ABC 中 $AB = AC = 10, BC = 12$，边内部点 $D \in AB, E \in AC$ 满足 $AD = DE = EC$。将 AD 约为 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2020-I-Q01] | Part A (Q1-Q5)

An isosceles triangle has $AB = AC$. Choose $D \in AC$ and $E \in AB$, both in the interiors of those sides, with $AE = ED = DB = BC$. Write $\angle ABC = m/n$ degrees in lowest terms. Determine $m + n$.

等腰三角形满足 $AB = AC$。在 AC, AB 的内部依次取点 D, E，使 $AE = ED = DB = BC$。若 $\angle ABC$ 的度数约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2021-II-Q05] | Part A (Q1-Q5)

For each $s > 0$, consider all obtuse triangles of area s having sides of lengths 4 and 10. The values of s for which at least one exists and all are congruent form $[a, b)$. Find $a^2 + b^2$.

对每个 $s > 0$，考虑面积为 s 且有两边长 $4, 10$ 的全部钝角三角形。使其非空且全部全等的 s 构成区间 $[a, b)$。求 $a^2 + b^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2023-II-Q03] | Part A (Q1-Q5)

In a right isosceles triangle ABC with right angle at A, an interior point P has $AP = 10$ and $\angle PAB = \angle PBC = \angle PCA$. Find the triangle area.

等腰直角三角形 ABC 的直角在 A。内点 P 满足 $AP = 10$ 且 $\angle PAB = \angle PBC = \angle PCA$。求三角形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q10] | Part B (Q6-Q10)

Isosceles triangle ABC has $AC = BC$ and $C = 106^\circ$. Interior point M satisfies $\angle MAC = 7^\circ$ and $\angle MCA = 23^\circ$. Find $\angle CMB$ in degrees.

等腰三角形 ABC 满足 $AC = BC, C = 106^\circ$。内点 M 满足 $\angle MAC = 7^\circ, \angle MCA = 23^\circ$，求 $\angle CMB$ 度数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-II-Q07] | Part B (Q6-Q10)

In rhombus $ABCD$, the circumradii of ABD and ACD are 12.5 and 25. Determine the rhombus area.

菱形 $ABCD$ 中，三角形 ABD, ACD 的外接圆半径分别为 $12.5, 25$，求菱形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-I-Q07] | Part B (Q6-Q10)

Quadrilateral $ABCD$ has $BC = 8, CD = 12, DA = 10$ and interior angles $A = B = 60^\circ$. If $AB = p + \sqrt{q}$ for positive integers p, q, find $p + q$.

四边形 $ABCD$ 中 $BC = 8, CD = 12, DA = 10$，内角 $A = B = 60^\circ$。若 $AB = p + \sqrt{q}$，其中 p, q 为正整数，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-I-Q10] | Part B (Q6-Q10)

An isosceles trapezoid has longer base AD, base angles 60°, and diagonals $10\sqrt{21}$. A point E has $EA = 10\sqrt{7}, ED = 30\sqrt{7}$. If F is the perpendicular foot from C to AD and $EF = m\sqrt{n}$ with squarefree n, find $m + n$.

等腰梯形长底为 AD，底角六十度，对角线长 $10\sqrt{21}$。点 E 满足 $EA = 10\sqrt{7}, ED = 30\sqrt{7}$。$F$ 为 C 到 AD 的垂足，若 $EF = m\sqrt{n}$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2013-I-Q09] | Part B (Q6-Q10)

Fold an equilateral paper triangle ABC of side twelve so A lands at T on BC with $BT = 9$. If the crease length is $m\sqrt{p}/n$, with coprime m, n and squarefree p, find $m + n + p$.

将边长十二的等边纸三角形折叠，使 A 落在 BC 上点 T，且 $BT = 9$。若折痕长为 $m\sqrt{p}/n$，m, n 互质、p 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2000-I-Q14] | Part C (Q11-Q15)

In isosceles ABC with $B = C$, choose $P \in AC, Q \in AB$ so $AP = PQ = QB = BC$. Let $r = \angle ACB / \angle APQ$. Find $\lfloor 1000r \rfloor$.

等腰三角形 ABC 满足 $B = C$，在 AC, AB 上取 P, Q，使 $AP = PQ = QB = BC$。令 $r = \angle ACB / \angle APQ$，求 $\lfloor 1000r \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2003-I-Q12] | Part C (Q11-Q15)

Convex quadrilateral $ABCD$ has $A = C$, $AB = CD = 180$, unequal AD, BC, and perimeter 640. Find $\lfloor 1000 \cos A \rfloor$.

凸四边形 $ABCD$ 满足 $A = C, AB = CD = 180, AD \neq BC$，周长为 640，求 $\lfloor 1000 \cos A \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2005-II-Q14] | Part C (Q11-Q15)

Triangle ABC has $AB = 13, BC = 15, CA = 14$. On BC, let $CD = 6$ and choose E with $\angle BAE = \angle CAD$. If $BE = p/q$ is reduced, find q.

三角形 ABC 中 $AB = 13, BC = 15, CA = 14$。$D, E$ 在 BC 上，$CD = 6$ 且 $\angle BAE = \angle CAD$。若 $BE = p/q$ 为最简分数，求 q。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2006-II-Q15] | Part C (Q11-Q15)

Real numbers satisfy $x = \sqrt{y^2 - 1/16} + \sqrt{z^2 - 1/16}$, $y = \sqrt{z^2 - 1/25} + \sqrt{x^2 - 1/25}$, $z = \sqrt{x^2 - 1/36} + \sqrt{y^2 - 1/36}$. If $x + y + z = m/\sqrt{n}$ with squarefree n, find $m + n$.

实数满足 $x = \sqrt{y^2 - 1/16} + \sqrt{z^2 - 1/16}$，$y = \sqrt{z^2 - 1/25} + \sqrt{x^2 - 1/25}$，$z = \sqrt{x^2 - 1/36} + \sqrt{y^2 - 1/36}$。若和为 $m/\sqrt{n}$ 且 n 无平方素因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2007-II-Q11] | Part C (Q11-Q15)

A radius-72 cylinder rolls without slipping along level ground, over a fixed parallel radius-24 cylinder, and back onto the ground. After exactly one full revolution it has its original circumferential contact point down again. The net horizontal displacement is $a\pi + b\sqrt{c}$, c squarefree. Find $a + b + c$.

半径七十二的圆柱在水平地面无滑动滚动，越过固定且平行的半径二十四圆柱后回到地面。恰转一整周，原周面接地点再次朝下，水平净位移为 $a\pi + b\sqrt{c}$，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2008-II-Q14] | Part C (Q11-Q15)

For $a \geq b > 0$, the system $a^2 + y^2 = b^2 + x^2 = (a - x)^2 + (b - y)^2$ has a solution with $0 \leq x < a, 0 \leq y < b$. If the largest possible ratio is $\rho = a/b$ and $\rho^2 = m/n$ reduced, find $m + n$.

对 $a \geq b > 0$，方程组 $a^2 + y^2 = b^2 + x^2 = (a - x)^2 + (b - y)^2$ 有解满足 $0 \leq x < a, 0 \leq y < b$。若最大比值为 $\rho = a/b$，其平方为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q12] | Part C (Q11-Q15)

Two noncongruent integer-sided isosceles triangles have equal area and perimeter, and their base lengths are in ratio $8:7$. Find the least common perimeter possible.

两个不全等整边等腰三角形面积、周长均相等，底边长之比为 $8:7$。求最小的共同周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2010-II-Q14] | Part C (Q11-Q15)

In right triangle ABC, $\angle C = 90^\circ, \angle A < 45^\circ, AB = 4$. A point P on AB satisfies $CP = 1$ and $\angle APC = 2\angle ACP$. If $AP/BP = p + q\sqrt{r}$ with positive integers p, q and squarefree r, find $p + q + r$.

直角三角形 ABC 中，$\angle C = 90^\circ, \angle A < 45^\circ, AB = 4$。边 AB 上点 P 满足 $CP = 1$ 和 $\angle APC = 2\angle ACP$。若 $AP/BP = p + q\sqrt{r}$，p, q 为正整数、r 无平方因子，求 $p + q + r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2018-II-Q12] | Part C (Q11-Q15)

Convex quadrilateral $ABCD$ has $AB = CD = 10, BC = 14, AD = 2\sqrt{65}$. Its diagonals meet at P and obey $[APB] + [CPD] = [BPC] + [APD]$. Find its area.

凸四边形 $ABCD$ 中 $AB = CD = 10, BC = 14, AD = 2\sqrt{65}$。对角线交于 P，且 $[APB] + [CPD] = [BPC] + [APD]$，求四边形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

[AIME-2025-II-Q12] | Part C (Q11-Q15)

A simple nonconvex polygon $A_1 \cdots A_{11}$ has perimeter 20. For $i = 2, \ldots, 10$, triangle $A_i A_1 A_{i+1}$ has area 1 and angle at A_1 with cosine $12/13$. If $A_1A_2 + A_1A_{11} = (m\sqrt{n} - p)/q$ in reduced squarefree form, find $m + n + p + q$.

简单非凸十一边形 $A_1 \cdots A_{11}$ 周长为 20。对 $i = 2$ 至 10，三角形 $A_i A_1 A_{i+1}$ 面积为 1，且在 A_1 的角余弦为 $12/13$。若 $A_1A_2 + A_1A_{11} = (m\sqrt{n} - p)/q$ 为约简且根号内无平方因子的形式，求 $m + n + p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-II-Q05] | Part A (Q1-Q5)

Triangle ABC has angles $A = 84^\circ, B = 60^\circ, C = 36^\circ$, and side midpoints $D \in BC, E \in CA, F \in AB$. Their circle meets BD, AE, AF again at G, H, J. The six cyclic points occur as G, D, E, H, J, F. Find the sum of consecutive minor-arc measures $\widehat{DE} + 2\widehat{HJ} + 3\widehat{FG}$.

三角形 ABC 的角为 $84^\circ, 60^\circ, 36^\circ$，$D, E, F$ 分别为 BC, CA, AB 中点。其外接圆与 BD, AE, AF 再交于 G, H, J，六点沿圆依次为 G, D, E, H, J, F。求相邻劣弧度数 $\widehat{DE} + 2\widehat{HJ} + 3\widehat{FG}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-II-Q06] | Part B (Q6-Q10)

A square $ABCD$ is inscribed in a circle. Another square $EFGH$ has E, F on segment CD and G, H on the circle. Its area ratio to $ABCD$ is $m/n < 1$ reduced. Find $10n + m$.

正方形 $ABCD$ 内接于圆。另一正方形 $EFGH$ 的 E, F 位于线段 CD，G, H 位于圆上。其面积与原正方形之比为最简分数 $m/n < 1$，求 $10n + m$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2011-II-Q10] | Part B (Q6-Q10)

A radius-25 circle has intersecting chords of lengths $30, 14$ with midpoints 12 apart. Their intersection is P, and the center is O. Express $OP^2 = m/n$ in lowest terms and find $(m + n) \bmod 1000$.

半径二十五的圆中，两条长为三十、十四的弦交于 P，两弦中点相距十二。圆心为 O。若 $OP^2 = m/n$ 已最简，求 $m + n$ 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2013-II-Q08] | Part B (Q6-Q10)

A cyclic hexagon has successive sides $22, 22, 20, 22, 22, 20$. If its circumradius is $p + \sqrt{q}$ for positive integers, find $p + q$.

一个圆内接六边形的边依次为 $22, 22, 20, 22, 22, 20$。若外接圆半径为 $p + \sqrt{q}$，求正整数 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2015-I-Q06] | Part B (Q6-Q10)

On one circle, A, B, C, D, E occur consecutively at equal arc intervals along a minor arc. On a second circle centered at C, E, F, G, H, I, A occur consecutively at equal arc intervals along a minor arc. Given $\angle ABD - \angle AHG = 12^\circ$, determine $\angle BAG$ in degrees.

第一圆的小弧上，A, B, C, D, E 依次等弧分布。第二圆以 C 为圆心，其小弧上 E, F, G, H, I, A 依次等弧分布。已知 $\angle ABD - \angle AHG = 12^\circ$，求 $\angle BAG$ 的度数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-II-Q10] | Part B (Q6-Q10)

Triangle ABC is cyclic. Along side AB, the points occur as A, P, Q, B with $AP = 4, PQ = 3, QB = 6$. Rays CP, CQ reach the circumcircle again at S, T. Given $AS = 7, BT = 5$, express $ST = m/n$ in lowest terms and find $m + n$.

三角形 ABC 内接于圆。边 AB 上点序为 A, P, Q, B，$AP = 4, PQ = 3, QB = 6$。射线 CP, CQ 再次交外接圆于 S, T。已知 $AS = 7, BT = 5$，将 $ST = m/n$ 约为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2023-II-Q09] | Part B (Q6-Q10)

Two intersecting circles meet at P, Q. Their common external tangent nearer P touches them at A, B. A line through P parallel to AB meets the circles again at X, Y, with $PX = 10, PY = 14, PQ = 5$. The area of trapezoid $XABY$ is $m\sqrt{n}$ with squarefree n. Find $m + n$.

两圆交于 P, Q，离 P 较近的公外切线分别切于 A, B。过 P 作 AB 的平行线，再交两圆于 X, Y，其中 $PX = 10, PY = 14, PQ = 5$。梯形 $XABY$ 面积为 $m\sqrt{n}$，n 无平方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2024-I-Q10] | Part B (Q6-Q10)

Triangle ABC has $AB = 5, BC = 9, CA = 10$. Tangents to its circumcircle at B, C meet at D, and segment AD meets the circle again at P. Express $AP = m/n$ in lowest terms and find $m + n$.

三角形 ABC 中 $AB = 5, BC = 9, CA = 10$。外接圆在 B, C 处的切线交于 D，线段 AD 与圆再交于 P。将 AP 化为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-I-Q06] | Part B (Q6-Q10)

An isosceles trapezoid of area 72 has an incircle of radius 3. Its unequal parallel sides have lengths r, s. Find $r^2 + s^2$.

等腰梯形面积为 72，内切圆半径为 3，两条不等的平行边长为 r, s。求 $r^2 + s^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2026-II-Q10] | Part B (Q6-Q10)

In triangle ABC with $AB = 200, AC = 225$, the internal angle bisector AD meets BC at D. A circle through A tangent to BC at D meets AB, AC again at interior points E, F. If AE, AF, BD, CD are positive integers, maximize BC.

三角形 ABC 中 $AB = 200, AC = 225$，内角平分线 AD 交 BC 于 D。过 A 且在 D 与 BC 相切的圆再与 AB, AC 的内部交于 E, F。若 AE, AF, BD, CD 均为正整数，求最大 BC。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-I-Q13] | Part C (Q11-Q15)

In one circle the chord for an arc of d degrees has length 22, and the chord for $2d$ degrees is 20 longer than that for $3d$ degrees, with $0 < d < 120$. Write the $3d$ chord length as $-m + \sqrt{n}$ for positive integers m, n. Find $m + n$.

同一圆中，d 度弧的弦长为 22，$2d$ 度弧的弦比 $3d$ 度弧的弦长 20，且 $0 < d < 120$。将 $3d$ 弦长写为 $-m + \sqrt{n}$，求正整数 m, n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2002-I-Q13] | Part C (Q11-Q15)

Triangle ABC has medians $AD = 18, CE = 27$ and side $AB = 24$. Extend median CE past E to meet the circumcircle again at F. If $[AFB] = m\sqrt{n}$ with squarefree n, find $m + n$.

三角形 ABC 中中线 $AD = 18, CE = 27$，且 $AB = 24$。将 CE 越过 E 延长，再交外接圆于 F。若 $[AFB] = m\sqrt{n}$，n 无平方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2005-I-Q15] | Part C (Q11-Q15)

Triangle ABC has $AB = 20$. Its incircle cuts the median from C into three equal segments. If the area is $m\sqrt{n}$ with squarefree n, find $m + n$.

三角形 ABC 中 $AB = 20$，内切圆将从 C 出发的中线分为三等段。若面积为 $m\sqrt{n}$ 且 n 无平方素因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2009-I-Q15] | Part C (Q11-Q15)

Triangle ABC has $AB = 10, BC = 14, CA = 16$. For interior $D \in BC$, let I_B, I_C be incenters of ABD, ACD. The circles through B, I_B, D and C, I_C, D meet again at P. The largest area $[BPC]$ is $a - b\sqrt{c}$ in squarefree form. Find $a + b + c$.

三角形 ABC 边长为十、十四、十六。D 在 BC 内，I_B, I_C 为 ABD, ACD 内心。圆 BI_BD, CI_CD 再交于 P。BPC 面积最大值为 $a - b\sqrt{c}$，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2010-II-Q15] | Part C (Q11-Q15)

Triangle ABC has $AC = 13, BC = 14, AB = 15$. Let M, N be the midpoints of AC, AB, and let the internal bisectors from B, C meet those sides at D, E. The circles through A, M, N and A, D, E meet again at P. Ray AP meets BC at Q. If $BQ/CQ = m/n$ in lowest terms, find $m - n$.

三角形 ABC 中 $AC = 13, BC = 14, AB = 15$，$M, N$ 为 AC, AB 中点，B, C 的内角平分线分别交这两边于 D, E。圆 AMN 与圆 ADE 再交于 P，射线 AP 交 BC 于 Q。将 BQ/CQ 约为 m/n，求 $m - n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2012-II-Q15] | Part C (Q11-Q15)

Triangle ABC with sides $AB = 5, BC = 7, AC = 3$ has circumcircle ω. The A-bisector meets BC at D and ω again at E. The circle of diameter DE meets ω again at F. Reduce $AF^2 = m/n$ and find $m + n$.

三角形 ABC 的边长为 $AB = 5, BC = 7, AC = 3$，外接圆为 ω。A 的角平分线交 BC 于 D、再交 ω 于 E。以 DE 为直径的圆再交 ω 于 F。将 AF^2 约为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2014-I-Q15] | Part C (Q11-Q15)

In triangle ABC, $AB = 3, BC = 4, AC = 5$. A circle passes through B, meets AB again at E, BC again at D, and AC at distinct F, G, with all named points on the side segments. Given $EF = DF, DG/EG = 3/4$, express $DE = a\sqrt{b}/c$ with coprime a, c and squarefree b, and find $a + b + c$.

三角形 ABC 的边为 $AB = 3, BC = 4, AC = 5$。过 B 的圆再交 AB, BC 于 E, D，交 AC 于不同的 F, G，各点均在对应边线段上。若 $EF = DF, DG/EG = 3/4$，将 DE 写为 $a\sqrt{b}/c$，a, c 互质、b 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-I-Q15] | Part C (Q11-Q15)

Two intersecting circles meet at X, Y and share an external tangent touching them at A, B. The tangent line is nearer X than Y. A circle through A, B meets the first circle again at D and the second again at C, with C, Y, D collinear. Given $XC = 67, XY = 47, XD = 37$, find AB^2.

两相交圆交于 X, Y，一条外公切线分别切于 A, B，且此切线距 X 比距 Y 近。过 A, B 的另一圆再次交第一圆于 D、第二圆于 C，并且 C, Y, D 共线。已知 $XC = 67, XY = 47, XD = 37$，求 AB^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-I-Q15] | Part C (Q11-Q15)

Four distinct stick lengths form three noncongruent convex cyclic quadrilaterals, all with circumradius one and common area K. Their acute diagonal angles have sines $2/3, 3/5, 6/7$. Reduce $K = m/n$ and find $m + n$.

四根不同长度的杆可构成三个不全等凸圆内接四边形，外接圆半径均为一，公共面积为 K。三个对角线锐夹角的正弦分别为 $2/3, 3/5, 6/7$。将 K 约为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-II-Q14] | Part C (Q11-Q15)

The incircle of ABC touches BC at X. Line AX meets it again at $Y \neq X$. The tangent at Y meets side segments AB, AC at P, Q. Given $AP = 3, PB = 4, AC = 8$, reduce $AQ = m/n$ and find $m + n$.

三角形内切圆切 BC 于 X，直线 AX 再交圆于 $Y \neq X$。Y 处切线交边线段 AB, AC 于 P, Q。已知 $AP = 3, PB = 4, AC = 8$，将 AQ 约为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q13] | Part C (Q11-Q15)

Triangle ABC has $AB = 4, BC = 5, CA = 6$. Points D, E on ray AB satisfy $AB < AD < AE$. The circles ACD, EBC meet again at $F \neq C$ with $DF = 2, EF = 7$. Write $BE = (a + b\sqrt{c})/d$ reduced with squarefree c, and find $a + b + c + d$.

三角形 ABC 边长为 $AB = 4, BC = 5, CA = 6$。射线 AB 上 $AB < AD < AE$，圆 ACD, EBC 再交于 $F \neq C$，且 $DF = 2, EF = 7$。将 $BE = (a + b\sqrt{c})/d$ 约简，求 $a + b + c + d$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q15] | Part C (Q11-Q15)

A chord AB of circle ω contains P, with $AP = 5, PB = 3$. Inner circles through A, P and through B, P are each internally tangent to ω, and meet again at Q. Their common chord line meets ω at X, Y, with $XY = 11$. Write $PQ^2 = m/n$ reduced and find $m + n$.

圆 ω 的弦 AB 上有 P，$AP = 5, PB = 3$。分别过 A, P 及 B, P 的两内圆与大圆内切，并再交于 Q。公共弦直线与大圆交于 X, Y，$XY = 11$。若 $PQ^2 = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-II-Q15] | Part C (Q11-Q15)

In acute triangle ABC, let P be the foot from C to AB and Q the foot from B to AC. Their line meets the circumcircle at X, Y, with $XP = 10, PQ = 25, QY = 15$. Write $AB \cdot AC = m\sqrt{n}$ with squarefree n and find $m + n$.

锐角三角形 ABC 中，P 为 C 到 AB 垂足，Q 为 B 到 AC 垂足。PQ 所在直线与外接圆交于 X, Y，$XP = 10, PQ = 25, QY = 15$。若 $AB \cdot AC = m\sqrt{n}$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2020-II-Q15] | Part C (Q11-Q15)

For an acute scalene triangle ABC, tangents to its circumcircle at B, C meet at T. Drop perpendiculars from T to the lines AB, AC, with feet X, Y. Given $TB = TC = 16$, $BC = 22$, and $TX^2 + TY^2 + XY^2 = 1143$, find XY^2.

锐角不等边三角形 ABC 的外接圆在 B, C 处的切线交于 T。从 T 向直线 AB, AC 作垂线，垂足为 X, Y。已知 $TB = TC = 16$、$BC = 22$、$TX^2 + TY^2 + XY^2 = 1143$，求 XY^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-I-Q11] | Part C (Q11-Q15)

A cyclic quadrilateral has successive side lengths $4, 5, 6, 7$. Project each vertex perpendicularly onto the other diagonal, obtaining $A_1, C_1 \in BD$ and $B_1, D_1 \in AC$. If the perimeter of $A_1B_1C_1D_1$ is m/n reduced, find $m+n$.

圆内接四边形 $ABCD$ 连续边长为 $4, 5, 6, 7$。将 A, C 向 BD 投影得 A_1, C_1，将 B, D 向 AC 投影得 B_1, D_1。若 $A_1B_1C_1D_1$ 周长约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-II-Q14] | Part C (Q11-Q15)

An acute triangle has circumcenter O and centroid G. The tangent at A meets the perpendicular to GO through G at X; let $Y = XG \cap BC$. Given $\angle B : \angle C : \angle XOY = 13 : 2 : 17$, write $\angle A = m/n$ degrees reduced and find $m + n$.

锐角三角形外心为 O、重心为 G。A 处切线与过 G 且垂直 GO 的直线交于 X，令 $Y = XG \cap BC$。已知 $\angle B : \angle C : \angle XOY = 13 : 2 : 17$。若 $\angle A$ 的度数约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-I-Q11] | Part C (Q11-Q15)

In parallelogram $ABCD$ with acute $\angle BAD$, a circle is tangent to side segments DA, AB, BC. Its intersections with diagonal AC occur in order A, P, Q, C, with $AP = 3, PQ = 9, QC = 16$. Write the area as $m\sqrt{n}$, where n is squarefree, and find $m + n$.

平行四边形 $ABCD$ 的 $\angle BAD$ 为锐角。一个圆与边线段 DA, AB, BC 相切，并与对角线 AC 依次交于 P, Q；A, P, Q, C 共线且依此排列，$AP = 3, PQ = 9, QC = 16$。面积写成 $m\sqrt{n}$，其中 n 无平方因子。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-II-Q15] | Part C (Q11-Q15)

Externally tangent circles have centers O_1, O_2 with $O_1O_2 = 15$. A third circle through both centers meets the first at B, C and the second at A, D. The cyclic order A, B, O_1, C, D, O_2 forms a convex hexagon. Given $AB = 2, CD = 16$, find its area.

两外切圆的圆心为 O_1, O_2，$O_1O_2 = 15$。第三圆经过两圆心，与第一圆交于 B, C，与第二圆交于 A, D。六点按 A, B, O_1, C, D, O_2 顺序构成凸六边形。已知 $AB = 2, CD = 16$，求面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2011-I-Q04] | Part A (Q1-Q5)

Triangle ABC has $AB = 125, AC = 117, BC = 120$. Let AL, BK be its internal angle bisectors. The perpendicular feet from C onto BK, AL are M, N. Find MN.

三角形 ABC 中 $AB = 125, AC = 117, BC = 120$，$AL, BK$ 为内角平分线。C 到 BK, AL 的垂足分别为 M, N，求 MN。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2011-II-Q04] | Part A (Q1-Q5)

In triangle ABC, $AB/AC = 20/11$. The A-bisector meets BC at D, and M is the midpoint of AD. Line BM meets AC at P. Reduce $CP/PA = m/n$ and find $m+n$.

三角形 ABC 中 $AB/AC = 20/11$，A 的角平分线交 BC 于 D，M 为 AD 中点。直线 BM 交 AC 于 P。将 CP/PA 约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-I-Q03] | Part A (Q1-Q5)

In isosceles trapezoid $ABCD$, $AB \parallel CD$, $AB = 500$, $CD = 650$, and $AD = BC = 333$. The internal bisectors at A, D meet at P; those at B, C meet at Q. Find PQ.

等腰梯形 $ABCD$ 中，$AB \parallel CD, AB = 500, CD = 650, AD = BC = 333$。$A, D$ 的内角平分线交于 P，B, C 的内角平分线交于 Q。求 PQ。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2001-II-Q07] | Part B (Q6-Q10)

Right triangle PQR has $PQ = 90, PR = 120, QR = 150$. Through its incircle, take the non-side tangent perpendicular to PR, meeting PR, QR at S, T; similarly take the non-side tangent perpendicular to PQ, meeting PQ, QR at U, V. The incenters of RST, QUV are distance $\sqrt{10n}$ apart. Find n.

直角三角形 PQR 边长为 $PQ = 90, PR = 120, QR = 150$。取内切圆的非原边切线，垂直 PR 并交 PR, QR 于 S, T；类似取垂直 PQ 的非原边切线交 PQ, QR 于 U, V。RST, QUV 两内心距为 $\sqrt{10n}$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2002-I-Q10] | Part B (Q6-Q10)

In right triangle ABC with right angle at B, let AD bisect A and meet BC at D. Take $E \in AB, F \in AC$ with $AE = 3, EB = 9, AF = 10, FC = 27$, and put $G = AD \cap EF$. Find the nearest integer to $[DCFG]$.

直角三角形 ABC 的直角在 B，角平分线 AD 交 BC 于 D。边上点满足 $AE = 3, EB = 9, AF = 10, FC = 27$，令 $G = AD \cap EF$。求四边形 $DCFG$ 面积最接近的整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2007-II-Q09] | Part B (Q6-Q10)

Rectangle $ABCD$ has $AB = 63, BC = 448$. Points $E \in AD, F \in BC$ satisfy $AE = CF = 84$. The incircles of BEF and DEF touch EF at P, Q. Find PQ.

矩形 $ABCD$ 满足 $AB = 63, BC = 448$，$E \in AD, F \in BC$，且 $AE = CF = 84$。三角形 BEF, DEF 的内切圆分别在 P, Q 与 EF 相切，求 PQ。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2016-I-Q06] | Part B (Q6-Q10)

In triangle ABC, the internal bisector from C meets AB at L and the circumcircle again at D. Its incenter is I. Given $LI = 2, LD = 3$, write $IC = p/q$ in lowest terms and find $p + q$.

三角形 ABC 中，C 的内角平分线交 AB 于 L，再交外接圆于 D，内心为 I。已知 $LI = 2, LD = 3$，将 $IC = p/q$ 约为最简分数，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2024-II-Q10] | Part B (Q6-Q10)

A triangle has incenter I, circumcenter O, inradius 6, and circumradius 13. If $IA \perp OI$, find $AB \cdot AC$.

三角形的内心为 I，外心为 O，内切圆半径 6，外接圆半径 13。若 $IA \perp OI$，求 $AB \cdot AC$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2026-II-Q08] | Part B (Q6-Q10)

An isosceles triangle has $AB = BC$ and incenter I. Both ABC and AIC have integer side lengths, and their perimeters are in ratio $125 : 6$. Find the least possible AB.

等腰三角形满足 $AB = BC$，内心为 I。ABC 与 AIC 的边长全为整数，周长比为 $125 : 6$。求最小的 AB。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2009-I-Q12] | Part C (Q11-Q15)

Right triangle ABC has legs $AC = 12, BC = 35$ and altitude CD to hypotenuse AB. The circle with diameter CD is tangent to AI, BI for a point I outside ABC, with the circle inside ABI. Write perimeter$(ABI)/AB = m/n$ in lowest terms and find $m + n$.

直角三角形 ABC 两直角边为十二、三十五，CD 为斜边高。以 CD 为直径的圆与 AI, BI 相切，I 在原三角形外且圆位于 ABI 内。若 ABI 周长与 AB 之比为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2010-I-Q15] | Part C (Q11-Q15)

Triangle ABC has $AB = 12, BC = 13, AC = 15$. A point M on AC makes the incircles of ABM and BCM have equal radii. Express $AM/CM = p/q$ in lowest terms and find $p + q$.

三角形 ABC 中 $AB = 12, BC = 13, AC = 15$。边 AC 上点 M 使三角形 ABM, BCM 的内切圆半径相等。若 $AM/CM = p/q$ 为最简分数，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2012-I-Q12] | Part C (Q11-Q15)

In right triangle ABC with right angle C, points A, D, E, B occur in that order on AB and CD, CE trisect the right angle. Given $DE/BE = 8/15$, write $\tan B = m\sqrt{p}/n$ with coprime m, n and squarefree p, and find $m + n + p$.

直角三角形 ABC 的直角在 C。斜边上点依次为 A, D, E, B，CD, CE 三等分直角。已知 $DE/BE = 8/15$，将 $\tan B$ 写为 $m\sqrt{p}/n$，m, n 互质、p 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2014-II-Q14] | Part C (Q11-Q15)

In triangle ABC, $AB = 10, A = 30^\circ, C = 45^\circ$. On line BC, let H be the altitude foot, D the internal bisector endpoint from A, and M the midpoint of BC. Let N be the midpoint of HM, and P on ray AD have perpendicular projection N. Reduce $AP^2 = m/n$ and find $m + n$.

三角形 ABC 中 $AB = 10, A = 30^\circ, C = 45^\circ$。直线 BC 上，H 为高足，D 为 A 内角平分线交点，M 为 BC 中点。N 为 HM 中点，射线 AD 上点 P 的垂足为 N。若 $AP^2 = m/n$ 已最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2015-I-Q11] | Part C (Q11-Q15)

Isosceles triangle ABC has positive integral sides and $AB = AC$. Its incenter is I, and $BI = 8$. Find the smallest possible perimeter.

等腰三角形 ABC 的三边均为正整数，$AB = AC$。内心为 I，且 $BI = 8$。求最小可能周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2018-I-Q13] | Part C (Q11-Q15)

Triangle ABC has $AB = 30, BC = 32, AC = 34$. For X strictly inside BC, let I_1, I_2 be the incenters of ABX, ACX. Find the least area of triangle AI_1I_2.

三角形 ABC 中 $AB = 30, BC = 32, AC = 34$。$X$ 在 BC 内部，I_1, I_2 分别为 ABX, ACX 的内心。求三角形 AI_1I_2 的最小面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2020-I-Q13] | Part C (Q11-Q15)

In ABC, $AB = 4, BC = 5, CA = 6$, and the internal bisector from A meets BC at D. The perpendicular bisector of AD meets the internal angle-bisector lines from B, C at E, F. If $[AEF] = m\sqrt{n}/p$ with $\gcd(m, p) = 1$ and n squarefree, find $m + n + p$.

在 $AB = 4, BC = 5, CA = 6$ 的三角形中，A 的内角平分线交 BC 于 D。AD 的垂直平分线与 B, C 的内角平分线所在直线分别交于 E, F。设 $[AEF] = m\sqrt{n}/p$，其中 m, p 互质且 n 无平方因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-I-Q14] | Part C (Q11-Q15)

A line through a boundary point of a triangle is called a perimeter bisector when it cuts the triangle into two polygons with equal perimeter. Triangle ABC has $BC = 219$ and positive integral lengths AB, AC. The perimeter bisectors through the midpoints of AB, AC meet at an acute angle of 30°. Find the triangle perimeter.

过三角形边界上一点的直线若将其分成两个周长相等的多边形，称为周长平分线。三角形 ABC 中 $BC = 219$，AB, AC 为正整数。过 AB, AC 中点的两条周长平分线所成锐角为 30°。求三角形周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

[AIME-2022-II-Q11] | Part C (Q11-Q15)

Convex quadrilateral $ABCD$ has $AB = 2, AD = 7, CD = 3$. The internal bisectors of the acute angles DAB and ADC both pass through the midpoint of BC. Find the square of its area.

凸四边形 $ABCD$ 满足 $AB = 2, AD = 7, CD = 3$。锐角 DAB, ADC 的内角平分线都经过 BC 中点。求四边形面积的平方。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-I-Q02] | Part A (Q1-Q5)

Twenty equal circles fill a rectangle in three staggered rows of $7, 6, 7$ circles. Each outer row is tangent to both vertical sides, each row has neighboring circles tangent, and each middle-row circle touches the two nearest circles in each outer row. The top and bottom circles touch the horizontal sides. If the rectangle aspect ratio is $(\sqrt{p} - q)/2$ with positive integers p, q, find $p + q$.

二十个等圆在矩形内排成三行，数目为 $7, 6, 7$，中间一行交错放置。外两行各与左右边相切；同行邻圆相切，中行每圆与上下行各两个最近圆相切；最上、最下圆分别切水平边。若长宽比为 $(\sqrt{p} - q)/2$，求正整数 p, q 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2005-I-Q01] | Part A (Q1-Q5)

Six equal circles form a cyclic ring, each touching its neighbors externally and a radius-30 circle internally. If K is the area inside the large circle but outside the six small disks, find $\lfloor K \rfloor$.

六个等圆首尾成环，相邻外切，且均与半径 30 的大圆内切。大圆内、六小圆盘外的面积为 K，求 $\lfloor K \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2009-II-Q05] | Part A (Q1-Q5)

An equilateral triangle is inscribed in a radius-ten circle. At its three vertices place internally tangent circles with radii $3, 2, 2$, respectively. A fourth circle is externally tangent to all three. If its radius is reduced m/n, find $m + n$.

半径十的圆内接等边三角形，在三个顶点分别作与大圆内切的半径三、二、二小圆。另一圆与三个小圆均外切，半径为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2000-II-Q10] | Part B (Q6-Q10)

A circle is inscribed in quadrilateral $ABCD$, touching AB at P and CD at Q. If $AP = 19, PB = 26, CQ = 37, QD = 23$, find the circle radius squared.

圆内切于四边形 $ABCD$，与 AB, CD 分别切于 P, Q。已知 $AP = 19, PB = 26, CQ = 37, QD = 23$，求半径平方。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2005-II-Q08] | Part B (Q6-Q10)

Two externally tangent circles of radii $4, 10$ are internally tangent to a third circle, all centers collinear. A common external tangent cuts a chord of the third circle of length $m\sqrt{n}/p$ in reduced squarefree form. Find $m+n+p$.

半径 $4, 10$ 的两圆外切，并均与第三圆内切，三圆心共线。两小圆的公外切线截大圆的弦长为约简的 $m\sqrt{n}/p$，求 $m+n+p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2006-II-Q09] | Part B (Q6-Q10)

Circles centered at $(0,0),(12,0),(24,0)$ have radii $1,2,4$. Choose a positive-slope common internal tangent of the first two, and a negative-slope common internal tangent of the last two. Their intersection has $x = p - q\sqrt{r}$ in squarefree form. Find $p+q+r$.

圆心为 $(0,0),(12,0),(24,0)$ 的三圆半径分别为一、二、四。取前两圆正斜率公内切线及后两圆负斜率公内切线，交点横坐标为 $p - q\sqrt{r}$，求 $p+q+r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2007-I-Q09] | Part B (Q6-Q10)

A right triangle has legs $30, 16$. On the hypotenuse's exterior side, two equal circles touch one another externally; one is tangent to the hypotenuse and the extension of the first leg, the other to the hypotenuse and the extension of the second leg. If their radius is reduced p/q, find $p+q$.

直角三角形两直角边长为三十、十六。在斜边外侧作两个等圆相互外切，并分别与斜边及两条直角边的向外延长线相切。半径为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2014-II-Q08] | Part B (Q6-Q10)

A radius-two circle has diameter AB. An inner circle D is tangent to it at A. A circle E is internally tangent to the outer circle, externally tangent to D, and tangent to segment AB. The radius of D is three times that of E and equals $\sqrt{m}-n$. Find $m+n$.

半径二的圆以 AB 为直径，内圆 D 与它在 A 内切。圆 E 与大圆内切、与 D 外切，并与直径线段相切。D 半径为 E 的三倍，且为 $\sqrt{m}-n$，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2022-I-Q08] | Part B (Q6-Q10)

An equilateral triangle ABC has circumradius 18. For each vertex, draw the circle tangent to both adjacent side segments and internally tangent to the circumcircle. The three circles meet pairwise at six points. The three intersections farther from the vertices form the smaller of two concentric equilateral triangles. Its side is $\sqrt{a}-\sqrt{b}$ for positive integers a, b. Find $a+b$.

等边三角形 ABC 的外接圆半径为 18。在每个顶点处作一个圆，与两条邻边线段相切并与外接圆内切。这三个圆两两交于六点；其中离三个原顶点较远的三个交点构成较小的等边三角形。其边长为 $\sqrt{a}-\sqrt{b}$，a, b 为正整数。求 $a+b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2022-II-Q07] | Part B (Q6-Q10)

Two circles of radii 6 and 24 touch externally. Their three common tangent lines bound a triangle. Find its area.

半径为 $6, 24$ 的两圆外切，三条公切线围成一个三角形。求其面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2024-I-Q08] | Part B (Q6-Q10)

Inside triangle ABC, a row of eight mutually consecutive tangent circles of radius 34 rests on BC, with the end circles touching AB and AC. A row of 2024 circles of radius 1 fits in the same way. If the triangle inradius is m/n in lowest terms, find $m+n$.

在三角形 ABC 内，八个半径 34 的圆依次相切且均与 BC 相切，首尾圆分别与 AB, AC 相切。同样方式也能排下 2024 个半径 1 的圆。内切圆半径为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2025-II-Q06] | Part B (Q6-Q10)

A radius-6 circle centered at A is internally tangent at B to a radius-15 circle. Let BC be a large-circle diameter, and D a large-circle point with $AD \perp BC$. A rectangle $EFGH$ is inscribed in the small circle, with $EF \perp BC$, C nearer GH than EF, and D nearer FG than EH. If triangles DGF, CHG have equal area, find $m+n$ for rectangle area m/n in lowest terms.

半径 6、圆心为 A 的圆与半径 15 的圆在 B 内切。BC 为大圆直径，D 在大圆上且 $AD \perp BC$。小圆内接矩形 $EFGH$ 满足 $EF \perp BC$，C 更靠近 GH，D 更靠近 FG。若 DGF, CHG 面积相等，将矩形面积化为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-II-Q14] | Part C (Q11-Q15)

Right triangle APM, right at A, has perimeter 152. A circle of radius 19 has center $O \in AP$ and is tangent to AM, PM. If $OP = m/n$ reduced, find $m + n$.

直角三角形 APM 的直角在 A，周长为 152。半径 19 的圆心 O 位于 AP 上，并与 AM, PM 相切。若 $OP = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2002-II-Q15] | Part C (Q11-Q15)

Two circles intersect at two points, one being $(9, 6)$, and their radii have product 68. Both the x-axis and $y = mx$ with $m > 0$ are common tangents. Write $m = a\sqrt{b}/c$ with b squarefree and $\gcd(a, c) = 1$. Find $a + b + c$.

两圆交于两点，其中一点为 $(9, 6)$，半径乘积为 68。x 轴及 $y = mx$（$m > 0$）均为两圆公切线。写 $m = a\sqrt{b}/c$，b 无平方因子且 a, c 互质，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2004-II-Q12] | Part C (Q11-Q15)

An isosceles trapezoid has bases $AB = 6, CD = 4$ and legs 5. Circles centered at A, B have radius 3; those at C, D have radius 2. A circle inside the trapezoid touches all four. If its radius is $(-k + m\sqrt{n})/p$ with squarefree n and $\gcd(k, p) = 1$, find $k + m + n + p$.

等腰梯形两底为 $AB = 6, CD = 4$，腰均为五。以 A, B 为心的圆半径三，以 C, D 为心的圆半径二。梯形内一圆与四圆都相切。半径写为 $(-k + m\sqrt{n})/p$，n 无平方因子且 k, p 互质，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2007-II-Q15] | Part C (Q11-Q15)

Inside a 13–14–15 triangle, three equal circles each touch the two sides at one vertex, and a fourth equal circle is externally tangent to all three. If their radius is reduced m/n, find $m+n$.

在边长十三、十四、十五的三角形内，三个等圆各与某顶点的两条邻边相切，第四个等圆分别与三者外切。若公共半径为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2008-II-Q11] | Part C (Q11-Q15)

In triangle ABC, $AB = AC = 100, BC = 56$. A radius-sixteen circle touches AC, BC. A second circle touches AB, BC, lies wholly inside the triangle, and is externally tangent to the first. If its radius is $m - n\sqrt{k}$ with squarefree k, find $m + nk$.

等腰三角形 ABC 中 $AB = AC = 100, BC = 56$。半径十六的圆与 AC, BC 相切，另一圆与 AB, BC 相切且完全在三角形内，并与前圆外切。其半径为 $m - n\sqrt{k}$，求 $m + nk$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2015-II-Q15] | Part C (Q11-Q15)

Circles of radii 1 and 4 touch externally at A. A common external tangent touches them at B, C, respectively. A line through A meets the small circle again at D and the large one at E, with B, C on the same side of this line. Suppose $[DBA] = [ACE] = m/n$ in lowest terms. Find $m + n$.

半径为 $1, 4$ 的两圆外切于 A，一条外公切线分别切两圆于 B, C。过 A 的直线再次交小圆、大圆于 D, E，且 B, C 在该直线同侧。若 $[DBA] = [ACE] = m/n$ 为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2019-I-Q11] | Part C (Q11-Q15)

An integer-sided isosceles triangle has $AB = AC$. A circle centered at its incenter is internally tangent to the BC-excircle and externally tangent to both other excircles. Find the least possible triangle perimeter.

整边等腰三角形 $AB = AC$，某以其内心为圆心的圆与 BC 旁切圆内切、与另两旁切圆外切。求最小周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2020-II-Q13] | Part C (Q11-Q15)

A convex pentagon has consecutive side lengths $5, 6, 6, 6, 7$ and a circle tangent to all five sides. Determine its area.

凸五边形的连续五边长为 $5, 6, 6, 6, 7$，且存在与五边均相切的内切圆。求面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.6 Tangent circles & common tangents

相切圆与公切线

[AIME-2021-I-Q13] | Part C (Q11-Q15)

Two intersecting circles have radii $961, 625$ and common chord line AB. Another circle is externally tangent to both, and line AB cuts it in a chord whose minor arc measures 120°. Find the distance between the first two centers.

半径为 $961, 625$ 的两圆相交于 A, B。第三圆与前两圆均外切，直线 AB 截第三圆所得弦的劣弧为 120°。求前两圆圆心距。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-I-Q02] | Part A (Q1-Q5)

For integers $0 < v < u$, let $A = (u, v)$. Reflect successively across $y = x$, the y-axis, the x-axis, and the y-axis to obtain B, C, D, E. If polygon $ABCDE$ has area 451, find $u + v$.

整数满足 $0 < v < u$，取 $A = (u, v)$。依次关于 $y = x$、y 轴、x 轴、y 轴反射，得到 B, C, D, E。若五边形面积为 451，求 $u + v$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2001-II-Q04] | Part A (Q1-Q5)

Points P, Q lie respectively on $8y = 15x$ and $10y = 3x$, with midpoint $(8, 6)$. If $PQ = m/n$ reduced, find $m + n$.

P, Q 分别位于 $8y = 15x$、$10y = 3x$，其中点为 $(8, 6)$。若 $PQ = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-I-Q04] | Part A (Q1-Q5)

In parallelogram $ABCD$, $M \in AB, N \in AD$ satisfy $AM/AB = 17/1000, AN/AD = 17/2009$. Let $P = AC \cap MN$. Find AC/AP.

平行四边形 $ABCD$ 中，$M \in AB, N \in AD$ 满足 $AM/AB = 17/1000, AN/AD = 17/2009$。令 $P = AC \cap MN$，求 AC/AP。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-II-Q03] | Part A (Q1-Q5)

Rectangle $ABCD$ has $AB = 100$ and midpoint E of AD. If $AC \perp BE$, find the greatest integer smaller than AD.

矩形 $ABCD$ 中 $AB = 100$，E 为 AD 中点。若 $AC \perp BE$，求小于 AD 的最大整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q02] | Part A (Q1-Q5)

Rectangle $ABCD$ has $AB = 12, BC = 10$. Interior points satisfy $BE = 9, DF = 8, BE \parallel DF, EF \parallel AB$, and line BE meets segment AD. Write $EF = m\sqrt{n} - p$ with squarefree n, and find $m + n + p$.

矩形 $ABCD$ 中 $AB = 12, BC = 10$。内部点 E, F 满足 $BE = 9, DF = 8, BE \parallel DF, EF \parallel AB$，且直线 BE 交线段 AD。若 $EF = m\sqrt{n} - p$ 且 n 无平方因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q03] | Part A (Q1-Q5)

Line L has slope $5/12$ through $A = (24, -1)$; line M is perpendicular to it through $B = (5, 6)$. Use L, M as new x, y axes, oriented positively toward A, B. Find the sum of the new coordinates of the old point $(-14, 27)$.

直线 L 过 $A = (24, -1)$、斜率为 $5/12$；直线 M 过 $B = (5, 6)$ 且垂直于 L。以两线为新横纵轴，正方向分别指向 A, B。求原坐标点 $(-14, 27)$ 的新坐标之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-II-Q02] | Part A (Q1-Q5)

In square $ABCD$, nonvertex points E on AD and F on BC satisfy $BE = EF = FD = 30$. Find the square area.

正方形 $ABCD$ 的边 AD, BC 上分别有非顶点 E, F，满足 $BE = EF = FD = 30$。求正方形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-II-Q04] | Part A (Q1-Q5)

Construct equilateral triangle ABC from $A=(1,0), B=(2,2\sqrt{3})$, choosing C in the first quadrant. Its center (x,y) has $xy=p\sqrt{q}/r$, with coprime p,r and squarefree q. Find $p+q+r$.

以 $A=(1,0), B=(2,2\sqrt{3})$ 作等边三角形，取第一象限顶点 C。中心为 (x,y)，若 $xy=p\sqrt{q}/r$，p,r 互质、q 无平方因子，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2015-I-Q04] | Part A (Q1-Q5)

Points A, B, C are collinear in that order with $AB = 16, BC = 4$. Equilateral triangles ABD, BCE lie on the same side of AC. Let M be the midpoint of AE and N the midpoint of CD. If triangle BMN has area x, find x^2.

A, B, C 依次共线，$AB = 16, BC = 4$。等边三角形 ABD, BCE 位于 AC 同侧。M, N 分别是 AE, CD 中点。若 BMN 面积为 x，求 x^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2018-II-Q04] | Part A (Q1-Q5)

A convex equiangular octagon has consecutive labels $CAROLINE$, with $CA = RO = LI = NE = \sqrt{2}$ and the other four sides one. Reconnect the same vertices in order $CORNELIA$, forming six bounded nonoverlapping triangles. If their total area is reduced a/b, find $a + b$.

凸等角八边形按 $CAROLINE$ 排列，$CA = RO = LI = NE = \sqrt{2}$，其余四边均为一。将原顶点改按 $CORNELIA$ 顺序连接，围成六个不重叠的有界三角形区域。若总面积为最简分数 a/b，求 $a + b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

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[AIME-2019-II-Q01] | Part A (Q1-Q5)

On the same side of $AB = 9$, distinct points C, D satisfy $AC = BD = 17, BC = AD = 10$. The overlapping triangle regions ABC, BAD have area reduced m/n. Find $m + n$.

在线段 $AB = 9$ 同侧有不同点 C, D，满足 $AC = BD = 17, BC = AD = 10$。两三角形区域重叠面积为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2020-II-Q04] | Part A (Q1-Q5)

A clockwise rotation through $m \in (0, 180)$ degrees about (x, y) maps $A = (0, 0), B = (0, 12), C = (16, 0)$ respectively to $(24, 18), (36, 18), (24, 2)$. Find $m + x + y$.

绕 (x, y) 顺时针旋转 m 度，$0 < m < 180$，将 $(0, 0), (0, 12), (16, 0)$ 依次变为 $(24, 18), (36, 18), (24, 2)$。求 $m + x + y$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2021-I-Q02] | Part A (Q1-Q5)

Rectangle $ABCD$ has $AB = 3, BC = 11$. A second rectangle, in cyclic order $AECF$, shares opposite vertices A, C and has $AF = 7, FC = 9$, with F on the opposite side of line AC from B. Its overlap with $ABCD$ has area m/n reduced. Find $m + n$.

矩形 $ABCD$ 中 $AB = 3, BC = 11$。另一矩形按 A, E, C, F 顺序取顶点，共用对角顶点 A, C，且 $AF = 7, FC = 9$；F 与 B 位于直线 AC 两侧。两矩形重叠面积约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2023-I-Q05] | Part A (Q1-Q5)

A square $ABCD$ is inscribed in a circle. A point P on that circle satisfies $PA\,PC = 56$ and $PB\,PD = 90$. Determine the square area.

正方形 $ABCD$ 内接于一圆，圆上点 P 满足 $PA\,PC = 56, PB\,PD = 90$。求正方形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2024-I-Q05] | Part A (Q1-Q5)

Rectangles $ABCD$ and $EFGH$ have consecutive side lengths $(107, 16)$ and $(184, 17)$. Along one line the points occur in order D, E, C, F; the rectangles lie on opposite sides of this line. If A, D, H, G are concyclic, determine CE.

矩形 $ABCD, EFGH$ 的连续两边长分别为 $(107, 16)$、$(184, 17)$。D, E, C, F 依次共线，两矩形位于该线两侧。若 A, D, H, G 共圆，求 CE。

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[AIME-2025-II-Q01] | Part A (Q1-Q5)

Collinear points A, B, C, D, E, F occur in that order, with $AC = 26, BD = 22, CE = 31, DF = 33, AF = 73$. An off-line point G has $CG = 40, DG = 30$. Find $[BGE]$.

共线点 A, B, C, D, E, F 依次排列，$AC = 26, BD = 22, CE = 31, DF = 33, AF = 73$。线外点 G 满足 $CG = 40, DG = 30$。求 BGE 面积。

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[AIME-2026-II-Q03] | Part A (Q1-Q5)

A simple nonconvex pentagon $ABCDE$ has angles $A = E = 90^\circ, B = D = 45^\circ$, with B, C, D on the same side of AE. Given $DE < AB$, $AE = 20$, $BC = 14\sqrt{2}$, integer $AB < 2026$, and area divisible by 16, count possible values of AB.

简单非凸五边形 $ABCDE$ 中，$A = E = 90^\circ, B = D = 45^\circ$，$B, C, D$ 位于 AE 同侧。已知 $DE < AB, AE = 20, BC = 14\sqrt{2}$，$AB$ 为小于 2026 的整数，面积为 16 的整数倍。求 AB 的可能值个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2000-II-Q08] | Part B (Q6-Q10)

In trapezoid $ABCD$, BC is perpendicular to the parallel bases AB, CD, and diagonals AC, BD are perpendicular. Given $AB = \sqrt{11}, AD = \sqrt{1001}$, determine BC^2.

梯形 $ABCD$ 中，BC 垂直两底 AB, CD，对角线 AC, BD 也互相垂直。已知 $AB = \sqrt{11}, AD = \sqrt{1001}$，求 BC^2。

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[AIME-2004-II-Q07] | Part B (Q6-Q10)

Fold rectangle $ABCD$ so B lands at $B' \in AD$. The crease meets AB, CD at E, F, with $AE = 8, EB = 17, CF = 3$. If the rectangle perimeter is m/n reduced, find $m + n$.

矩形 $ABCD$ 折叠后 B 与 AD 上点 B' 重合，折痕交 AB, CD 于 E, F。已知 $AE = 8, EB = 17, CF = 3$。若矩形周长约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2005-I-Q10] | Part B (Q6-Q10)

Triangle ABC has area 70, $B = (12, 19)$, $C = (23, 20)$, and $A = (p, q)$. Its median to BC lies on a line of slope -5. Maximize $p + q$.

三角形 ABC 面积为 70，$B = (12, 19), C = (23, 20), A = (p, q)$。到 BC 的中线所在直线斜率为 -5，求 $p + q$ 最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2006-I-Q10] | Part B (Q6-Q10)

Eight radius-$1/2$ disks have centers $(i+1/2, j+1/2)$ for $i, j \in \{0,1,2\}$ except $(i,j) = (2,2)$. A slope-three line bisects their total area. Write it as $ax = by + c$ in relatively prime positive integers and find $a^2+b^2+c^2$.

八个半径 $1/2$ 的圆盘，圆心为 $(i+1/2, j+1/2)$，其中 $i, j = 0,1,2$，但去掉 $(2,2)$。一条斜率三的直线平分总面积，写成互质正整数方程 $ax = by + c$，求 $a^2+b^2+c^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2006-II-Q06] | Part B (Q6-Q10)

In unit square $ABCD$, points $E \in BC, F \in CD$ form equilateral triangle AEF. A smaller axis-parallel square inside $ABCD$ has vertex B and its opposite vertex on AE. Its side is $(a - \sqrt{b})/c$ in squarefree form. Find $a + b + c$.

单位正方形 $ABCD$ 中，$E \in BC, F \in CD$ 使 AEF 为等边三角形。内部另一边平行的小正方形以 B 为一顶点、对顶点在 AE 上，边长为 $(a - \sqrt{b})/c$，求 $a + b + c$。

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[AIME-2009-II-Q10] | Part B (Q6-Q10)

Points A, B, C have $AB = 5, BC = 12, AC = 13$. A fourth point D satisfies $\angle BAD = \angle CAD$ and $\angle ACB = \angle DCB$, with nondegenerate rays. If $AD = p\sqrt{r}/q$ is reduced and r squarefree, find $p + q + r$.

$AB = 5, BC = 12, AC = 13$，另有点 D 满足 $\angle BAD = \angle CAD, \angle ACB = \angle DCB$，各射线不退化。若 $AD = p\sqrt{r}/q$ 为约简形式，求 $p + q + r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2010-II-Q09] | Part B (Q6-Q10)

In regular hexagon $ABCDEF$, let G, H, I, J, K, L be the midpoints of AB, BC, CD, DE, EF, FA. Segments AH, BI, CJ, DK, EL, FG enclose a smaller regular hexagon. Its area divided by the original area is m/n in lowest terms. Find $m+n$.

正六边形 $ABCDEF$ 各边 AB, BC, CD, DE, EF, FA 的中点依次为 G, H, I, J, K, L。线段 AH, BI, CJ, DK, EL, FG 围成一个较小正六边形，面积比为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2013-I-Q07] | Part B (Q6-Q10)

A rectangular box measures 12 by 16 by m/n, with reduced positive m/n. The centers of three faces meeting at one corner form a triangle of area thirty. Find $m+n$.

长方体的三尺寸为十二、十六及最简正分数 m/n。同一顶点处三个面的中心构成面积三十的三角形，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2014-I-Q10] | Part B (Q6-Q10)

A unit disk rolls without slipping outside a fixed radius-five disk, initially tangent at marked point A. Let E be the fixed center. Stop after the rolling disk rotates once relative to the ground, and let B be the new position of its marked point. Reduce $\sin^2 \angle BEA = m/n$ and find $m+n$.

单位圆盘沿固定半径五的圆盘外侧无滑动滚动，初始切点标为 A，固定圆心为 E。当活动圆盘相对地面自转一整圈时停止，标记点到达 B。将 $\sin^2 \angle BEA$ 约为 m/n，求 $m+n$。

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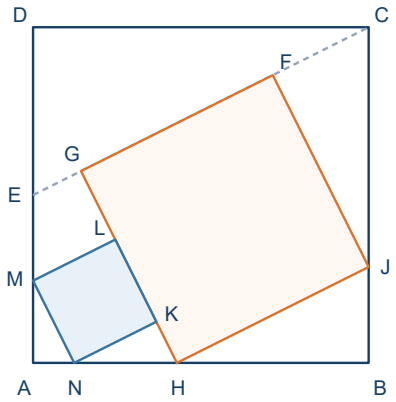
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[AIME-2015-I-Q07] | Part B (Q6-Q10)

In square $ABCD$, E is the midpoint of AD. Square $FGHJ$ has F, G on CE, H on AB, and J on BC. Square $KLMN$ has K, L on GH, M on AD, and N on AB; vertex names for each square follow its boundary. Given $[KLMN] = 99$, find $[FGHJ]$.

正方形 $ABCD$ 中，E 是 AD 中点。正方形 $FGHJ$ 的 F, G 在 CE 上、H 在 AB 上、J 在 BC 上。正方形 $KLMN$ 的 K, L 在 GH 上、M 在 AD 上、N 在 AB 上；各正方形顶点均按边界顺序命名。已知 $KLMN$ 面积为 99，求 $FGHJ$ 面积。



Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2016-II-Q07] | Part B (Q6-Q10)

Two concentric, side-parallel squares have areas 2016 and a smaller positive integer. A third square has one vertex on each side of the outer square, while each vertex of the inner square lies on a side of the third. Find the difference between the largest and smallest integral areas of the third square.

两个同心且对应边平行的正方形面积分别为 2016 和某个较小正整数。第三个正方形的四个顶点分别在外正方形四边上，内正方形每个顶点又在第三个正方形的一边上。求第三正方形面积的最大、最小整数值之差。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2017-II-Q10] | Part B (Q6-Q10)

Rectangle $ABCD$ has $AB = 84, AD = 42$. Let M be the midpoint of AD, N the point on AB with $AN = 28$, and $O = CM \cap DN$. A segment BP, with P on the boundary of quadrilateral $BCON$, splits its area equally. Find $[CDP]$.

矩形 $ABCD$ 中 $AB = 84, AD = 42$。M 为 AD 中点，N 在 AB 上且 $AN = 28$，$O = CM \cap DN$。线段 BP 将四边形 $BCON$ 面积平分，P 在其边界上。求 $[CDP]$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2018-I-Q08] | Part B (Q6-Q10)

An equiangular convex hexagon $ABCDEF$ has $AB = 6, BC = 8, CD = 10, DE = 12$. If d is the largest diameter of a disk contained in it, find d^2.

凸等角六边形 $ABCDEF$ 满足 $AB = 6, BC = 8, CD = 10, DE = 12$。设可完全放入其中的圆盘最大直径为 d，求 d^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2018-II-Q09] | Part B (Q6-Q10)

From $[0,27]\times[0,23]$, remove each corner right triangle with horizontal leg eight and vertical leg six, obtaining octagon $ABCDEFGH$ starting at $A=(0,6)$ and $B=(8,0)$ around the boundary. Let J be the midpoint of AH. Fan-triangulate from J into seven triangles using the other six vertices B through G. Find the area of their centroid polygon.

从矩形 $[0,27]\times[0,23]$ 各角去掉水平直角边八、竖直直角边六的三角形，得到依次以 $A=(0,6), B=(8,0)$ 开始的八边形 $ABCDEFGH$。J 为 AH 中点，从 J 连至 B 至 G 将区域分为七个三角形，求各重心组成多边形的面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2021-I-Q09] | Part B (Q6-Q10)

In an isosceles trapezoid $ABCD$ with parallel bases $AB < CD$ and legs $AD = BC$, the distances from A to the lines BC, CD, BD are $15, 18, 10$. If its area is K, determine $\sqrt{2}K$.

等腰梯形 $ABCD$ 的底边 $AB \parallel CD$、$AB < CD$，腰 $AD = BC$。点 A 到直线 BC, CD, BD 的距离依次为 $15, 18, 10$。若面积为 K，求 $\sqrt{2}K$。

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[AIME-2023-I-Q08] | Part B (Q6-Q10)

A rhombus $ABCD$ has acute angle BAD. Point P lies on its incircle and is at distances $9, 5, 16$ from the lines DA, AB, BC, respectively. Find the rhombus perimeter.

菱形 $ABCD$ 的 $\angle BAD$ 为锐角。内切圆上点 P 到直线 DA, AB, BC 的距离依次为 $9, 5, 16$。求菱形周长。

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[AIME-2025-I-Q08] | Part B (Q6-Q10)

For real k, the system $|25+20i-z|=5$, $|z-4-k|=|z-3i-k|$ has exactly one complex solution. The sum of all such k is m/n in lowest terms. Find $m+n$.

实数 k 使复数方程组 $|25+20i-z|=5, |z-4-k|=|z-3i-k|$ 恰有一解。所有这样的 k 之和为最简分数 m/n，求 $m+n$。

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[AIME-2026-I-Q10] | Part B (Q6-Q10)

Triangle ABC has $AB = 13, BC = 14, CA = 15$. Rotate it about its circumcenter to $A'B'C'$ so $A'C' \perp BC$ and A', B lie on opposite sides of $B'C'$. Find the nearest integer to area $AA'CC'BB'$.

三角形 ABC 边长为 $AB = 13, BC = 14, CA = 15$，绕外心旋转得 $A'B'C'$，使 $A'C' \perp BC$，且 A', B 位于 $B'C'$ 两侧。求六边形 $AA'CC'BB'$ 面积最接近的整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2000-I-Q13] | Part C (Q11-Q15)

A vehicle starts at the crossing of perpendicular straight roads. It travels at speed 50 miles/hour on roads and 14 miles/hour elsewhere in the plane. The area reachable within six minutes is m/n square miles in lowest terms. Find $m+n$.

车辆从两条垂直直路交点出发，公路速度为每小时 50 英里，平面其他地方速度为每小时 14 英里。六分钟内可达区域面积约分为 m/n 平方英里，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2000-II-Q11] | Part C (Q11-Q15)

An isosceles trapezoid has lattice vertices, $A = (20, 100), D = (21, 107)$, with AB, CD its only parallel sides and no side horizontal or vertical. Sum the absolute values of every possible distinct slope of AB, write the result m/n reduced, and find $m + n$.

等腰梯形顶点均为整点，$A = (20, 100), D = (21, 107)$；仅 AB, CD 平行，所有边均不水平也不竖直。将 AB 所有不同可能斜率的绝对值相加，约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2002-I-Q11] | Part C (Q11-Q15)

Inside the cube $[0,12]^3$, a light ray starts at $(0,0,0)$ in direction $(12,7,5)$ and reflects specularly at every face. At its first subsequent arrival at a cube vertex, its traveled length is $m\sqrt{n}$ with squarefree n. Find $m+n$.

在立方体 $[0,12]^3$ 内，光线从 $(0,0,0)$ 沿 $(12,7,5)$ 方向出发，在每个面镜面反射。首次再次到达立方体顶点时，路程为 $m\sqrt{n}$，其中 n 无平方因子，求 $m+n$。

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坐标、向量与变换

[AIME-2003-I-Q15] | Part C (Q11-Q15)

In ABC, $AB = 360, BC = 507, CA = 780$. Let M be the midpoint of CA, and let BD be the internal bisector of B meeting CA. Through D, the perpendicular to BD meets BC at F and BM at E. If $DE/EF = m/n$ reduced, find $m + n$.

三角形 ABC 边长为 $AB = 360, BC = 507, CA = 780$。$M$ 为 CA 中点，B 的内角平分线交 CA 于 D。过 D 作 BD 垂线交 BC 于 F、交 BM 于 E。若 $DE/EF = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2003-II-Q11] | Part C (Q11-Q15)

In right triangle ABC, $AC = 7, BC = 24$ and C is right. Let M be the midpoint of AB. Choose D on the same side of AB as C with $AD = BD = 15$. Write $[CDM] = m\sqrt{n}/p$ reduced with squarefree n, and find $m + n + p$.

直角三角形 ABC 中 $AC = 7, BC = 24$，直角在 C。M 为斜边 AB 中点，D 与 C 位于 AB 同侧且 $AD = BD = 15$。将 $[CDM]$ 写为最简 $m\sqrt{n}/p$，n 无平方因子，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2003-II-Q14] | Part C (Q11-Q15)

A convex equilateral hexagon $ABCDEF$ has $A=(0,0), B=(b,2)$, $\angle FAB = 120^\circ$, and opposite sides parallel: $AB \parallel DE, BC \parallel EF, CD \parallel FA$. Its six y-coordinates are $0, 2, 4, 6, 8, 10$ in some order. If area $= m\sqrt{n}$ with squarefree n, find $m+n$.

凸等边六边形 $ABCDEF$ 满足 $A=(0,0), B=(b,2), \angle FAB = 120^\circ$，且 $AB \parallel DE, BC \parallel EF, CD \parallel FA$。六个纵坐标恰为 $0, 2, 4, 6, 8, 10$。面积写为 $m\sqrt{n}$，n 无平方因子，求其和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2004-II-Q15] | Part C (Q11-Q15)

A 1024×1 strip is divided into unit squares numbered from the left. Ten times, fold its right half over and on top of its left half. In the final stack, how many squares are below the square originally numbered 942?

长一千零二十四、宽一的纸条分为单位方格，从左编号。连续十次将右半折到左半上方。最终堆叠中，原第九百四十二格下方有多少格?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2005-I-Q14] | Part C (Q11-Q15)

Points $A=(0,12), B=(10,9), C=(8,0), D=(-4,7)$ lie on different sides of one square. If its area is K, find $10K \bmod 1000$.

点 $A=(0,12), B=(10,9), C=(8,0), D=(-4,7)$ 分别位于同一正方形的不同边上。若其面积为 K，求 $10K$ 除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2005-II-Q12] | Part C (Q11-Q15)

In a side-900 square with center O, points A, E, F, B occur in order along one side, $EF = 400$, $AE < BF$, and $\angle EOF = 45^\circ$. If $BF = p + q\sqrt{r}$ in squarefree form, find $p + q + r$.

边长九百的正方形中心为 O，某边上依次为 A, E, F, B，$EF = 400, AE < BF, \angle EOF = 45^\circ$。若 $BF = p + q\sqrt{r}$，r 无平方素因子，求 $p + q + r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2006-I-Q14] | Part C (Q11-Q15)

A rigid tripod has three length-five legs with equal mutual angles and initially stands with its top four feet high. One leg loses its bottom foot while the mutual angles remain fixed. The new height is $h = m/\sqrt{n}$ with squarefree n. Find $\lfloor m + \sqrt{n} \rfloor$.

刚性三脚架三腿均长五英尺，两两夹角相同，原顶高四。某一腿底部断去一英尺而夹角保持不变。新高度为 $m/\sqrt{n}$，n 无平方素因子，求 $\lfloor m + \sqrt{n} \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2011-I-Q14] | Part C (Q11-Q15)

In regular octagon $A_1 \cdots A_8$, let M_i be the midpoint of A_iA_{i+1} for odd i. From these four points draw inward rays R_i with each consecutive pair perpendicular. Let $B_i = R_i \cap R_{i+2}$, using indices modulo eight. If $B_1B_3 = A_1A_2$, write $\cos(2\angle A_3M_3B_1) = m - \sqrt{n}$ with positive integers m, n, and find $m + n$.

正八边形 $A_1 \cdots A_8$ 中，奇数 i 对应的 M_i 为边 A_iA_{i+1} 中点。从四点向内作射线 R_i，相邻射线互相垂直，记 $B_i = R_i \cap R_{i+2}$，下标模八。若 $B_1B_3 = A_1A_2$，将 $\cos(2\angle A_3M_3B_1)$ 写成 $m - \sqrt{n}$，求 $m + n$。

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[AIME-2011-II-Q13] | Part C (Q11-Q15)

In a side-12 square $ABCD$, point P is on AC with $AP > CP$. The circumcenters O_1, O_2 of ABP, CDP satisfy $\angle O_1PO_2 = 120^\circ$. If $AP = \sqrt{a} + \sqrt{b}$, find $a + b$.

边长十二的正方形 $ABCD$ 中，P 在对角线 AC 上且 $AP > CP$。ABP, CDP 的外心分别为 O_1, O_2，满足 $\angle O_1PO_2 = 120^\circ$。若 $AP = \sqrt{a} + \sqrt{b}$，求 $a + b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2012-I-Q13] | Part C (Q11-Q15)

An equilateral triangle has one vertex on each of three concentric circles of radii $3, 4, 5$. Its greatest area is $a + (b/c)\sqrt{d}$, with coprime positive b, c and squarefree d. Find $a + b + c + d$.

一个等边三角形的三个顶点分别在半径为三、四、五的同心圆上。其最大面积为 $a + (b/c)\sqrt{d}$，正整数 b, c 互质且 d 无平方因子，求四者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2012-II-Q13] | Part C (Q11-Q15)

Equilateral ABC has side $\sqrt{111}$. Points D_1, D_2 both satisfy $AD_i = \sqrt{111}, BD_i = \sqrt{11}$. On each base AD_i construct both equilateral triangles, giving four apexes $E_1, \ldots, E_4$. Find $\sum_{k=1}^{4} CE_k^2$.

等边三角形 ABC 边长为 $\sqrt{111}$。两点 D_1, D_2 均满足 $AD_i = \sqrt{111}, BD_i = \sqrt{11}$。在每条 AD_i 两侧分别作等边三角形，四顶点为 E_1 至 E_4。求 $\sum CE_k^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2013-I-Q12] | Part C (Q11-Q15)

Triangle PQR has angles $P = 75^\circ, Q = 60^\circ$. A unit regular hexagon $ABCDEF$ lies inside it, with AB on PQ, CD on QR, and another vertex on RP. Express the triangle area as $(a + b\sqrt{c})/d$, with coprime a, d and squarefree c, and find $a + b + c + d$.

三角形 PQR 的角为 $75^\circ, 60^\circ$。单位边长正六边形 $ABCDEF$ 位于内部，AB 在 PQ 上，CD 在 QR 上，另有一顶点在 RP 上。若三角形面积为 $(a + b\sqrt{c})/d$，a, d 互质、c 无平方因子，求四者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2013-II-Q13] | Part C (Q11-Q15)

An isosceles triangle has $AC = BC$. Point D on BC obeys $CD = 3BD$, and E is the midpoint of AD. Given $CE = \sqrt{7}, BE = 3$, write the triangle area $m\sqrt{n}$ with squarefree n and find $m + n$.

等腰三角形满足 $AC = BC$，边 BC 上点 D 满足 $CD = 3BD$，E 为 AD 中点。已知 $CE = \sqrt{7}, BE = 3$，若面积为 $m\sqrt{n}$ 且 n 无平方因子，求 $m + n$。

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[AIME-2014-I-Q13] | Part C (Q11-Q15)

On the respective sides AB, BC, CD, DA of a square lie E, F, G, H. The segments EG, FH are perpendicular, both of length 34, and meet at P. The corner-region areas $AEPH, BFPE, CGPF, DHPG$ are in ratio $269 : 275 : 405 : 411$. Find the square area.

正方形四边 AB, BC, CD, DA 上分别有 E, F, G, H，EG, FH 垂直且均长三十四，交于 P。四角区域 $AEPH, BFPE, CGPF, DHPG$ 面积比为 $269 : 275 : 405 : 411$。求正方形面积。

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[AIME-2014-II-Q11] | Part C (Q11-Q15)

Triangle RED has $RD = 1, \angle R = 75^\circ, \angle E = 45^\circ$. Let M be the midpoint of RD and take C on ED so $RC \perp EM$. Extend DE past E to A with $CA = AR$. If $AE = (a - \sqrt{b})/c$ with coprime a, c, find $a + b + c$.

三角形 RED 中 $RD = 1, \angle R = 75^\circ, \angle E = 45^\circ$，$M$ 为 RD 中点。在 ED 上取 C 使 $RC \perp EM$。延长 DE 过 E 至 A，使 $CA = AR$。若 $AE = (a - \sqrt{b})/c$、a, c 互质，求三者和。

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[AIME-2015-II-Q11] | Part C (Q11-Q15)

Acute triangle ABC has circumcenter O. The line through O perpendicular to OB meets lines AB, BC at P, Q. Given $AB = 5, BC = 4, BQ = 9/2$, express $BP = m/n$ in lowest terms and find $m + n$.

锐角三角形 ABC 的外心为 O。过 O 作垂直 OB 的直线，交直线 AB, BC 于 P, Q。已知 $AB = 5, BC = 4, BQ = 9/2$。将 $BP = m/n$ 约为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2016-I-Q14] | Part C (Q11-Q15)

At each lattice point, center a circle of radius $1/10$ and an axis-aligned closed square of side $1/5$. The segment from $(0,0)$ to $(1001,429)$ intersects m squares and n circle circumferences, with tangency included. Find $m+n$.

每个整点为圆心作半径 $1/10$ 的圆周，并以该点为中心作边长 $1/5$、边平行坐标轴的闭正方形。从 $(0,0)$ 到 $(1001,429)$ 的线段与 m 个正方形、n 个圆周相交，切触也计入。求 $m+n$。

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[AIME-2017-I-Q15] | Part C (Q11-Q15)

A right triangle has legs $5, 2\sqrt{3}$ and hypotenuse $\sqrt{37}$. Among equilateral triangles with one vertex on each side segment, write the smallest area as $m\sqrt{p}/n$, where m, n are coprime positive integers and p is squarefree. Find $m+n+p$.

直角三角形的两直角边为 $5, 2\sqrt{3}$，斜边为 $\sqrt{37}$。在每条边线段上各取一顶点构成等边三角形，将最小面积写为 $m\sqrt{p}/n$，其中 m, n 为互质正整数，p 无平方因子。求 $m+n+p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2017-II-Q15] | Part C (Q11-Q15)

Tetrahedron $ABCD$ has opposite edge pairs $AD = BC = 28$, $AC = BD = 44$, $AB = CD = 52$. Minimize $AX + BX + CX + DX$ over all spatial points X. If the minimum is $m\sqrt{n}$ with squarefree positive integer n, find $m + n$.

四面体 $ABCD$ 的对棱分别满足 $AD = BC = 28, AC = BD = 44, AB = CD = 52$。在空间任意点 X 中最小化 $AX + BX + CX + DX$。若最小值为 $m\sqrt{n}$，n 为无平方因子的正整数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2019-II-Q11] | Part C (Q11-Q15)

Triangle ABC has $AB = 7, BC = 8, CA = 9$. One circle passes through B and is tangent to line AC at A; another passes through C and is tangent to line AB at A. Their second intersection is K. Write $AK = m/n$ reduced and find $m + n$.

三角形 ABC 中 $AB = 7, BC = 8, CA = 9$。一圆过 B 并在 A 与直线 AC 相切，另一圆过 C 并在 A 与 AB 相切。再交点为 K，若 $AK = m/n$ 最简，求 $m + n$。

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[AIME-2019-II-Q13] | Part C (Q11-Q15)

A regular octagon is inscribed in a circle of area one. For an interior point P, the curvilinear regions formed by P, A_1, A_2 and their minor arc, and by P, A_3, A_4 and their minor arc, have areas $1/7, 1/9$. The analogous P, A_6, A_7 region has area $1/8 - \sqrt{2}/n$. Find positive integer n.

正八边形内接于面积一的圆，圆内点 P 与 A_1, A_2 及其劣弧围成区域面积 $1/7$，与 A_3, A_4 对应区域面积 $1/9$。P, A_6, A_7 对应面积为 $1/8 - \sqrt{2}/n$，求正整数 n。

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[AIME-2020-I-Q15] | Part C (Q11-Q15)

An acute triangle ABC has circumcircle ω and orthocenter H. The tangent at H to the circle through H, B, C cuts ω at X, Y. Suppose $HA = 3, HX = 2, HY = 6$. If $[ABC] = m\sqrt{n}$ with n squarefree, find $m + n$.

锐角三角形 ABC 的外接圆为 ω，垂心为 H。过 H, B, C 的圆在 H 处的切线交 ω 于 X, Y。已知 $HA = 3, HX = 2, HY = 6$。设面积为 $m\sqrt{n}$，n 无平方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2020-II-Q12] | Part C (Q11-Q15)

An m-row, n-column grid, with odd $m, n > 1$, is numbered consecutively left to right in each row, starting at the top. Label 200 lies in row 1, and label 2000 in row m. Count ordered pairs (m, n) for which the line through those two cell centers crosses the interior of cell 1099.

一个 m 行 n 列单位方格阵中，$m, n > 1$ 均为奇数，从顶行起逐行由左向右连续编号。200 号在顶行，2000 号在底行。求使这两格中心连线穿过 1099 号格内部的有序对 (m, n) 数量。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2021-II-Q12] | Part C (Q11-Q15)

A convex quadrilateral has consecutive side lengths $5, 6, 9, 7$ and area 30. Its diagonals meet at acute angle θ. Write $\tan\theta = m/n$ reduced and find $m+n$.

凸四边形连续边长为 $5, 6, 9, 7$，面积为 30。两对角线的锐夹角为 θ。将 $\tan\theta$ 约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2023-I-Q12] | Part C (Q11-Q15)

An equilateral triangle ABC has side 55. On BC, CA, AB lie D, E, F, with $BD = 7, CE = 30, AF = 40$. An interior point P satisfies $\angle AEP = \angle BFP = \angle CDP = \theta$. Find $\tan^2\theta$.

边长为 55 的等边三角形 ABC 中，D, E, F 分别在 BC, CA, AB 上，且 $BD = 7, CE = 30, AF = 40$。内点 P 满足 $\angle AEP = \angle BFP = \angle CDP = \theta$。求 $\tan^2\theta$。

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[AIME-2023-II-Q12] | Part C (Q11-Q15)

Triangle ABC has sides $AB = 13, BC = 14, CA = 15$. Let M be the midpoint of BC, and let line AM meet the circumcircle again at P, beyond M. The unique Q on AM satisfying $\angle PBQ = \angle PCQ$ has $AQ = m/\sqrt{n}$, with coprime positive integers m, n. Find $m + n$.

三角形 ABC 的边长为 $AB = 13, BC = 14, CA = 15$。$M$ 为 BC 中点，直线 AM 在 M 外侧再交外接圆于 P。线段 AM 上唯一满足 $\angle PBQ = \angle PCQ$ 的点为 Q。若 $AQ = m/\sqrt{n}$，正整数 m, n 互质，求 $m + n$。

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[AIME-2025-I-Q12] | Part C (Q11-Q15)

In the plane $x+y+z=75$, the inequalities $x-yz<y-zx<z-xy$ define three convex regions, only one bounded. If its area is $a\sqrt{b}$ with squarefree b, find $a+b$.

平面 $x+y+z=75$ 内，满足 $x-yz<y-zx<z-xy$ 的点组成三个凸区域，恰有一个有界。其面积为 $a\sqrt{b}$，b 无平方素因子，求 $a+b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2026-I-Q14] | Part C (Q11-Q15)

An equiangular pentagon has sum of squared side lengths 308 and sum of squared diagonals 800. Its squared perimeter is $m\sqrt{n}$, with squarefree n. Find $m+n$.

等角五边形的边长平方和为 308，对角线长平方和为 800。周长平方为 $m\sqrt{n}$，其中 n 无平方素因子，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2001-I-Q05] | Part A (Q1-Q5)

An equilateral triangle lies on $x^2 + 4y^2 = 4$, has vertex $(0, 1)$, and has an altitude on the y-axis. Its squared side length is m/n reduced. Find $m + n$.

等边三角形内接于 $x^2 + 4y^2 = 4$，一个顶点为 $(0, 1)$，一条高位于 y 轴。边长平方约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-I-Q09] | Part B (Q6-Q10)

A rhombus has vertices on $x^2/20 - y^2/24 = 1$ and its diagonals meet at the origin. Find the greatest real number strictly smaller than BD^2 for every such rhombus $ABCD$.

菱形 $ABCD$ 的顶点在双曲线 $x^2/20 - y^2/24 = 1$ 上，两对角线交于原点。求对所有这种菱形均严格小于 BD^2 的最大实数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2025-I-Q09] | Part B (Q6-Q10)

Rotate $y = x^2 - 4$ counterclockwise by 60° about the origin. The unique fourth-quadrant intersection of the original curve and its image has $y = (a - \sqrt{b})/c$, with positive integers and coprime a, c. Find $a + b + c$.

将抛物线 $y = x^2 - 4$ 绕原点逆时针旋转 60°。原曲线与像曲线在第四象限的唯一交点，其纵坐标为 $(a - \sqrt{b})/c$，其中为正整数且 a, c 互质，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2026-II-Q06] | Part B (Q6-Q10)

Sum all radii r for which a circle centered at $(4, 39)$ is tangent at some point to the parabola $2y = x^2 - 8x + 12$.

求所有使圆心 $(4, 39)$ 的圆与抛物线 $2y = x^2 - 8x + 12$ 在至少一点相切的半径 r 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2005-II-Q15] | Part C (Q11-Q15)

Circles are given by $x^2+y^2+10x-24y-87=0$ and $x^2+y^2-10x-24y+153=0$. Find the least positive slope a of a line $y=ax$ containing the center of a circle internally tangent to the first and externally tangent to the second. If $a^2=p/q$ is reduced, find $p+q$.

两圆方程为 $x^2+y^2+10x-24y-87=0$、$x^2+y^2-10x-24y+153=0$。直线 $y=ax$ 包含某圆圆心，该圆与第一圆内切、第二圆外切。求最小正斜率 a，若 $a^2=p/q$ 最简，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2021-I-Q15] | Part C (Q11-Q15)

For positive integers k, consider $y = x^2 - k$ and $x = 2(y-20)^2 - k$. Among k for which they have four distinct intersections lying on a circle of radius at most 21, add the least and greatest values.

对正整数 k，考虑 $y = x^2 - k$ 与 $x = 2(y-20)^2 - k$。要求两曲线有四个不同交点，且这些点位于半径不超过 21 的同一圆上。求符合条件的最小与最大 k 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2022-II-Q12] | Part C (Q11-Q15)

For real x, y and parameters $a > 4, b > 1$, suppose $x^2/a^2 + y^2/(a^2 - 16) = 1$ and $(x - 20)^2/(b^2 - 1) + (y - 11)^2/b^2 = 1$. Determine the minimum of $a + b$.

实数 x, y 及参数 $a > 4, b > 1$ 满足 $x^2/a^2 + y^2/(a^2 - 16) = 1$，$(x - 20)^2/(b^2 - 1) + (y - 11)^2/b^2 = 1$。求 $a + b$ 的最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-II-Q12] | Part C (Q11-Q15)

Let $O=(0,0), A=(1/2,0), B=(0,\sqrt{3}/2)$. Among all unit segments joining the positive coordinate axes, there is exactly one interior point C of AB lying on no other such segment. Write $OC^2=p/q$ in lowest terms and find $p+q$.

令 $O=(0,0), A=(1/2,0), B=(0,\sqrt{3}/2)$。考虑所有连接两正坐标轴且长为 1 的线段。在 AB 内有唯一一点 C 不属于其他这种线段。若 $OC^2=p/q$ 为最简分数，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-II-Q03] | Part A (Q1-Q5)

A right square pyramid has base edge 6 and volume 54. All five vertices lie on one sphere. If its radius is reduced m/n, compute $m+n$.

正四棱锥的底边长为 6、体积为 54。五个顶点共球，球半径为最简分数 m/n。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q03] | Part A (Q1-Q5)

A radius-200 solid hemisphere rests on its equatorial disk. A radius-42 sphere rests on that disk at a point P and must fit inside the hemisphere. If the area proportion of allowed P is p/q in lowest terms, find $p+q$.

半径 200 的实心上半球以赤道圆盘为底。一半径 42 的球在圆盘点 P 处与底面相切并须完全位于半球内。允许的 P 的面积占圆盘面积之比为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-I-Q06] | Part B (Q6-Q10)

Two holes of radii 1 and 2 are cut in a plane board, their centers 7 apart. Identical spheres rest in the holes on the same side and touch each other. For their common radius r, write $r^2 = m/n$ reduced and find $m+n$.

平面板上两圆孔半径为 $1, 2$，孔心距为 7。两个等半径球在板的同侧分别置于圆孔中，且互相相切。若球半径平方约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-II-Q07] | Part B (Q6-Q10)

Two congruent solid cones have radius 3 and height 8. Their axes meet perpendicularly at an interior point O that is 3 from each base plane. A sphere lies in their intersection. If the greatest possible radius squared is m/n reduced, find $m+n$.

两个全等实心圆锥的底面半径为 3、高为 8，轴线在内部点 O 垂直相交，且 O 距两底面均为 3。球同时包含在两个圆锥内。若最大球半径平方约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2021-II-Q10] | Part B (Q6-Q10)

Three spheres of radii $36, 36, 13$ are pairwise externally tangent and each tangent to two intersecting planes $\mathcal{P}, \mathcal{Q}$. Let their intersection line be ℓ. The radius-13 sphere touches $\mathcal{P}$ at U. If $\operatorname{dist}(U, \ell) = m/n$ reduced, find $m + n$.

半径为 $36, 36, 13$ 的三个球两两外切，且每球均与相交平面 $\mathcal{P}, \mathcal{Q}$ 相切。设交线为 ℓ，小球与 $\mathcal{P}$ 切于 U。若 U 到 ℓ 的距离约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-I-Q10] | Part B (Q6-Q10)

Spheres of radii $11, 13, 19$ are externally tangent in pairs. A plane cuts them in equal-radius circles with centers A, B, C corresponding to those radii. All sphere centers are on one side of the plane. Given $AB^2 = 560$, calculate AC^2.

半径分别为 $11, 13, 19$ 的三个球两两外切，同一平面截出三个等半径圆，其圆心依次为 A, B, C。三个球心位于平面同侧。已知 $AB^2 = 560$，求 AC^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2024-II-Q08] | Part B (Q6-Q10)

A torus is generated by rotating a radius-3 circle about a coplanar axis 6 units from its center. It touches a radius-11 sphere along a circle, once while inside and once while outside the sphere. If the two contact-circle radii satisfy $r_i - r_o = m/n$ in lowest terms, find $m+n$.

将半径为 3 的圆绕其平面内距圆心 6 的轴旋转得到圆环面。该环面分别从半径 11 球的内部、外部沿圆周相切，相切圆半径为 r_i, r_o。若 $r_i - r_o = m/n$ 为最简分数，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2000-II-Q12] | Part C (Q11-Q15)

Points A, B, C lie on a radius-20 sphere centered at O, with $AB = 13, BC = 14, CA = 15$. The distance from O to the filled triangle ABC is $m\sqrt{n}/k$, with n squarefree and $\gcd(m, k) = 1$. Find $m + n + k$.

半径 20、球心 O 的球面上三点满足 $AB = 13, BC = 14, CA = 15$。$O$ 到实心三角形 ABC 的距离为 $m\sqrt{n}/k$，n 无平方因子且 m, k 互质，求 $m + n + k$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2001-I-Q12] | Part C (Q11-Q15)

Find the inradius r of the tetrahedron with vertices $(6,0,0),(0,4,0),(0,0,2),(0,0,0)$. If $r = m/n$ reduced, find $m+n$.

四面体顶点为 $(6,0,0),(0,4,0),(0,0,2),(0,0,0)$，内切球半径 r 约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2011-I-Q13] | Part C (Q11-Q15)

A cube of side 10 is entirely above a plane. Its nearest vertex A has three adjacent vertices at heights $10, 11, 12$. If the height of A is $(r - \sqrt{s})/t$ for positive integers r, s, t, find $r + s + t$.

边长十的立方体悬在平面上方，最近顶点 A 的三个相邻顶点高度分别为 $10, 11, 12$。若 A 的高度为 $(r - \sqrt{s})/t$，r, s, t 为正整数，求 $r + s + t$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2016-II-Q14] | Part C (Q11-Q15)

Equilateral triangle ABC has side600. Points P, Q lie on opposite sides of its plane, each equidistant from A, B, C. The half-planes PAB, QAB form a 120° dihedral angle. A sphere centered at O passes through A, B, C, P, Q with radius d. Find d.

等边三角形 ABC 边长600。点 P, Q 位于其平面两侧，分别到 A, B, C 等距，半平面 PAB, QAB 所成二面角为 120°。以 O 为心的球经过 A, B, C, P, Q，半径为 d。求 d。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q12] | Part C (Q11-Q15)

In a plane, right triangle ABC has $AB = 6, AC = 4, A = 90^\circ$. Reflect its centroid across BC to D. Spheres above the plane, tangent at A, B, C, D, have radii $1, 2, 3, r$. A distinct second plane is tangent to all four, with their centers on one side. If $r = m/n$ in lowest terms, find $m + n$.

平面内直角三角形 ABC 满足 $AB = 6, AC = 4, \angle A = 90^\circ$。重心关于 BC 对称得 D。四球位于该平面同侧并分别在 A, B, C, D 相切，半径为 $1, 2, 3, r$，且均在另一不同公切平面的同侧。将 r 化为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-II-Q12] | Part C (Q11-Q15)

A tetrahedron has two faces with side lengths $5\sqrt{10}, 5\sqrt{10}, 10$ and two with $5\sqrt{10}, 5\sqrt{10}, 18$. Its circumsphere and insphere centers are S, R. Write $RS = m/n$ in lowest terms and find $m + n$.

四面体两面边长为 $5\sqrt{10}, 5\sqrt{10}, 10$，另两面为 $5\sqrt{10}, 5\sqrt{10}, 18$。外接球、内切球圆心为 S, R。将 RS 化为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-II-Q02] | Part A (Q1-Q5)

A cube has vertices $(7, 12, 10), (8, 8, 1), (11, 3, 9)$ among its eight vertices. Find its surface area.

立方体八顶点中包含 $(7, 12, 10), (8, 8, 1), (11, 3, 9)$，求表面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q05] | Part A (Q1-Q5)

Take all points inside or at Euclidean distance at most 1 from a $3 \times 4 \times 5$ solid box. Its volume is $(m + n\pi)/p$ with positive integers and $\gcd(n, p) = 1$. Find $m + n + p$.

取 $3 \times 4 \times 5$ 实心长方体内部及到它欧氏距离不超过 1 的全部点。体积为 $(m + n\pi)/p$，参数为正整数且 n, p 互质，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q04] | Part A (Q1-Q5)

The four face centers of a regular tetrahedron form another tetrahedron. If their volume ratio is m/n reduced, find $m+n$.

正四面体四个面心组成另一个四面体。若小大体积比约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-II-Q05] | Part A (Q1-Q5)

A radius-6 circular cylinder is cut by one plane perpendicular to its axis and another making 45° with that plane. The planes' intersection line touches the cylinder at just one point. The wedge between the cuts has volume $n\pi$. Find n.

半径六的圆柱被一垂直轴线的平面及与它成 45° 的另一平面切割，两平面交线与圆柱恰有一个公共点。两切面间楔形体积为 $n\pi$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q05] | Part A (Q1-Q5)

A cone of base radius r and height h rolls on its side without slipping around its fixed vertex. Its first return to the initial position requires seventeen complete spins about its own axis. Express $h/r = m\sqrt{n}$ with squarefree n and find $m+n$.

底半径 r、高 h 的圆锥侧卧无滑动滚动，顶点固定，首次回原位共绕自身轴转十七整周。若 $h/r = m\sqrt{n}$ 且 n 无平方素因子，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2012-II-Q05] | Part A (Q1-Q5)

A side-15 square is concentric and parallel-sided inside a side-40 square. Join each outer side midpoint to the two nearest inner vertices. Cut out the inner square and four attached triangles, and fold along the inner sides into a square pyramid. Find its volume.

边长十五的正方形与边长四十的外正方形同心且各边平行。将外正方形各边中点连至内正方形最近的两个顶点，剪下内部正方形及四个三角形并折成正四棱锥，求体积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2016-I-Q04] | Part A (Q1-Q5)

A right prism has regular hexagonal bases of edge length 12 and height h. Vertex A and its three neighboring prism vertices form a tetrahedron. The dihedral angle between its base-plane face and its face opposite A is 60°. Find h^2.

直六棱柱的底面为边长 12 的正六边形，高为 h。一个顶点 A 及与它相邻的三个棱柱顶点组成四面体。位于底面内的面与 A 的对面所成二面角为 60°。求 h^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2017-I-Q04] | Part A (Q1-Q5)

A tetrahedron has an isosceles base of side lengths $20, 20, 24$; each edge joining its apex to a base vertex has length25. Its volume is $m\sqrt{n}$ with squarefree positive integer n. Find $m + n$.

四面体的等腰底面三边为 $20, 20, 24$，顶点到底面三个顶点的棱均长25。体积为 $m\sqrt{n}$，n 为无平方因子的正整数。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2000-I-Q08] | Part B (Q6-Q10)

A sealed right circular cone has height 12 and base radius 5. Liquid is 9 deep with the apex down. After turning the apex upward, its depth is $m - n\sqrt[3]{p}$, with positive integers m, n, p and cube-free p. Find $m + n + p$.

密封直圆锥高 12、底面半径 5。尖端向下时液深为 9。倒置为尖端向上后，液深为 $m - n\sqrt[3]{p}$，其中 p 无立方因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2003-I-Q06] | Part B (Q6-Q10)

Add the areas of all triangles formed by three distinct vertices of a unit cube. If the result is $m + \sqrt{n} + \sqrt{p}$ for integers m, n, p, find $m + n + p$.

将单位立方体任意三个不同顶点构成的三角形面积全部相加。若结果写为 $m + \sqrt{n} + \sqrt{p}$，其中参数为整数，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2005-II-Q10] | Part B (Q6-Q10)

The face centers of a regular octahedron are the vertices of a cube. If octahedron volume divided by cube volume is reduced m/n, find $m+n$.

正八面体的八个面心构成一个立方体的顶点。八面体与该立方体体积比为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2012-I-Q08] | Part B (Q6-Q10)

Label a unit cube by $A=(0,0,0), B=(1,0,0), C=(1,1,0), D=(0,1,0)$ and E, F, G, H directly above them. A plane through D and the midpoints of AB, CG divides the cube. If the larger volume is reduced p/q, find $p+q$.

单位立方体的底面顶点为 $A=(0,0,0), B=(1,0,0), C=(1,1,0), D=(0,1,0)$，$E, F, G, H$ 分别在其正上方。过 D 及 AB, CG 中点的平面将其分成两部分，若较大体积为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-II-Q09] | Part B (Q6-Q10)

A cylindrical barrel of radius 4 ft and height 10 ft is full of water. An 8-ft-edge solid cube is lowered centrally into it, with one body diagonal vertical, until the cube rests on the rim. If the displaced water volume is v, find v^2.

半径 4 英尺、高 10 英尺的圆柱桶装满水。边长 8 英尺的实心立方体沿桶轴下放，一条体对角线保持竖直，直至卡在桶口。若排水体积为 v，求 v^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2021-I-Q06] | Part B (Q6-Q10)

From one cube vertex A, the adjacent vertices are B, C, D, and the opposite vertex is G. A point P has distances $PB = 60\sqrt{10}, PC = 60\sqrt{5}, PD = 120\sqrt{2}, PG = 36\sqrt{7}$. Determine PA.

立方体中，与顶点 A 相邻的三点为 B, C, D，对顶点为 G。一点 P 满足 $PB = 60\sqrt{10}, PC = 60\sqrt{5}, PD = 120\sqrt{2}, PG = 36\sqrt{7}$，求 PA。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q12] | Part C (Q11-Q15)

Start with a volume-1 regular tetrahedron $\mathcal{P}_0$. To form each next solid, attach an outward regular tetrahedron on the midpoint triangle of every triangular face, replacing that midpoint patch. If $\mathrm{vol}(\mathcal{P}_3) = m/n$ reduced, find $m+n$.

从体积为 1 的正四面体 $\mathcal{P}_0$ 出发。每次在每个三角面中点三角形上向外附加正四面体，替换原中部面片，得到下一个立体。若 $\mathcal{P}_3$ 体积约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2001-II-Q15] | Part C (Q11-Q15)

A cube of side 8 has opposite vertices A, E. On the three edges from E, take I, J, K at distance 2 from E. Drill the triangular tunnel bounded by the planes through IJ, JK, KI parallel to AE. The remaining solid has total surface area, including tunnel walls, $m + n\sqrt{p}$ with squarefree p. Find $m + n + p$.

棱长为 8 的立方体有对顶点 A, E。在 E 出发的三条棱上取 I, J, K，均距 E 为 2。以过 IJ, JK, KI 且平行体对角线 AE 的三个平面为壁钻出三角隧道。剩余立体包括隧道壁的总表面积为 $m + n\sqrt{p}$，p 无平方因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2002-I-Q15] | Part C (Q11-Q15)

A six-faced polyhedron has faces $ABCD, ABFG, CDE, ADEG, BCEF, EFG$. The first is a square of side 12; $ABFG$ is an isosceles trapezoid with $AB \parallel GF$, $AG = BF = 8$, $GF = 6$; and $CE = DE = 14$. The distance from E to square face $ABCD$ is 12. Given $EG^2 = p - q\sqrt{r}$ with positive integers and squarefree r, find $p + q + r$.

六面多面体的面为 $ABCD, ABFG, CDE, ADEG, BCEF, EFG$。其中 $ABCD$ 为边长 12 的正方形；$ABFG$ 为等腰梯形，$AB \parallel GF, AG = BF = 8, GF = 6$；$CE = DE = 14$。$E$ 到正方形面 $ABCD$ 的距离为 12。若 $EG^2 = p - q\sqrt{r}$，各参数为正整数且 r 无平方因子，求其和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q11] | Part C (Q11-Q15)

A height-4, radius-3 cone is painted over its lateral surface and base, then cut parallel to the base into a small cone C and a frustum F. The ratio of their originally painted areas equals their volume ratio, both $k = m/n$ reduced. Find $m + n$.

高四、底半径三的圆锥原侧面与底面全部涂漆，再平行底面切成小圆锥 C 和圆台 F。两者原涂漆面积比等于体积比，均为最简 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-I-Q14] | Part C (Q11-Q15)

A taut rope of length 20 runs from a ground-level point on a radius-8 vertical cylinder, around its wall, then to an endpoint at height 4 and horizontal distance 4 from the cylinder surface. Its wall-contact length is $(a-\sqrt{b})/c$, where a, b, c are positive integers and c is prime. Find $a+b+c$.

长二十的绷紧绳从半径八的竖直圆柱底部地面点出发，绕柱壁后伸向高四、水平距柱面四的端点。贴柱壁段长为 $(a-\sqrt{b})/c$，参数为正整数且 c 为质数，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2004-II-Q11] | Part C (Q11-Q15)

A cone has base radius 600 and height $200\sqrt{7}$. Two surface points lie on opposite generators and are at distances 125 and $375\sqrt{2}$ from the apex. Find their shortest surface-path distance.

圆锥底半径六百、高 $200\sqrt{7}$。两侧面点位于正相对母线上，距顶点分别为 $125, 375\sqrt{2}$，求沿侧面的最短路长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2007-I-Q13] | Part C (Q11-Q15)

A square pyramid has all eight edges length four. A plane passes through the midpoints of one lateral edge AE and base edges BC, CD. If its cross-sectional area is $\sqrt{p}$, find p.

正方锥的八条棱都长四。过侧棱 AE、底边 BC, CD 的中点作平面，截面面积为 $\sqrt{p}$，求 p。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2008-I-Q15] | Part C (Q11-Q15)

From each corner of a side-100 square, mark points $\sqrt{17}$ along its two edges and cut segments from those points to a common point on the inward diagonal, with angle 60° between the cuts. Fold along lines joining neighboring inner cut vertices and tape each pair of cut edges together to form a tray. Its height is $\sqrt[n]{m}$ with $m < 1000$ free of nth-power prime factors. Find $m + n$.

边长一百的正方纸每角沿两边各取距角 $\sqrt{17}$ 的点，向内对角线上同一点切两刀，两切线夹角六十度。沿相邻内切点连线折起，并粘合各角切口，形成托盘。高度为 $\sqrt[n]{m}$，$m < 1000$ 且无素数的 n 次方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2010-I-Q11] | Part C (Q11-Q15)

Revolve the region $|8-x|+y \le 10, 3y-x \ge 15$ about the line $3y-x=15$. Its volume is $m\pi/(n\sqrt{p})$, with m, n coprime and p squarefree. Find $m+n+p$.

将满足 $|8-x|+y \le 10, 3y-x \ge 15$ 的平面区域绕直线 $3y-x=15$ 旋转。体积为 $m\pi/(n\sqrt{p})$，m, n 互质，p 无平方因子。求 $m+n+p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2015-I-Q15] | Part C (Q11-Q15)

A solid right circular cylinder has radius 6 and height 8. Two points A, B on the rim of one base cut off a minor arc of 120°. A plane through A, B and the cylinder center produces a flat cross-section of area $a\pi + b\sqrt{c}$, with integral a, b, c and squarefree c. Find $a + b + c$.

实心直圆柱半径为 6、高为 8。同一底面边缘上两点 A, B 所夹小弧为 120°。过 A, B 和圆柱中心的平面截得平面图形，其面积为 $a\pi + b\sqrt{c}$，其中 a, b, c 为整数，c 无平方因子。求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2023-I-Q13] | Part C (Q11-Q15)

Two noncongruent parallelepipeds have all six faces rhombi with diagonal lengths $\sqrt{21}, \sqrt{31}$. If their larger-to-smaller volume ratio is the reduced fraction m/n, find $m+n$.

两个不全等平行六面体的每个面都是对角线长为 $\sqrt{21}, \sqrt{31}$ 的菱形。较大体积与较小体积之比为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2023-II-Q14] | Part C (Q11-Q15)

A cube has parallel edges AB, CD and face diagonals AC, BD. The rectangle $ABDC$ lies in a vertical plane. The heights of A, B, C, D above a horizontal floor are $0, 2, 8, 10$ meters. Water fills the cube up to height 7 meters. If its volume is m/n cubic meters in lowest terms, find $m + n$.

立方体的 AB, CD 为平行棱，AC, BD 为面对角线，矩形 $ABDC$ 所在平面竖直。A, B, C, D 离水平地面的高度分别为 $0, 2, 8, 10$ 米。水面高度为 7 米。若水量为最简分数 m/n 立方米，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 2.10 Volumes & solid geometry

体积与立体几何

[AIME-2024-I-Q14] | Part C (Q11-Q15)

Tetrahedron $ABCD$ has opposite-edge pairs $AB = CD = \sqrt{41}$, $AC = BD = \sqrt{80}$, $AD = BC = \sqrt{89}$. Its interior point equidistant from all faces has distance $m\sqrt{n}/p$, with coprime m, p and squarefree n. Find $m + n + p$.

四面体 $ABCD$ 的对棱分别满足 $AB = CD = \sqrt{41}, AC = BD = \sqrt{80}, AD = BC = \sqrt{89}$。内部到四面等距的点，其共同距离为 $m\sqrt{n}/p$，其中 m, p 互质且 n 无平方素因子。求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2000-I-Q01] | Part A (Q1-Q5)

Find the first positive exponent n for which every factorization $10^n = ab$ into positive integers has a zero digit in at least one factor.

求最小正整数 n，使任意正整数分解 $10^n = ab$ 中，至少一个因数的十进制表示含数字 0。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2002-II-Q05] | Part A (Q1-Q5)

Add all positive integers $a = 2^n 3^m$, with nonnegative integers n, m, for which $a^6 \nmid 6^a$.

求所有正整数 $a = 2^n 3^m$ 之和，其中 n, m 为非负整数，且 $a^6 \nmid 6^a$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q04] | Part A (Q1-Q5)

Let N be the trailing-zero count of $\prod_{k=1}^{100} k!$. Find $N \bmod 1000$.

令 N 为 $\prod_{k=1}^{100} k!$ 的十进制末尾零数，求其除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-II-Q03] | Part A (Q1-Q5)

Find the exponent of three in $P = 1 \cdot 3 \cdot 5 \cdots 199$.

求奇数乘积 $P = 1 \cdot 3 \cdot 5 \cdots 199$ 中素因子三的指数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2007-I-Q01] | Part A (Q1-Q5)

Count positive perfect squares below 10^6 divisible by 24.

求小于 10^6 且能被 24 整除的正平方数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2010-II-Q03] | Part A (Q1-Q5)

Let $K=\prod_{1\le a<b\le 20}(b-a)$. Find the exponent of 2 in its prime factorization.

令 $K=\prod_{1\le a<b\le 20}(b-a)$，求其素因数分解中 2 的指数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2015-I-Q03] | Part A (Q1-Q5)

Find the prime p for which $16p + 1$ is a positive integer cube.

求使 $16p + 1$ 为正整数立方的质数 p。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2019-II-Q04] | Part A (Q1-Q5)

Roll a fair six-sided die four times. If the chance that the product is square is reduced m/n, find $m+n$.

公平六面骰掷四次，乘积为平方数的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2020-II-Q01] | Part A (Q1-Q5)

How many ordered pairs of positive integers (m, n) satisfy $m^2 n = 20^{20}$?

正整数有序对 (m, n) 满足 $m^2 n = 20^{20}$，共有多少组?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2021-I-Q03] | Part A (Q1-Q5)

How many positive integers below 1000 are a difference of two integer powers of 2?

小于 1000 的正整数中，有多少个可写为两个 2 的整数次幂之差?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2023-I-Q04] | Part A (Q1-Q5)

Sum all positive integers m for which $13!/m$ is an integer square. If this sum is $2^a3^b5^c7^d11^e13^f$, find the total of its six exponents.

将所有满足 $13!/m$ 为完全平方整数的正整数 m 相加。若和为 $2^a3^b5^c7^d11^e13^f$，求六个指数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q09] | Part B (Q6-Q10)

A geometric sequence has positive integer first term a and ratio r. If the sum of base-eight logarithms of its first twelve terms is 2006, count possible ordered pairs (a, r).

等比数列首项 a、公比 r 均为正整数，前十二项的以八为底的对数之和为 2006。求 (a, r) 个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2009-II-Q07] | Part B (Q6-Q10)

Reduce $\sum_{i=1}^{2009}(2i-1)!!/(2i)!!$ to lowest terms. If its denominator is $2^a b$ with odd b, find $ab/10$.

将 $\sum_{i=1}^{2009}(2i-1)!!/(2i)!!$ 约成最简分数，若分母为 $2^a b$ 且 b 奇，求 $ab/10$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2016-II-Q08] | Part B (Q6-Q10)

Count unordered sets of three distinct positive integers whose product is $11 \cdot 21 \cdot 31 \cdot 41 \cdot 51 \cdot 61$.

求由三个不同正整数组成、乘积为 $11 \cdot 21 \cdot 31 \cdot 41 \cdot 51 \cdot 61$ 的无序集合数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2019-I-Q07] | Part B (Q6-Q10)

Positive integers x, y satisfy $\log_{10} x + 2\log_{10}\gcd(x,y) = 60$ and $\log_{10} y + 2\log_{10}\operatorname{lcm}(x,y) = 570$. If m, n count prime factors of x, y with multiplicity, find $3m + 2n$.

正整数 x, y 满足 $\log_{10} x + 2\log_{10}\gcd(x,y) = 60$，$\log_{10} y + 2\log_{10}\operatorname{lcm}(x,y) = 570$。若按重数计素因子数为 m, n，求 $3m + 2n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2020-I-Q10] | Part B (Q6-Q10)

Find the smallest $m+n$ for positive integers m, n such that $\gcd(m+n, 210) = 1$, $n^n \mid m^m$, and $n \nmid m$.

正整数 m, n 满足 $\gcd(m+n, 210) = 1$、$n^n \mid m^m$，但 $n \nmid m$。求 $m+n$ 的最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2000-I-Q11] | Part C (Q11-Q15)

Let S sum all reduced fractions a/b whose positive numerator and denominator both divide 1000. Find $\lfloor S/10 \rfloor$.

将所有分子分母均为 1000 正约数的最简正分数 a/b 相加，得 S。求 $\lfloor S/10 \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-I-Q13] | Part C (Q11-Q15)

For even $x > 0$, let $g(x)$ be its largest dividing power of two, and define $S_n = \sum_{k=1}^{2^{n-1}} g(2k)$. Find the greatest $n < 1000$ for which S_n is square.

对正偶数 x，令 $g(x)$ 为整除它的最大二次幂，$S_n = \sum_{k=1}^{2^{n-1}} g(2k)$。求使 S_n 为平方数的最大 $n < 1000$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2006-II-Q14] | Part C (Q11-Q15)

Let S_n sum the reciprocals of every nonzero decimal digit occurring in the integers from one through 10^n, inclusive. Find the least positive n making S_n integral.

对一至 10^n（含端点）所有整数中出现的每个非零十进制数字取倒数并相加，记为 S_n。求使其为整数的最小正整数 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.1 Prime factorization & valuations

质因数分解与赋值

[AIME-2016-I-Q12] | Part C (Q11-Q15)

Find the least positive integer m such that $m^2 - m + 11$ has at least four prime factors, counted with multiplicity.

求使 $m^2 - m + 11$ 至少有四个质因数（按重数计）的最小正整数 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2003-I-Q01] | Part A (Q1-Q5)

Express $((3!)!)!/3! = k\,n!$ with positive integers k, n and maximal n. Find $k+n$.

将 $((3!)!)!/3!$ 写成 $k\,n!$，k, n 为正整数且 n 尽量大，求 $k+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2007-II-Q02] | Part A (Q1-Q5)

Count positive integer triples (a, b, c) with $a \mid b$, $a \mid c$ and $a + b + c = 100$.

求满足 $a \mid b, a \mid c, a + b + c = 100$ 的正整数有序三元组数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2015-II-Q01] | Part A (Q1-Q5)

Let N be the smallest positive integer that equals 78% of an integer and also 116% of an integer. Find $N \bmod 1000$.

最小正整数 N 同时是某整数的 78% 和另一整数的 116%。求 N 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2017-I-Q02] | Part A (Q1-Q5)

The integers $702, 787, 855$ all leave the same positive remainder r modulo a positive integer m. The integers $412, 722, 815$ all leave the same positive remainder $s \neq r$ modulo a positive integer n. Find $m + n + r + s$.

$702, 787, 855$ 除以正整数 m 后均余正整数 r；$412, 722, 815$ 除以正整数 n 后均余正整数 $s \neq r$。求 $m + n + r + s$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2019-II-Q03] | Part A (Q1-Q5)

Count positive integer seven-tuples satisfying $abc = 70, cde = 71, efg = 72$.

求满足 $abc = 70, cde = 71, efg = 72$ 的正整数七元组数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2020-I-Q04] | Part A (Q1-Q5)

Consider positive integers whose decimal expansion ends in 2020. Require that deleting these four digits leaves a positive divisor of the original integer. Sum the decimal digits of every integer satisfying this requirement.

考虑十进制末四位为 2020 的正整数，删去末四位后所得正整数须整除原数。求所有符合条件的数的数位和之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2022-II-Q01] | Part A (Q1-Q5)

Adults initially constitute $5/12$ of a concert audience. A bus adds 50 people, after which the adult proportion is $11/25$. What is the smallest possible final number of adults?

音乐会观众中成人原占 $5/12$。一辆巴士又带来 50 人，成人比例变为 $11/25$。求最后成人数的最小可能值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2025-II-Q02] | Part A (Q1-Q5)

Sum the positive integers n satisfying $(n+2) \mid 3(n+3)(n^2+9)$.

求所有满足 $(n+2) \mid 3(n+3)(n^2+9)$ 的正整数 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2021-I-Q10] | Part B (Q6-Q10)

Start with $a_1 = 2020/2021$. Whenever $a_k = m/n$ is in lowest positive terms, define $a_{k+1} = (m+18)/(n+19)$. Add all positive indices j for which $a_j = t/(t+1)$ for a positive integer t.

从 $a_1 = 2020/2021$ 开始。若 $a_k = m/n$ 已约分，定义 $a_{k+1} = (m+18)/(n+19)$。求所有满足 $a_j = t/(t+1)$（t 为正整数）的下标 j 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

[AIME-2025-II-Q07] | Part B (Q6-Q10)

Choose a uniformly random subset of the positive divisors of 2025. If the probability that it is nonempty and has least common multiple 2025 is m/n in lowest terms, find $m+n$.

在 2025 的所有正约数组成的集合中等可能选一个子集。该子集非空且最小公倍数为 2025 的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2007-I-Q04] | Part A (Q1-Q5)

Three coplanar circular orbits about one star have periods $60, 84, 140$ years, with uniform motion in the same direction. Starting with the star and all planets collinear, how many years until their next collinearity?

三颗行星在同一平面绕恒星做同向匀速圆周运动，周期为六十、八十四、一百四十年。现四者共线，求下次共线的间隔年数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2010-I-Q02] | Part A (Q1-Q5)

Find $\prod_{j=1}^{999}(10^j - 1)$ modulo 1000.

求 $\prod_{j=1}^{999}(10^j - 1)$ 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2025-I-Q05] | Part A (Q1-Q5)

Let N count the eight-digit permutations of $1, 2, \ldots, 8$ divisible by 22. Compute $N - 2025$.

将 $1, 2, \ldots, 8$ 各使用一次排成八位数，其中能被 22 整除的数有 N 个，求 $N - 2025$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-I-Q09] | Part B (Q6-Q10)

Three painters cover respectively pickets $1, 1+h, 1+2h, \ldots$; $2, 2+t, 2+2t, \ldots$; and $3, 3+u, 3+2u, \ldots$. For positive integers h, t, u, require every positive-numbered picket painted exactly once. Add all resulting integers $100h + 10t + u$.

三人分别刷编号为 $1, 1+h, 1+2h, \ldots$；$2, 2+t, 2+2t, \ldots$；$3, 3+u, 3+2u, \ldots$ 的栅栏。正整数 h, t, u 须使每根正编号栅栏恰被刷一次。求所有对应整数 $100h + 10t + u$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-II-Q07] | Part B (Q6-Q10)

Find the least positive k such that $1^2 + 2^2 + \cdots + k^2$ is divisible by 200.

求最小正整数 k，使 $1^2 + 2^2 + \cdots + k^2$ 被 200 整除。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2011-I-Q07] | Part B (Q6-Q10)

How many positive integers m admit nonnegative integers $x_0, x_1, \ldots, x_{2011}$ with $m^{x_0} = \sum_{k=1}^{2011} m^{x_k}$?

求正整数 m 的个数，使存在非负整数 $x_0, x_1, \ldots, x_{2011}$ 满足 $m^{x_0} = \sum_{k=1}^{2011} m^{x_k}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-I-Q10] | Part B (Q6-Q10)

Take all nonnegative perfect squares ending in decimal digits 256, remove their last three digits, and order the resulting integers increasingly. Find the tenth number modulo 1000.

将所有十进制末三位为 256 的非负完全平方数删去末三位，再将所得整数递增排列。求第十个数模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2014-I-Q08] | Part B (Q6-Q10)

A positive integer N and its square have the same last four decimal digits $abcd$, with $a \neq 0$. Find the three-digit integer $\overline{abc}$.

正整数 N 与其平方末四位相同，均为 $abcd$，且 a 非零，求三位数 $\overline{abc}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2017-I-Q09] | Part B (Q6-Q10)

Set $a_{10} = 10$ and $a_n = 100a_{n-1} + n$ for integers $n > 10$. Find the smallest $n > 10$ such that $99 \mid a_n$.

令 $a_{10} = 10$，整数 $n > 10$ 时 $a_n = 100a_{n-1} + n$。求使 $99 \mid a_n$ 的最小 $n > 10$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2017-II-Q08] | Part B (Q6-Q10)

Count positive integers $n < 2017$ for which $\sum_{j=0}^{6} n^j/j!$ is an integer.

求使 $\sum_{j=0}^{6} n^j/j!$ 为整数的正整数 $n < 2017$ 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2020-II-Q10] | Part B (Q6-Q10)

Add all positive integers n for which the remainder of $\sum_{j=1}^{n} j^3$ upon division by $n+5$ is 17.

求所有正整数 n 之和，使 $1^3+\cdots+n^3$ 除以 $n+5$ 的余数为 17。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2022-II-Q08] | Part B (Q6-Q10)

Count positive integers $n \leq 600$ that are uniquely identified, among all positive integers, by the triple $(\lfloor n/4 \rfloor, \lfloor n/5 \rfloor, \lfloor n/6 \rfloor)$.

有多少个不超过 600 的正整数 n，其三元组 $(\lfloor n/4 \rfloor, \lfloor n/5 \rfloor, \lfloor n/6 \rfloor)$ 能在所有正整数中唯一确定它?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2023-I-Q07] | Part B (Q6-Q10)

Count the positive integers below 1000 whose five remainders on division by $2, 3, 4, 5, 6$ are pairwise different.

求小于 1000 的正整数中，除以 $2, 3, 4, 5, 6$ 所得五个余数两两不同的数的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2024-II-Q07] | Part B (Q6-Q10)

Find the greatest four-digit integer N such that replacing any single digit by 1 always produces a multiple of 7. If $N = 1000Q + R$ with $0 \le R < 1000$, determine $Q + R$.

求最大的四位数 N，使将其中任意一位改为 1 后所得数均为 7 的倍数。设 $N = 1000Q + R, 0 \le R < 1000$，求 $Q + R$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2026-I-Q08] | Part B (Q6-Q10)

Let N be the number of divisors of 17017^{17} congruent to 5 modulo 12. Find $N \bmod 1000$.

设 N 为 17017^{17} 的正约数中模 12 余 5 的个数，求 N 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2002-I-Q14] | Part C (Q11-Q15)

A set of distinct positive integers contains 1, has maximum 2002, and has an integer mean after any one element is deleted. What is its largest possible size?

互异正整数集合含 1，最大元为 2002，删去任意一个元素后其余数平均值均为整数。求集合最大可能元素数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2011-II-Q14] | Part C (Q11-Q15)

Count permutations $(a_1, \ldots, a_{30})$ of $1, \ldots, 30$ for which $a_{n+m} \equiv a_n \pmod{m}$ whenever both indices are in range, for each $m = 2, 3, 5$. Report the count modulo 1000.

排列 $(a_1, \ldots, a_{30})$ 使用一至三十各一次。对 $m = 2, 3, 5$ 及所有有效下标，要求 $a_{n+m} \equiv a_n \pmod{m}$。求排列数模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-I-Q15] | Part C (Q11-Q15)

For $1 < n < 1000$, seats around a circle are numbered modulo n. A rearrangement sends person k to seat ak for some positive integer a, and for every pair, neither new clockwise or counterclockwise gap equals either original gap. Count the possible n.

$1 < n < 1000$ 个圆桌座位按模 n 编号。某次换座使原第 k 人坐到 ak，a 为正整数；对任意两人，新顺逆时针两种间隔人数均不同于原来的任一种间隔人数。求可行的 n 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2012-II-Q12] | Part C (Q11-Q15)

A positive integer is called p-safe when its distance from every integer multiple of p exceeds two. Count integers at most 10000 that are simultaneously 7-, 11-, and 13-safe.

若正整数与 p 的每个整数倍的距离都大于二，称为 p 安全数。求不超过一万且同时对七、十一、十三安全的整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-I-Q11] | Part C (Q11-Q15)

Let N be divisible by $14, 15, 16$. For three integers $0 < x < y < z < 14$, division of N by each leaves remainder three. Find the sum of distinct prime factors of the least such positive N.

正整数 N 被十四、十五、十六整除，且存在 $0 < x < y < z < 14$，分别用三数除 N 均余三。求最小可行 N 的不同素因子之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-I-Q15] | Part C (Q11-Q15)

Count increasing arithmetic progressions $0 \le A < B < C \le 99$ for which some prime p has residues $A \equiv a, B \equiv b, C \equiv c \pmod{p}$ with $0 \le b < a < c < p$ and (b, a, c) also an arithmetic progression.

求递增等差数列 $0 \le A < B < C \le 99$ 的个数，使存在素数 p 及余数 $A \equiv a, B \equiv b, C \equiv c \pmod{p}$，满足 $0 \le b < a < c < p$，且 (b, a, c) 也成等差数列。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2013-II-Q14] | Part C (Q11-Q15)

For $n > 1$, let $F(n)$ be the largest remainder of n upon division by an integer $1 \leq k \leq n/2$. Find $\sum_{n=20}^{100} F(n)$ modulo 1000.

$n > 1$ 时，令 $F(n)$ 为用整数 $1 \leq k \leq n/2$ 除 n 所得余数的最大值。求 $n = 20$ 至一百的 $F(n)$ 之和模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2019-II-Q14] | Part C (Q11-Q15)

Unlimited stamps of denominations $5, n, n+1$ have largest nonrepresentable positive amount 91. Sum all positive integers n with this property.

有无限枚面值 $5, n, n+1$ 的邮票，不能凑出的最大正金额为九十一。求所有正整数 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

[AIME-2021-II-Q13] | Part C (Q11-Q15)

Find the least positive n with $2^n + 5^n \equiv n \pmod{1000}$.

求满足 $2^n + 5^n \equiv n \pmod{1000}$ 的最小正整数 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2017-I-Q03] | Part A (Q1-Q5)

Let d_n be the last decimal digit of $n(n+1)/2$. Compute $\sum_{n=1}^{2017} d_n$ modulo1000.

令 d_n 为 $n(n+1)/2$ 的十进制个位数字，求 $\sum_{n=1}^{2017} d_n$ 模1000的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2018-II-Q02] | Part A (Q1-Q5)

Let $a_0 = 2, a_1 = 5, a_2 = 8$. For $n > 2$, let a_n be the least nonnegative residue of $4(a_{n-1} + a_{n-2} + a_{n-3})$ modulo eleven. Find $a_{2018}a_{2020}a_{2022}$.

令 $a_0 = 2, a_1 = 5, a_2 = 8$。$n > 2$ 时，a_n 为 $4(a_{n-1} + a_{n-2} + a_{n-3})$ 模十一的最小非负余数。求 $a_{2018}a_{2020}a_{2022}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2001-II-Q10] | Part B (Q6-Q10)

How many distinct positive multiples of 1001 have form $10^j - 10^i$ for integers $0 \leq i < j \leq 99$?

有多少个不同的 1001 的正倍数可写为 $10^j - 10^i$，其中 $0 \leq i < j \leq 99$ 为整数?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2021-II-Q09] | Part B (Q6-Q10)

Count ordered pairs $1 \le m, n \le 30$ with $\gcd(2^m + 1, 2^n - 1) > 1$.

求 $1 \le m, n \le 30$ 中满足 $\gcd(2^m + 1, 2^n - 1) > 1$ 的有序对数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2011-I-Q11] | Part C (Q11-Q15)

Let R be the set of distinct residues of 2^n modulo 1000 for all integers $n \geq 0$. Find $\sum_{r \in R} r \pmod{1000}$.

令 R 为所有非负整数 n 的 2^n 模 1000 的不同余数组成的集合，求其元素和模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2018-I-Q11] | Part C (Q11-Q15)

Find the least positive n for which the last two base-143 digits of 3^n are 01.

求最小正整数 n，使 3^n 的一百四十三进制末两位为 01。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2019-I-Q14] | Part C (Q11-Q15)

Find the smallest odd prime divisor of $2019^8 + 1$.

求 $2019^8 + 1$ 的最小奇素因子。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2020-I-Q12] | Part C (Q11-Q15)

Let n be the least positive exponent with $3^3 5^5 7^7 \mid 149^n - 2^n$. How many positive divisors does n have?

令 n 为满足 $3^3 5^5 7^7 \mid 149^n - 2^n$ 的最小正整数。求 n 的正约数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2023-II-Q15] | Part C (Q11-Q15)

For each $n \geq 1$, let a_n be the smallest positive multiple of 23 congruent to 1 modulo 2^n. Count $n \leq 1000$ for which $a_n = a_{n+1}$.

对每个正整数 n，令 a_n 为满足 $a_n \equiv 1 \pmod{2^n}$ 的最小正 23 的倍数。求 $n \leq 1000$ 中使 $a_n = a_{n+1}$ 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2024-I-Q13] | Part C (Q11-Q15)

Let p be the smallest prime for which p^2 divides n^4+1 for some integer n. Determine the smallest positive m with $p^2 \mid m^4+1$.

设 p 为使某个整数 n 满足 $p^2 \mid n^4+1$ 的最小素数。求满足 $p^2 \mid m^4+1$ 的最小正整数 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

[AIME-2025-I-Q15] | Part C (Q11-Q15)

Count ordered triples $1 \le a, b, c \le 3^6$ such that $3^7 \mid a^3 + b^3 + c^3$, and give the count modulo 1000.

求满足 $1 \le a, b, c \le 3^6$ 及 $3^7 \mid a^3 + b^3 + c^3$ 的有序三元组数，并给出除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-II-Q02] | Part A (Q1-Q5)

Count all integer-coordinate points (x, y) satisfying $x^2 - y^2 = 2000^2$.

求满足 $x^2 - y^2 = 2000^2$ 的整数点 (x, y) 个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-II-Q03] | Part A (Q1-Q5)

Increasing positive integers a, b, c form a geometric sequence, satisfy $\log_6 a + \log_6 b + \log_6 c = 6$, and have $b - a$ a perfect square. Find $a + b + c$.

递增正整数 a, b, c 构成等比数列，满足 $\log_6 a + \log_6 b + \log_6 c = 6$，且 $b - a$ 为完全平方数。求三数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q01] | Part A (Q1-Q5)

Three positive integers have product N equal to six times their sum, and one integer equals the sum of the other two. Add all possible distinct values of N.

三个正整数乘积为 N，且等于三数之和的六倍，其中一数又等于另两数之和。求 N 所有不同可能值之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q03] | Part A (Q1-Q5)

A rectangular block made of unit cubes is viewed with three adjacent faces visible. Exactly 231 unit cubes touch none of these faces. Find the least possible total number of cubes.

单位立方体拼成的长方体可见三个相邻面，恰有二百三十一个单元不接触任何可见面。求单元总数最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-I-Q04] | Part A (Q1-Q5)

A band has five members left over after making a square formation. It can instead form a complete rectangle with seven more rows than columns. Find the largest possible band size.

乐队排成正方阵后多出五人，但可排成行数比列数多七的完整长方阵。求最多人数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q03] | Part A (Q1-Q5)

Positive integer cycling, jogging and swimming rates b, j, s satisfy $2b + 3j + 4s = 74$ and $4b + 2j + 3s = 91$. Find $b^2 + j^2 + s^2$.

骑车、慢跑、游泳的正整数速度 b, j, s 满足 $2b + 3j + 4s = 74, 4b + 2j + 3s = 91$，求三者平方和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-I-Q04] | Part A (Q1-Q5)

Positive integers x, y satisfy $x^2 + 84x + 2008 = y^2$. Find $x + y$.

正整数 x, y 满足 $x^2 + 84x + 2008 = y^2$，求 $x + y$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q04] | Part A (Q1-Q5)

In an eating contest the first-place child eats n grapes, and place k eats $n+2-2k$, a nonnegative count. The total eaten is 2009. Minimize n.

吃葡萄比赛第一名吃 n 个，第 k 名吃 $n+2-2k$ 个且非负，总数为 2009。求最小 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2010-I-Q05] | Part A (Q1-Q5)

Positive integers obey $a > b > c > d$, $a + b + c + d = 2010$ and $a^2 - b^2 + c^2 - d^2 = 2010$. How many values of a are possible?

正整数满足 $a > b > c > d, a + b + c + d = 2010, a^2 - b^2 + c^2 - d^2 = 2010$。求 a 的可能值数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2012-II-Q01] | Part A (Q1-Q5)

Count positive integer pairs (m, n) solving $20m + 12n = 2012$.

求正整数解 (m, n) 的个数，使 $20m + 12n = 2012$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2021-I-Q05] | Part A (Q1-Q5)

Consider strictly increasing three-term arithmetic progressions of integers. Require that the sum of the three squared terms equals the middle term times the square of the common difference. Sum the last terms of all such progressions.

考虑严格递增的三项整数等差数列，要求三项平方和等于中项乘公差的平方。求所有此类数列末项之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2025-I-Q04] | Part A (Q1-Q5)

Count integer pairs $-100 \leq x, y \leq 100$ satisfying $12x^2 - xy - 6y^2 = 0$.

求满足 $-100 \leq x, y \leq 100$ 及 $12x^2 - xy - 6y^2 = 0$ 的整数有序对数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2026-I-Q04] | Part A (Q1-Q5)

How many integers at most 100 can be written as $a+b+ab$ with distinct positive integers a, b?

不超过 100 的整数中，有多少个可写成 $a+b+ab$，其中 a, b 是不同的正整数?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2000-I-Q06] | Part B (Q6-Q10)

Count integer pairs $0 < x < y < 10^6$ such that $(x+y)/2 = \sqrt{xy} + 2$.

求整数对 $0 < x < y < 10^6$ 的数量，使 $(x+y)/2 = \sqrt{xy} + 2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2002-I-Q08] | Part B (Q6-Q10)

Among nondecreasing positive-integer sequences with $a_n = a_{n-1} + a_{n-2}$ for $n > 2$, find the least k that can be a_9 for two different sequences.

非降正整数数列满足 $a_n = a_{n-1} + a_{n-2}$，$n > 2$。求最小 k，使至少两个不同数列的第九项均为 k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q07] | Part B (Q6-Q10)

On segment AC, point B satisfies $AB = 9, BC = 21$. A point D off line AC has equal integer lengths $AD = CD$ and integer BD. Sum all possible perimeters of triangle ACD.

线段 AC 上点 B 满足 $AB = 9, BC = 21$。点 D 不在直线 AC 上，且 $AD = CD$ 为整数、BD 也为整数。求 ACD 所有可能周长之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-I-Q08] | Part B (Q6-Q10)

Four increasing positive integers have their first three terms arithmetic and their last three terms geometric. The fourth exceeds the first by 30. Find their sum.

四个递增正整数的前三项等差、后三项等比，第四项比第一项大 30。求四数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2003-II-Q10] | Part B (Q6-Q10)

Two positive integers differ by 60, and the sum of their square roots equals the square root of a nonsquare integer. Maximize their sum.

两正整数差为 60，其平方根之和等于某非完全平方整数的平方根，求两整数和的最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q06] | Part B (Q6-Q10)

Three monkeys successively take portions of a banana pile. The first keeps $3/4$ of its portion and splits the rest between the others; the second keeps $1/4$; the third takes the remainder and keeps $1/12$, again splitting the rest equally. Every received amount is an integer, and final totals have ratio $3:2:1$. Find the least initial pile size.

三猴依次取香蕉。第一猴留所取的四分之三，余数平分给另两猴；第二猴留四分之一并平分余数；第三猴取全部剩余，留十二分之一并平分余数。每次所得均为整数，最后总数比为 $3:2:1$。求最小初始总数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-II-Q09] | Part B (Q6-Q10)

Let m, n count positive integer solutions of $4x + 3y + 2z = 2009$ and $4x + 3y + 2z = 2000$, respectively. Find $(m - n) \bmod 1000$.

令 m, n 分别为 $4x + 3y + 2z = 2009$ 和 $4x + 3y + 2z = 2000$ 的正整数解数，求差除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q06] | Part B (Q6-Q10)

Find the smallest positive N such that the thousand consecutive integers $1000N, \ldots, 1000N + 999$ contain no perfect square.

求最小正整数 N，使从 $1000N$ 至 $1000N + 999$ 的一千个连续整数中没有完全平方数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2013-II-Q07] | Part B (Q6-Q10)

Clerks sort 1775 files at thirty files per clerk-hour. Equal numbers are reassigned after hours one, two, and three, and the work finishes after three hours ten minutes. How many files were sorted during the first ninety minutes?

若干职员每人每小时整理三十份文件，共整理一千七百七十五份。在第一、二、三小时末各调走相同人数，三小时十分完成。求前九十分钟整理份数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2015-II-Q08] | Part B (Q6-Q10)

Positive integers a, b satisfy $(ab+1)/(a+b) < 3/2$. Write the maximum of $(a^3b^3+1)/(a^3+b^3)$ as reduced p/q and find $p+q$.

正整数 a, b 满足 $(ab+1)/(a+b) < 3/2$。将 $(a^3b^3+1)/(a^3+b^3)$ 的最大值写成最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-II-Q06] | Part B (Q6-Q10)

Find the sum of positive integers n for which $n^2 + 85n + 2017$ is an integer square.

求使 $n^2 + 85n + 2017$ 为整数平方的所有正整数 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2004-II-Q14] | Part C (Q11-Q15)

For how many positive integers n can plus signs inserted between a string of n sevens give sum 7000?

对于多少个正整数 n，在连续 n 个七组成的数位串之间插加号后能得到和七千?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2005-II-Q13] | Part C (Q11-Q15)

An integer-coefficient polynomial satisfies $P(17)=10, P(24)=17$. If $P(n)=n+3$ has two distinct integer solutions n_1, n_2, find $n_1 n_2$.

整系数多项式满足 $P(17)=10, P(24)=17$。若 $P(n)=n+3$ 有两个不同整数解 n_1, n_2，求其乘积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2008-II-Q15] | Part C (Q11-Q15)

Find the largest integer n whose square is a difference of consecutive integer cubes and for which $2n + 79$ is square.

求最大整数 n，使 n^2 为两个连续整数立方之差，且 $2n + 79$ 为平方数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2009-I-Q14] | Part C (Q11-Q15)

For 350 entries $a_i \in \{1,2,3,4\}$, let $S_t = \sum_i a_i^t$. Given $S_1 = 513, S_4 = 4745$, minimize S_2.

三百五十个数 $a_i \in \{1,2,3,4\}$，令 $S_t = \sum_i a_i^t$。已知 $S_1 = 513, S_4 = 4745$，求 S_2 最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2011-I-Q15] | Part C (Q11-Q15)

For some integer m, the polynomial $x^3 - 2011x + m$ has three integer roots a, b, c. Find $|a| + |b| + |c|$.

某整数 m 使 $x^3 - 2011x + m$ 有三个整数根 a, b, c。求 $|a| + |b| + |c|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

[AIME-2017-I-Q13] | Part C (Q11-Q15)

For integer $m \geq 2$, let $Q(m)$ be the smallest positive integer such that every integer $n \geq Q(m)$ admits an integer cube k^3 satisfying $n < k^3 \leq mn$. Find $\sum_{m=2}^{2017} Q(m)$ modulo1000.

对整数 $m \geq 2$，令 $Q(m)$ 为最小正整数，使每个整数 $n \geq Q(m)$ 都有整数立方 k^3 满足 $n < k^3 \leq mn$。求 $\sum_{m=2}^{2017} Q(m)$ 模1000的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-I-Q01] | Part A (Q1-Q5)

Add all two-digit positive integers divisible by both of their decimal digits.

求所有能被自身两个十进制数位分别整除的两位正整数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-II-Q01] | Part A (Q1-Q5)

Let N be the largest decimal integer for which every overlapping two-digit block is a two-digit perfect square. Find its first three digits.

设 N 是使每个重叠的两位数位块均为两位完全平方数的最大十进制整数，求其前三位。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2002-I-Q03] | Part A (Q1-Q5)

Jane is now 25, and Dick is older. Count positive integer pairs (d, n), where d is Dick's present age and, after n years, both ages are two-digit integers with reversed decimal digits.

Jane 现年 25，Dick 比她年长。设 Dick 现龄为正整数 d；正整数 n 年后，两人年龄均为两位数且数位相反。求有序对 (d, n) 数量。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2002-II-Q01] | Part A (Q1-Q5)

Three-digit decimal integers x, y have reversed digit strings. How many distinct values can $|x - y|$ take?

三位十进制整数 x, y 的数位顺序相反。$|x - y|$ 可以取多少个不同值?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2003-II-Q02] | Part A (Q1-Q5)

Let N be the largest positive multiple of 8 with no repeated decimal digit. Find $N \bmod 1000$.

设 N 为十进制各位不重复的最大正的八倍数，求 N 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2004-I-Q01] | Part A (Q1-Q5)

A four-digit number has consecutive decreasing digits. Add all distinct possible remainders on division by 37.

四位数的数位从左到右为连续递减整数，求除以 37 时所有不同可能余数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2006-I-Q03] | Part A (Q1-Q5)

Find the least positive integer whose value becomes one twenty-ninth as large when its leftmost decimal digit is deleted.

求删去十进制最左一位后数值恰变为原数二十九分之一的最小正整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2008-II-Q04] | Part A (Q1-Q5)

Express 2008 uniquely as a sum of distinct nonnegative powers of three with coefficients 1 or -1. Find the sum of the exponents used.

将 2008 唯一表示为三的不同非负整数次幂的和，各系数为正一或负一，求所用指数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2009-I-Q01] | Part A (Q1-Q5)

Call a three-digit decimal integer geometric if its three distinct digits form a geometric progression in order. Find the largest minus the smallest such integer.

若三位十进制正整数的三个互异数字依次成等比数列，称其为等比数。求最大者减最小者。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2010-II-Q01] | Part A (Q1-Q5)

Let N be the largest multiple of 36 whose digits are distinct and all even. Find $N \bmod 1000$.

设 N 为数字互不相同且全部为偶数的最大 36 的倍数，求其除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-I-Q01] | Part A (Q1-Q5)

Count decimal three-digit integers $\overline{abc}$ with $a, c \neq 0$ such that both $\overline{abc}$ and its reversal $\overline{cba}$ are divisible by 4. Digits may repeat.

求十进制三位数 $\overline{abc}$ 的个数，其中 a, c 非零，数字可重复，且原数与倒序数 $\overline{cba}$ 均被四整除。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-I-Q05] | Part A (Q1-Q5)

Let B contain the binary strings of length thirteen with eight ones and five zeros, interpreted as integers and allowing initial zeros. Among ordered subtractions of two elements of B, how many give 1?

集合 B 由长度十三、恰有八个一和五个零的二进制串表示的整数组成，允许前导零。将两元素作有序减法，结果为一的有多少次?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2013-I-Q02] | Part A (Q1-Q5)

Count five-digit decimal multiples of five whose first and last digits agree and whose digit sum is divisible by five.

求五位十进制数的个数，使其被五整除、首尾数字相同、各位数字和也被五整除。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2015-II-Q03] | Part A (Q1-Q5)

Find the least positive multiple of 17 whose decimal digit sum is 17.

求十进制数位和为 17 的最小正 17 的倍数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2017-I-Q05] | Part A (Q1-Q5)

A positive rational has base-eight expansion $ab.cd$ and base-twelve expansion $bb.ba$, with all four digits a, b, c, d nonzero. Find the decimal three-digit numeral abc.

一个正有理数的八进制表示为 $ab.cd$，十二进制表示为 $bb.ba$，其中 a, b, c, d 均为非零数字。求十进制三位数 abc。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2017-II-Q04] | Part A (Q1-Q5)

How many positive integers at most2017 have ternary representations using only digits1 and2?

不超过2017的正整数中，三进制表示仅使用数字1、2的有多少个?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2018-I-Q02] | Part A (Q1-Q5)

An integer has representations $\overline{abc}_{14} = \overline{acb}_{15} = \overline{acac}_{6}$, with $a > 0$. Find the integer in decimal.

某整数满足 $\overline{abc}_{14} = \overline{acb}_{15} = \overline{acac}_{6}$，且 $a > 0$，求其十进制值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2018-II-Q03] | Part A (Q1-Q5)

Find the sum of integer bases $b < 1000$ for which $\overline{36}_b$ is a square and $\overline{27}_b$ is a cube. Digits must be valid in base b.

求小于一千的整数进位制底数 b 之和，使 $\overline{36}_b$ 为完全平方、$\overline{27}_b$ 为完全立方，各数字须合法。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2019-I-Q01] | Part A (Q1-Q5)

Let $N = \sum_{k=1}^{321}(10^k - 1)$, the sum of the numbers with one through 321 consecutive nines. Find its decimal digit sum.

令 $N = \sum_{k=1}^{321}(10^k - 1)$，即从一位九到三百二十一位全九之和，求十进制数字和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2020-I-Q03] | Part A (Q1-Q5)

A number has digit strings abc in base 11 and $1bca$ in base 8; repeated letters indicate the same digit, and distinct letters need not differ. Find its least decimal value.

某正整数的十一进制表示为 abc，八进制表示为 $1bca$；相同字母代表同一数位，不同字母可相等。求该整数最小的十进制值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2020-II-Q05] | Part A (Q1-Q5)

Let $f(n)$ be the sum of the base-4 digits of n, and let $g(n)$ be the sum of the base-8 digits of $f(n)$. For the smallest positive N whose $g(N)$ has a hexadecimal digit A, B, C, D, E, or F, find $N \bmod 1000$.

令 $f(n)$ 为 n 的四进制数位和，$g(n)$ 为 $f(n)$ 的八进制数位和。设 N 是使 $g(N)$ 的十六进制表示出现 A 至 F 中某个数字的最小正整数，求 N 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2022-I-Q02] | Part A (Q1-Q5)

A decimal three-digit numeral abc denotes the same integer as the base-nine numeral bca. Digits may repeat. Find the decimal integer.

十进制三位数 abc 与九进制数 bca 表示同一整数，数字可以重复。求这个十进制整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2023-II-Q02] | Part A (Q1-Q5)

Determine the largest positive integer below 1000 whose representations in bases ten and eight are both palindromes.

求小于 1000 的最大正整数，使其十进制、八进制表示均为回文。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2024-II-Q03] | Part A (Q1-Q5)

Fill a 2×3 array with decimal digits, allowing leading zeroes. The two three-digit row values must sum to 999, and the three two-digit column values to 99. Count the arrays.

用十进制数字填满 2×3 方格，允许前导零。两行从左至右读出的数之和为 999，三列从上至下读出的数之和为 99。求填法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2025-I-Q01] | Part A (Q1-Q5)

Sum the integer bases $b > 9$ for which the base-b number 17_b divides 97_b.

求所有使 $17_b \mid 97_b$ 成立的整数进制 $b > 9$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2026-I-Q02] | Part A (Q1-Q5)

Count positive decimal palindromes containing no digit zero and having digit sum 13.

求不含数字零且数字和为 13 的正整数十进制回文数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2026-II-Q04] | Part A (Q1-Q5)

For a positive decimal integer n, reinterpret its digits in base b, where b is one more than its largest digit, to obtain $f(n)$. Count $1 \leq n < 1000$ with $f(n) = n$.

将正整数 n 的十进制数字串改按 b 进制解释，得到 $f(n)$，其中 b 为最大数字加一。求 $1 \leq n < 1000$ 中 $f(n) = n$ 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2001-I-Q08] | Part B (Q6-Q10)

Call $N > 0$ special when reading its base-7 digit string as a decimal integer gives $2N$. Find the greatest special integer.

若把正整数 N 的七进制数位串当成十进制整数读取，结果为 $2N$，则称其特殊。求最大的特殊整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2006-II-Q07] | Part B (Q6-Q10)

Count positive ordered pairs $a + b = 1000$ for which neither decimal representation contains zero.

求正整数有序对 $a + b = 1000$ 且两数十进制都不含零的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2012-II-Q07] | Part B (Q6-Q10)

Order increasingly all positive integers whose binary expansion contains exactly eight ones. Find the thousandth integer modulo 1000.

将二进制表示恰有八个一的正整数递增排列，求第一千个数模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2015-I-Q08] | Part B (Q6-Q10)

Let $s(n)$ be the decimal digit sum. Find the least positive integer n for which $s(n) = s(n + 864) = 20$.

令 $s(n)$ 表示十进制数位和。求满足 $s(n) = s(n + 864) = 20$ 的最小正整数 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2016-I-Q08] | Part B (Q6-Q10)

Permute digits 1 through 9 into three consecutive three-digit numerals and add them. Among permutations whose total ends in 0, let m be the smallest total and n the number of permutations attaining it. Find $|m-n|$.

将数字 1 至 9 排列，按顺序分成三个三位数并相加。在总和末位为 0 的排列中，最小总和为 m，达到此和的排列数为 n。求 $|m-n|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2000-II-Q14] | Part C (Q11-Q15)

Use factorial digits $k = \sum_{i=1}^{j} i! f_i$ with $0 \le f_i \le i$ and $f_j > 0$. For $k = \sum_{h=1}^{125} (-1)^{h+1} (16h)!$, find $\sum_{i=1}^{j} (-1)^{i+1} f_i$.

阶乘进位制写作 $k = \sum_{i=1}^{j} i! f_i$，其中 $0 \le f_i \le i, f_j > 0$。若 $k = \sum_{h=1}^{125} (-1)^{h+1} (16h)!$，求 $\sum_{i=1}^{j} (-1)^{i+1} f_i$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2003-I-Q13] | Part C (Q11-Q15)

Let N count integers $1 \leq k \leq 2003$ whose standard binary expansion contains more ones than zeros. Find $N \bmod 1000$.

设 N 为 1 至 2003 中标准二进制表示内一的个数多于零的整数数目，求 N 模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2014-II-Q15] | Part C (Q11-Q15)

Let $p(k)$ be the least prime not dividing positive k, and let $X(k)$ be the product of all smaller primes, using the empty product one. Starting $x_0 = 1$, define $x_{n+1} = x_n p(x_n)/X(x_n)$. Find the first positive index t with $x_t = 2090$.

设 $p(k)$ 为不整除正整数 k 的最小素数，$X(k)$ 为所有更小素数的乘积，空积取一。由 $x_0 = 1, x_{n+1} = x_n p(x_n)/X(x_n)$ 定义序列，求首次 $x_t = 2090$ 的正下标 t。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

[AIME-2024-II-Q14] | Part C (Q11-Q15)

Call an integer special in base $b \geq 2$ if it has exactly two base-b digits and their sum equals its square root. Find the least base with more than ten special integers.

若正整数在 $b \geq 2$ 进制中恰有两位，且两位数字之和等于该数的平方根，则称其为特殊数。求使特殊数超过十个的最小进制 b。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2014-II-Q04] | Part A (Q1-Q5)

Decimal digits a, b, c, with $a \neq 0$, satisfy $0.\overline{ab} + 0.\overline{abc} = 33/37$. Find $\overline{abc}$.

十进数字 a, b, c，其中 a 非零，满足 $0.\overline{ab} + 0.\overline{abc} = 33/37$，求三位数 $\overline{abc}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2023-II-Q05] | Part A (Q1-Q5)

For positive rationals r, compare the sums of numerator and denominator in the reduced fractions for r and $55r$. Let S be the rationals for which these two sums agree. If $\sum_{r\in S} r = p/q$ in lowest terms, find $p+q$.

将正有理数 r 及 $55r$ 分别约为最简分数；两分数的分子分母之和相等时，将 r 收入集合 S。若 S 各元素之和为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2006-I-Q06] | Part B (Q6-Q10)

Sum all real numbers $0.\overline{abc}$ whose three decimal digits a, b, c are pairwise distinct, allowing $a = 0$.

求所有循环小数 $0.\overline{abc}$ 的和，其中三位数字两两不同，允许 $a = 0$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2003-I-Q14] | Part C (Q11-Q15)

Find the smallest denominator n of a reduced fraction $0 < m/n < 1$ whose decimal expansion contains consecutive digits 251 in that order.

求最简分数 $0 < m/n < 1$ 的最小分母 n，使其十进制小数展开中连续出现数位 251。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

[AIME-2022-I-Q13] | Part C (Q11-Q15)

Consider all nonzero four-digit repeating decimals $0.\overline{abcd}$, allowing leading zero digits and shorter actual periods. Let N be the number of distinct positive numerators occurring after reduction to lowest terms. Find $N \bmod 1000$.

考虑所有非零四位循环小数 $0.\overline{abcd}$，允许循环块以零开头，也允许实际周期小于四。约为最简分数后，共有 N 种不同的正分子。求 N 模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2000-II-Q04] | Part A (Q1-Q5)

Find the smallest positive integer having exactly six odd positive divisors and twelve even positive divisors.

求恰有六个奇正约数和十二个偶正约数的最小正整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q02] | Part A (Q1-Q5)

How many positive differences k make the arithmetic sequence $1, 1+k, 1+2k, \ldots$ contain 2005?

有多少个正公差 k 使等差数列 $1, 1+k, 1+2k, \ldots$ 包含 2005?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q03] | Part A (Q1-Q5)

Count positive integers having exactly three proper divisors, all smaller than 50. A proper divisor excludes the integer itself.

求恰有三个真约数且三个都小于 50 的正整数个数，真约数不包括该数本身。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-II-Q04] | Part A (Q1-Q5)

Count positive integers dividing at least one of $10^{10}, 15^7, 18^{11}$.

求至少整除 $10^{10}, 15^7, 18^{11}$ 中一个数的正整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2014-I-Q03] | Part A (Q1-Q5)

Count rational numbers strictly between zero and one whose reduced numerator and denominator sum to 1000.

求零与一之间的有理数个数，使其最简分数的分子分母之和为一千。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2004-I-Q08] | Part B (Q6-Q10)

A regular n-pointed star is a closed route through n coplanar vertices with no three collinear, equal edge lengths and equal vertex angles. Each turn is counterclockwise and less than 180°, and every edge crosses some other edge away from endpoints. Up to similarity, how many such stars have $n = 1000$?

规则 n 角星由平面中无三点共线的 n 顶点组成闭合路线，各边等长、各顶角相等，每次逆时针转向且小于 180°，每边均须与另一边在非端点处相交。相似图形视为同类，求 $n = 1000$ 时类别数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2004-II-Q08] | Part B (Q6-Q10)

How many divisors of 2004^{2004} have exactly 2004 positive divisors of their own?

2004^{2004} 的正约数中，有多少个自身恰有二千零四个正约数?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2019-I-Q09] | Part B (Q6-Q10)

Let $\tau(n)$ count positive divisors. Sum the six least positive integers satisfying $\tau(n)+\tau(n+1)=7$.

令 $\tau(n)$ 为正约数数，求满足 $\tau(n)+\tau(n+1)=7$ 的最小六个正整数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2019-II-Q09] | Part B (Q6-Q10)

Sum all positive integers $n < 2019$ having exactly twenty positive divisors and divisible by twenty, obtaining S. Find $S/20$.

将小于 2019、恰有二十个正约数且能被二十整除的所有正整数相加得 S，求 $S/20$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2026-II-Q09] | Part B (Q6-Q10)

For $S = \sum_{k\ge 1} 1/(10^k - 1)$, find $\lfloor 10^{100} S \rfloor \bmod 1000$.

令 $S = \sum_{k\ge 1} 1/(10^k - 1)$，求 $\lfloor 10^{100} S \rfloor$ 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q12] | Part C (Q11-Q15)

Let $\tau(n)$ count positive divisors and $S(n)=\sum_{j=1}^{n}\tau(j)$. Among $1\le n\le 2005$, let a,b count odd and even values of $S(n)$. Find $|a-b|$.

令 $\tau(n)$ 为正约数个数，$S(n)=\sum_{j=1}^{n}\tau(j)$。在 $1\le n\le 2005$ 中，$S(n)$ 奇偶数目分别为 a,b，求 $|a-b|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2006-II-Q13] | Part C (Q11-Q15)

How many integers $N < 1000$ admit a representation as a sum of j consecutive positive odd integers for exactly five different positive lengths j?

有多少个整数 $N < 1000$ 能恰好对五种不同正整数长度 j 写成 j 个连续正奇数之和?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2016-II-Q11] | Part C (Q11-Q15)

Call a positive integer N k-nice if a^k has exactly N positive divisors for some positive integer a. How many positive integers below1000 are neither7-nice nor8-nice?

若存在正整数 a，使 a^k 恰有 N 个正约数，则称正整数 N 为 k-nice。求小于1000的正整数中，既非7-nice也非8-nice的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2021-I-Q14] | Part C (Q11-Q15)

Write $\sigma(u)$ for the sum of positive divisors of u. Find the least positive n such that $\sigma(a^n) \equiv 1 \pmod{2021}$ for every positive integer a. Add the prime factors of n, counted with their multiplicities.

令 $\sigma(u)$ 为 u 的正约数和。求使每个正整数 a 均满足 $\sigma(a^n) \equiv 1 \pmod{2021}$ 的最小正整数 n，并求 n 的质因数之和，按重数计算。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2003-II-Q03] | Part A (Q1-Q5)

Count seven-letter words over $\{A, B, C\}$ that contain none of the adjacent blocks AB, BC, CA.

用 A, B, C 组成七字母串，禁止相邻块 AB, BC, CA，共有多少个?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-II-Q04] | Part A (Q1-Q5)

How many positive decimal integers below 10000 use at most two distinct digit values?

小于一万的正十进制整数中，有多少个最多使用两种不同数位?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2005-I-Q05] | Part A (Q1-Q5)

Stack four identical gold and four identical silver coins. Each has one engraved face, independently orientable up or down. Count distinguishable stacks with no adjacent engraved faces facing one another.

将四枚相同金币与四枚相同银币叠成一柱，每枚仅一面有雕刻，可朝上或朝下。求相邻两枚雕刻面不相对的不同叠法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2006-II-Q04] | Part A (Q1-Q5)

Count permutations of $1, \ldots, 12$ satisfying $a_1 > a_2 > \cdots > a_6 < a_7 < \cdots < a_{12}$.

求一至十二的排列中满足 $a_1 > a_2 > \cdots > a_6 < a_7 < \cdots < a_{12}$ 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2007-II-Q01] | Part A (Q1-Q5)

Form five-character strings from the multiset $\{A, I, M, E, 2, 0, 0, 7\}$, never exceeding the supplied multiplicities. If there are N distinct strings, find $N/10$.

从多重集合 $\{A, I, M, E, 2, 0, 0, 7\}$ 中取字符排成长五的串，各字符不超过原有次数。若不同串数为 N，求 $N/10$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2011-I-Q05] | Part A (Q1-Q5)

Label a regular nonagon with digits 1 through 9, each once, so every three consecutive labels have sum divisible by 3. Count arrangements up to rotation, with reflections distinct.

将数字一至九各用一次标在正九边形顶点，使任意连续三顶点之和可被三整除。旋转视为相同，反射仍区分，求排列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2012-I-Q03] | Part A (Q1-Q5)

Nine distinct diners order three meals of each of three types. Count the assignments of these meal types to diners for which exactly one diner gets the requested type.

九名不同食客各点三种餐之一，每种恰有三人。求分配三类餐品的方法数，使恰一人收到自己所点的餐。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2019-I-Q04] | Part A (Q1-Q5)

A team starts eleven fixed players from twenty-two. It may make zero through three sequential substitutions: any current player leaves, an unused substitute enters, and removed players never return. An entering substitute may later leave. Count substitution sequences modulo 1000.

二十二人队先固定十一人上场，可依次换零至三次，每次任选当前一人离场、未上过场的替补一人进入，离场者不得返回，进入者之后也可被换下。求换人序列数除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2019-II-Q05] | Part A (Q1-Q5)

Four distinguishable ambassadors and their four distinct paired advisors sit in twelve numbered circular chairs. Ambassadors use even chairs, each advisor sits adjacent to their ambassador, and no chair is shared. Find the seating count modulo 1000.

四名不同大使及各自不同顾问坐十二个编号圆桌座位，大使只能坐偶数位，顾问须邻其大使，不得重座。求安排数除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2020-I-Q05] | Part A (Q1-Q5)

Permute the six distinct labels $1, 2, 3, 4, 5, 6$. How many permutations permit deletion of one label so that the remaining sequence is strictly increasing or strictly decreasing?

排列 $1, 2, 3, 4, 5, 6$ 六个不同标号。若删去一个标号后，其余五个数可严格递增或严格递减，求这样的排列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2021-II-Q03] | Part A (Q1-Q5)

For a permutation $(x_1, \ldots, x_5)$ of $1, 2, 3, 4, 5$, form $\sum_{i=1}^{5} x_i x_{i+1} x_{i+2}$ with cyclic indices. How many permutations make this sum divisible by 3?

将 $1, 2, 3, 4, 5$ 排成 $(x_1, \ldots, x_5)$，下标循环使用。求使 $\sum_{i=1}^{5} x_i x_{i+1} x_{i+2}$ 被 3 整除的排列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2025-I-Q03] | Part A (Q1-Q5)

Nine distinct people each choose one of three flavors. All flavors must occur, with first-flavor count greater than second-flavor count greater than third-flavor count. Find the number of assignments modulo 1000.

九个不同的人各选三种口味之一，每种都至少有人选择，且第一种人数大于第二种，第二种大于第三种。求合法选择数除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2004-I-Q06] | Part B (Q6-Q10)

Count four-digit integers with distinct digits satisfying $a_1 < a_2 > a_3 < a_4$.

求四个数位互异且满足 $a_1 < a_2 > a_3 < a_4$ 的四位整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2023-II-Q07] | Part B (Q6-Q10)

Color each vertex of a regular twelve-gon red or blue. Among the 2^{12} labeled colorings, how many contain no rectangle whose four vertices share one color?

将正十二边形每个顶点涂成红或蓝。2^{12} 种按位置区分的染色中，不存在四顶点同色矩形的共有多少种?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2025-I-Q10] | Part B (Q6-Q10)

Fill a 3×9 grid with digits $1, \ldots, 9$ so each row and each of the three consecutive 3×3 blocks contains every digit once. If the number of fillings is $\prod p_j^{e_j}$, find $\sum p_j e_j$.

用数字 1 至 9 填满 3×9 方格，每行及依次排列的三个 3×3 宫都恰含每个数字一次。将填法数分解为 $\prod p_j^{e_j}$，求 $\sum p_j e_j$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.1 Restricted permutations & arrangements

受限排列

[AIME-2026-I-Q07] | Part B (Q6-Q10)

Count onto maps $\pi : \{1, \ldots, 6\} \to \{1, \ldots, 6\}$ satisfying $\pi^6(a) = a$ for all a.

求满射 $\pi : \{1, \ldots, 6\} \to \{1, \ldots, 6\}$ 的个数，使每个 a 均满足六次迭代 $\pi^6(a) = a$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2005-II-Q01] | Part A (Q1-Q5)

A deck of $n \geq 6$ distinct cards has six times as many six-card subsets as three-card subsets. Find n.

一副 $n \geq 6$ 张不同的牌，其六张子集数为三张子集数的六倍，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2012-II-Q03] | Part A (Q1-Q5)

Three departments each have two male and two female professors. Count six-person committees with two from each department and three of each gender.

三个系各有两男两女教授，求六人委员会数，要求每系两人，且男女各三人。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q01] | Part A (Q1-Q5)

On the boundary of a nondegenerate triangle, mark its three vertices and another $3, 4, 5$ points in the interiors of its three respective sides. How many positive-area triangles have all vertices among these15 points?

在非退化三角形边界上标记三个顶点，并在三条边内部分别再取 $3, 4, 5$ 个不同点。用这15点作顶点，可组成多少个正面积三角形?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2000-II-Q07] | Part B (Q6-Q10)

If $\sum_{k=2}^{9} 1/[k!(19-k)!] = N/(1!18!)$, find the greatest integer strictly below $N/100$.

若 $\sum_{k=2}^{9} 1/[k!(19-k)!] = N/(1!18!)$，求严格小于 $N/100$ 的最大整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2002-II-Q09] | Part B (Q6-Q10)

Let N be the number of unordered pairs of nonempty disjoint subsets of a ten-element set. Find $N \bmod 1000$.

设十元集合的两个非空不交子集组成的无序对数为 N，求 N 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2007-I-Q10] | Part B (Q6-Q10)

Shade twelve cells of a six-row, four-column grid so each row has two and each column three shaded cells. Find the number of shadings modulo 1000.

六行四列方格中涂十二格，使每行恰两格、每列恰三格，求涂法数除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2009-II-Q06] | Part B (Q6-Q10)

Let m count five-element subsets of $\{1,\ldots,14\}$ containing at least one consecutive pair. Find $m \bmod 1000$.

令 m 为 $\{1,\ldots,14\}$ 的五元素子集中至少含一对相邻整数者的个数，求其除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2010-I-Q07] | Part B (Q6-Q10)

For ordered subsets A, B, C of a seven-element set, require each pairwise intersection to have exactly one element and the triple intersection to be empty. Count these ordered triples modulo 1000.

七元集合的有序子集三元组 (A, B, C) 满足三个两两交集各含一元素，三者公共交集为空。求三元组数模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2010-II-Q08] | Part B (Q6-Q10)

Count ordered nonempty complementary subsets A, B of $\{1, \ldots, 12\}$ satisfying $|A| \notin A$ and $|B| \notin B$.

求 $\{1, \ldots, 12\}$ 的非空互补子集有序对 (A, B) 数，使 $|A| \notin A, |B| \notin B$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q07] | Part B (Q6-Q10)

Let $S = \sum_{a,b\geq 0,\ a+b\leq 6} \binom{6}{a}\binom{6}{b}\binom{6}{a+b}$. Find $S \bmod 1000$.

令 $S = \sum_{a,b\geq 0,\ a+b\leq 6} \binom{6}{a}\binom{6}{b}\binom{6}{a+b}$，求 S 模1000的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2020-I-Q07] | Part B (Q6-Q10)

A group contains 11 men and 12 women. Count all committees with exactly one more woman than man, allowing every possible size. If this count is N, find the sum of its distinct prime divisors.

一个团体有 11 名男性和 12 名女性。委员会须恰好多一名女性，人数不限。设委员会数为 N，求 N 的不同质因数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2021-II-Q06] | Part B (Q6-Q10)

Count ordered pairs of subsets (A, B) of a five-element set such that $|A||B| = |A \cap B||A \cup B|$. Equal subsets are permitted.

五元集合的子集有序对 (A, B) 满足 $|A||B| = |A \cap B||A \cup B|$，允许 $A = B$，共有多少对?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2022-II-Q10] | Part B (Q6-Q10)

Compute $\sum_{k=3}^{40} \binom{\binom{k}{2}}{2}$ modulo 1000.

求 $\sum_{k=3}^{40} \binom{\binom{k}{2}}{2}$ 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2024-II-Q06] | Part B (Q6-Q10)

A set A of positive integers has the property that exactly 2024 finite nonempty sets of positive integers have their maximum in A. Find the sum of the elements of A.

正整数集合 A 满足：最大元素属于 A 的有限非空正整数集合恰有 2024 个。求 A 中所有元素的和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2007-II-Q13] | Part C (Q11-Q15)

An eleven-row triangular array is filled bottom-up by adding the two entries immediately below each cell. Bottom entries are independently chosen zero or one. Count bottom rows whose top entry is divisible by three.

十一行三角阵自下向上填数，每格等于下方相邻两格之和。底行每格取零或一，求顶端数被三整除的底行填法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2017-I-Q12] | Part C (Q11-Q15)

Count subsets $S \subseteq \{1, \ldots, 10\}$ with no elements $a, b, c \in S$ satisfying $ab = c$; the three elements need not be distinct.

求 $\{1, \ldots, 10\}$ 的子集 S 中，不存在 $a, b, c \in S$ 使 $ab = c$ 的个数，其中三元素允许相同。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2023-II-Q11] | Part C (Q11-Q15)

Count the families containing sixteen distinct subsets of a five-element set such that every pair of family members has nonempty intersection.

一个五元集合的十六个不同子集组成一族，要求族中任意两个成员交集非空。求这样的集合族数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 4.2 Subsets & binomial identities

子集与二项式恒等式

[AIME-2026-I-Q13] | Part C (Q11-Q15)

For $0 \le r < 502$, define $S_r = \sum_{j \ge 0} \binom{10000}{502j+r}$, treating out-of-range coefficients as zero. How many of these 502 sums are divisible by 503?

对 $0 \le r < 502$，定义 $S_r = \sum_{j \ge 0} \binom{10000}{502j+r}$，越界二项式系数视为零。求这 502 个和中能被 503 整除的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2000-II-Q05] | Part A (Q1-Q5)

Choose five of eight distinct rings and place them on four labeled fingers of one hand, allowing empty fingers and distinguishing ring order on each finger. Let N count arrangements. Read its first three nonzero decimal digits from the left.

从八个不同戒指中选五个戴在一只手的四根有标识手指上，允许空指，每指戒指顺序不同视为不同。设排法数为 N，求从左起前三个非零十进制数字组成的数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2021-I-Q04] | Part A (Q1-Q5)

Count positive integer triples $a < b < c$ with $a + b + c = 66$.

求满足 $a < b < c$ 且 $a + b + c = 66$ 的正整数三元组个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2003-I-Q09] | Part B (Q6-Q10)

Count four-digit decimal integers whose first two digits sum to the same value as their last two digits.

求前两位数位和等于后两位数位和的四位十进制整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q09] | Part B (Q6-Q10)

Three ordered prices are integers from one to 9999, and their concatenated digits consist of four ones and three threes. Count all possible ordered price triples.

三件有序商品价格为一至 9999 的整数，三价格串联后的数字恰为四个一、三个三。求可能的有序价格三元组数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q10] | Part B (Q6-Q10)

Fifteen labeled circular seats hold five distinct members of each of groups M, V, E, with seat one occupied by M and seat fifteen by E. Clockwise changes of group are allowed only $M \to V, V \to E, E \to M$. If total arrangements are $N(5!)^3$, find N.

十五个有编号圆桌座位坐三组各五名不同成员，第一座为 M 组，第十五座为 E 组。按顺时针，换组只允许 $M \to V, V \to E, E \to M$。若总安排数为 $N(5!)^3$，求 N。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2011-II-Q07] | Part B (Q6-Q10)

Arrange five identical green marbles and some identical red marbles in a row, with equally many same-color and different-color adjacent pairs. Maximize the red count m, and let N count the valid rows at that maximum. Find $N \bmod 1000$.

将五颗相同绿珠与若干相同红珠排成一行，要求同色相邻对数与异色相邻对数相等。令最大红珠数为 m，达到此最大值的排列数为 N，求 N 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2025-II-Q10] | Part B (Q6-Q10)

Choose eight occupied chairs among sixteen in a row, with no occupant having occupied neighbors on both sides. If there are N chair subsets, find $N \bmod 1000$.

十六把椅子排成一行，选八把坐人，要求无人左右两侧都有人坐。若合法椅子子集数为 N，求其除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2008-II-Q12] | Part C (Q11-Q15)

Arrange ten identical blue and nine identical green flags in ordered stacks on two distinguishable poles, each nonempty, with no adjacent greens on a pole. Find the count modulo 1000.

将十面相同蓝旗、九面相同绿旗按次序挂在两根不同旗杆上，每杆非空且绿旗不相邻，求排法数除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2018-II-Q15] | Part C (Q11-Q15)

Count integer-valued functions on $\{0,1,\ldots,6\}$ with $f(0)=0, f(6)=12$ and $|x-y| \le |f(x)-f(y)| \le 3|x-y|$ for every pair of inputs.

求定义在 $\{0,1,\ldots,6\}$ 上的整数值函数数，满足 $f(0)=0, f(6)=12$，且任意输入对均有 $|x-y| \le |f(x)-f(y)| \le 3|x-y|$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2022-II-Q13] | Part C (Q11-Q15)

The expression $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$ agrees with an integer-coefficient polynomial $P(x)$ on $0<x<1$. Find the coefficient of x^{2022}.

表达式 $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$ 在 $0<x<1$ 上等于某整系数多项式 $P(x)$。求 x^{2022} 的系数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2001-II-Q02] | Part A (Q1-Q5)

All 2001 students study Spanish or French or both. Spanish enrollment is between 80% and 85%; French enrollment is between 30% and 40%. If the minimum and maximum possible overlap sizes are m, M, find $M - m$.

2001 名学生每人至少学习西班牙语或法语之一。西语人数在总数的 80% 至 85% 之间，法语人数在 30% 至 40% 之间。若双语人数最小、最大分别为 m, M，求 $M - m$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2003-I-Q03] | Part A (Q1-Q5)

For every two-element subset of $\{1, 2, 3, 5, 8, 13, 21, 34\}$, record its larger element. What is the sum of all recorded values?

对 $\{1, 2, 3, 5, 8, 13, 21, 34\}$ 的每个二元素子集记下较大元素，求所有记录之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2016-II-Q04] | Part A (Q1-Q5)

An integral $a \times b \times c$ box is assembled from unit cubes colored red, green or yellow. Each of its a layers perpendicular to the first dimension contains 9 red and 12 green cubes. Each of its b layers perpendicular to the second dimension contains 20 green and 25 yellow cubes. Find the least possible volume.

整数边长 $a \times b \times c$ 的长方体由红、绿、黄单位方块组成。垂直第一方向的每个切层均有 9 红、12 绿；垂直第二方向的每个切层均有 20 绿、25 黄。求最小可能体积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2017-II-Q01] | Part A (Q1-Q5)

Count subsets of $\{1, \ldots, 8\}$ that are contained in neither $\{1, 2, 3, 4, 5\}$ nor $\{4, 5, 6, 7, 8\}$.

求 $\{1, \ldots, 8\}$ 的子集中，既不包含于 $\{1, 2, 3, 4, 5\}$ 也不包含于 $\{4, 5, 6, 7, 8\}$ 的个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2023-I-Q03] | Part A (Q1-Q5)

Forty nonparallel lines are drawn in the plane. Exactly $3, 4, 5, 6$ points have respectively $3, 4, 5, 6$ concurrent lines, and no point has more than six. How many intersection points have exactly two incident lines?

平面内有四十条两两不平行直线。恰有 $3, 4, 5, 6$ 个交点分别通过 $3, 4, 5, 6$ 条直线，无点通过超过六条。求恰通过两条直线的交点数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-II-Q01] | Part A (Q1-Q5)

A town of 900 people has ownership totals $195, 367, 562, 900$ for four different items; everyone owns the fourth item. Exactly 437 people own two item types and 234 own three. How many own all four?

某镇 900 人对四类物品的拥有者总数依次为 $195, 367, 562, 900$，第四类人人都有。恰有 437 人拥有两类，234 人拥有三类。求四类全有的人数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2014-II-Q09] | Part B (Q6-Q10)

Ten labeled chairs form a circle. Count subsets containing a run of at least three consecutive selected chairs.

十个固定座位围成圆，求含至少连续三个被选座位的子集数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2018-I-Q09] | Part B (Q6-Q10)

Count four-element subsets of $\{1, \ldots, 20\}$ containing a distinct-element pair summing to sixteen and a distinct-element pair summing to twenty-four. The two pairs may overlap.

求一至二十的四元子集数，使其中有两个不同元素之和为十六，也有两个不同元素之和为二十四；两对可有公共元素。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2020-II-Q09] | Part B (Q6-Q10)

Six distinct people initially sit in the order $1, 2, 3, 4, 5, 6$. In how many new linear seating orders is every originally neighboring pair separated?

六人最初依次坐为 $1, 2, 3, 4, 5, 6$。重新排成一行时，原来相邻的任意两人都不得相邻，求排法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q06] | Part B (Q6-Q10)

How many integer pairs (a, b) make $3, 4, 5, a, b, 30, 40, 50$ strictly increasing while containing no four-term arithmetic progression, even as a nonconsecutive subsequence?

求整数对 (a, b) 的个数，使 $3, 4, 5, a, b, 30, 40, 50$ 严格递增，且任选四项均不能组成等差数列（不要求四项相邻）。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2010-II-Q11] | Part C (Q11-Q15)

Count the 3×3 zero-one matrices with exactly five ones in which at most one of the three rows, three columns, or two long diagonals is constant.

求恰含五个一、四个零的 3×3 矩阵数，使三行、三列、两条长对角线中至多一条上的三个元素全相等。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q12] | Part C (Q11-Q15)

For every ordered pair of equal-cardinality subsets $A, B \subseteq \{1, \ldots, n\}$, add $|A \cap B|$; call the sum S_n. Reduce S_{2022}/S_{2021} to p/q and find $(p+q) \bmod 1000$.

遍历 $\{1, \ldots, n\}$ 的所有等势子集有序对 (A, B)，将 $|A \cap B|$ 相加，所得为 S_n。将 S_{2022}/S_{2021} 约为 p/q，求 $p+q$ 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-I-Q11] | Part C (Q11-Q15)

Independently color each vertex of a regular octagon blue or red, each with probability $1/2$. Find $m+n$ when the probability that some rotation sends every blue vertex to a formerly red position is m/n in lowest terms.

正八边形各顶点独立地以 $1/2$ 概率染蓝或红。若存在一次旋转使所有蓝点落在原先红点的位置，则称染色成功。成功概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-I-Q06] | Part B (Q6-Q10)

A frog begins at zero. Each step goes to either the next strictly larger multiple of three or the next strictly larger multiple of thirteen. Count distinct coordinate sequences ending at 39.

青蛙从零开始，每步跳到严格更大的最近三倍数或最近十三倍数。求终止于三十九的不同坐标序列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-II-Q06] | Part B (Q6-Q10)

A decimal digit string is parity-monotonic when every odd digit is followed by a strictly larger digit and every even digit by a strictly smaller one. Count four-digit positive integers with this property.

若十进制数字串中每个奇数字后都接更大数字、每个偶数字后都接更小数字，称其为奇偶单调。求符合的四位正整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2013-II-Q09] | Part B (Q6-Q10)

Tile a seven-cell row using arbitrary positive integer-length tiles, each colored red, blue, or green. Tile boundaries distinguish tilings even when neighboring tiles share a color. Count tilings using all three colors, modulo 1000.

用正整数长度的一格宽瓷砖铺满七格行，每砖为红蓝绿之一。即使相邻砖同色，改变砖的分界仍视为不同。求三色均出现的铺法数模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q10] | Part B (Q6-Q10)

Count permutations $a_1, \ldots, a_7$ of $1, \ldots, 7$ satisfying $a_k \leq a_{k+1} + 2$ for all $k < 7$.

求 1 至 7 的排列 $a_1, \ldots, a_7$ 中，满足对所有 $k < 7$ 均有 $a_k \leq a_{k+1} + 2$ 的排列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q08] | Part B (Q6-Q10)

A frog starts at $(0, 0)$ and may move one or two units east or one or two units north each jump. Count ordered jump sequences ending at $(4, 4)$.

青蛙从原点出发，每次可向东一或二单位，或向北一或二单位。求到达 $(4, 4)$ 的有序跳跃序列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2024-I-Q06] | Part B (Q6-Q10)

Count shortest grid paths from $(0,0)$ to $(8,8)$ that use unit right/up steps and turn exactly four times.

在方格上从 $(0,0)$ 到 $(8,8)$，每步向右或向上一个单位。求恰好转向四次的最短路径条数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2000-I-Q15] | Part C (Q11-Q15)

A stack of 2000 distinctly labeled cards is processed by repeatedly putting its top card onto a table and moving the next card to the bottom of the stack. The table receives labels in order $1, 2, \ldots, 2000$. How many cards originally lay above label 1999?

一叠 2000 张不同标号牌反复执行：顶牌放到桌上，下一张顶牌移到牌堆底部。桌上得到的标号顺序为 $1, 2, \ldots, 2000$。原牌堆中标号 1999 上方有多少张牌？

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2001-I-Q14] | Part C (Q11-Q15)

Nineteen houses lie in a row. Choose which receive mail so that no adjacent pair both receive it and no three consecutive houses all receive none. How many patterns are possible?

十九户排成一行。选择当天收信的住户，要求相邻两户不能都收信，连续三户也不能都不收信。求可能模式数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2004-I-Q15] | Part C (Q11-Q15)

For positive integers define $f(1)=1$, $f(x)=x/10$ when $10 \mid x$, and $f(x)=x+1$ otherwise. Starting at x, let $d(x)$ count the terms up to and including the first 1. If M counts x with $d(x)=20$, find the sum of the distinct prime divisors of M.

对正整数定义 $f(1)=1$；若 $10 \mid x$，则 $f(x)=x/10$；否则 $f(x)=x+1$。从 x 开始迭代，令 $d(x)$ 为包括首次一在内的项数。若 $d(x)=20$ 的起点数为 M，求 M 的不同质因数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2005-I-Q13] | Part C (Q11-Q15)

A lattice path uses steps $R=(1,0), U=(0,1), D=(1,1)$, but may not have adjacent RU or UR. Count paths from $(0,0)$ to $(5,5)$.

格点路径允许步长 $R=(1,0), U=(0,1), D=(1,1)$，但不得相邻出现 RU 或 UR。求从 $(0,0)$ 到 $(5,5)$ 的路径数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2006-I-Q11] | Part C (Q11-Q15)

Stack cubes of side lengths $1, \ldots, 8$, using all once. A cube of length k may have immediately above it only a cube of side at most $k+2$. Find the number of towers modulo 1000.

将边长一至八的八个立方体各用一次叠成塔，边长 k 的上面紧邻的立方体边长至多 $k+2$。求塔数除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2008-I-Q11] | Part C (Q11-Q15)

Count length-fourteen words in A, B whose maximal A runs all have even positive length and maximal B runs all have odd positive length.

求长十四的 A, B 字符串数，使每个极大连续 A 段长为正偶数，每个极大连续 B 段长为正奇数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q12] | Part C (Q11-Q15)

How many length-10 strings over $\{A, B\}$ have every maximal run of identical letters of length at most 3?

由 A, B 组成的长度 10 字符串中，连续相同字符的每个极大段长度都不超过 3，这样的字符串有多少个?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2016-II-Q12] | Part C (Q11-Q15)

Six labeled regions form a ring: each region shares an edge only with its two neighbors in the six-cycle. Paint them using four available colors, with neighboring regions different. Rotations are distinct because the positions on the wall are fixed. Count the colorings.

六个固定位置区域围成一环，每区仅与六循环中的左右邻区共边。用四种颜色涂色，邻区颜色不得相同；墙上位置固定，旋转后的不同位置涂法不合并。求涂法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-I-Q14] | Part C (Q11-Q15)

A frog starts at S on the cycle with consecutive vertices $S, P_1, P_2, P_3, E, P_4, P_5$. At each non-$E$ vertex it chooses either adjacent vertex; reaching E ends the path. Count distinct paths reaching E within twelve jumps.

七边循环顶点依次为 $S, P_1, P_2, P_3, E, P_4, P_5$。青蛙从 S 出发，每次在非 E 点可跳至任一邻点，一到 E 即停止。求不超过十二步到达 E 的不同路径数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q11] | Part C (Q11-Q15)

Count permutations of $1, \ldots, 6$ such that for each $1 \leq k \leq 5$, the first k terms contain an element greater than k.

求一至六的排列数，使对每个 $1 \leq k \leq 5$，前 k 项中至少有一项大于 k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2019-II-Q12] | Part C (Q11-Q15)

A progressive sequence of positive integers is strictly increasing and each term divides the next. Count all finite progressive sequences whose sum is 360, allowing length one.

正整数有限序列严格递增且每项整除下一项，称为递进序列。求项和为三百六十的此类序列数，允许只有一项。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2023-I-Q11] | Part C (Q11-Q15)

How many subsets of $\{1, \ldots, 10\}$ contain exactly one unordered pair of consecutive integers?

$\{1, \ldots, 10\}$ 的子集中，恰含一对相邻整数的共有多少个?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-I-Q15] | Part C (Q11-Q15)

A cell loop is the one-cell-thick boundary of an axis-aligned $a \times b$ cell rectangle, with $a, b \geq 2$; when one dimension is two, its whole rectangle is the loop. Count partitions of a 10×10 grid into exactly five disjoint cell loops.

格子环指轴对齐 $a \times b$ 长方形（$a, b \geq 2$）的一格厚边框；若一边为二，则整个长方形都是环。求将 10×10 方格恰好分成五个互不重叠格子环的划分数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-II-Q14] | Part C (Q11-Q15)

For integers define $a \circ b = a - b$ when a is odd and b even, and $a \circ b = a + b$ otherwise. Count compositions $a_1 + \cdots + a_j = 12$ into positive integers whose left-associated $\circ$ value is zero.

对整数定义：若 a 为奇数且 b 为偶数，则 $a \circ b = a - b$，否则 $a \circ b = a + b$。求正整数序列之和为 12 且从左至右计算的 $\circ$ 值为零的序列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2002-I-Q05] | Part A (Q1-Q5)

How many distinct planar squares have at least two vertices among the vertices of a regular dodecagon?

平面中有多少个不同正方形，其至少两个顶点属于某正十二边形的顶点集?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2004-I-Q03] | Part A (Q1-Q5)

A convex polyhedron has 26 vertices, 60 edges, 24 triangular faces, and 12 quadrilateral faces. How many vertex-joining segments have endpoints neither adjacent nor on any common face?

凸多面体有二十六顶点、六十棱、二十四个三角面及十二个四边形面。连接两顶点且两点不相邻、也不在同一面上的线段有多少条?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2007-II-Q05] | Part A (Q1-Q5)

Count unit lattice squares whose interiors lie entirely in the first quadrant and strictly below the line $9x + 223y = 2007$.

求内部完全位于第一象限且位于直线 $9x + 223y = 2007$ 下方的单位格正方形数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2014-I-Q05] | Part A (Q1-Q5)

For the twelve vertices of a regular dodecagon, count subsets Q for which some circle has all of Q strictly inside and all remaining vertices strictly outside. Include the empty subset.

对正十二边形的十二顶点，求子集 Q 的个数，使存在一圆将 Q 全部严格包含于内、其余顶点全部严格留于外，包括空集。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2016-I-Q03] | Part A (Q1-Q5)

A regular icosahedron is oriented with top vertex T, bottom vertex B, and horizontal upper and lower five-vertex rings. Count simple edge paths from T to B that never move upward. Each ring is a cycle, each upper-ring vertex has two lower-ring neighbors, T joins all upper vertices, and B joins all lower vertices.

正二十面体以一顶点 T 朝上、对顶点 B 朝下，中间为水平的上、下两个五顶点环。求从 T 到 B、沿棱且从不向上、也不重复顶点的路径数。每个环为五边形循环，上环每个顶点连接下环两个顶点，T, B 分别连接各自邻近环的全部顶点。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q05] | Part A (Q1-Q5)

On a circle, 20 distinct points are labeled 1 through 20 in circular order. Join two points precisely when the absolute difference of their labels is prime. Count triangles whose three vertices are labeled points and whose three edges are drawn.

圆上 20 个不同点按顺序标为 1 至 20。两点标号差的绝对值为质数时连线。求三个顶点均为已标点、三条边均已画出的三角形数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q03] | Part A (Q1-Q5)

Color all twelve unit edges of a 2×2 square grid red or blue so each unit square has two edges of each color. Count the colorings.

将 2×2 方格的十二条单位边染红或蓝，使每个小正方形恰有两红边、两蓝边。求染色数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2008-II-Q10] | Part B (Q6-Q10)

On the sixteen lattice points $\{0,1,2,3\}^2$, a growing path visits distinct points with strictly increasing successive distances. If the maximum point count is m and exactly r paths attain it, find mr.

在十六个格点 $\{0,1,2,3\}^2$ 上，路径访问互异点且相邻距离严格递增。若最多可有 m 个点，达到者有 r 条，求 mr。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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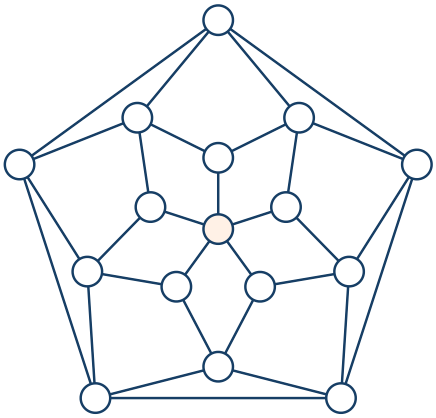
Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-I-Q07] | Part B (Q6-Q10)

Sixteen students stand at the vertices of the connected undirected graph shown. They share 3360 coins. Simultaneously each sends all their coins equally among their neighbors, after which every student has the original number again. Find the initial coin count at the central vertex.

十六名学生位于图示连通无向网络的顶点，共有 3360 枚硬币。所有人同时将自己的硬币平均送给各邻点学生，交换后每人硬币数与原来相同。求中心学生原有硬币数。



Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2018-I-Q07] | Part B (Q6-Q10)

A right prism has unit-side regular hexagonal bases and height two. Count isosceles triangles determined by three of its twelve vertices, including equilateral triangles.

直六棱柱底面为单位边长正六边形，高为二。求十二顶点任取三点构成的等腰三角形数，包括等边三角形。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q09] | Part B (Q6-Q10)

Two parallel lines contain respectively 7 and 5 marked points. Draw every segment between points on different lines, along with both full lines. Assume every interior crossing involves exactly two drawn segments. How many bounded planar regions result?

两条平行线上分别有 7 个和 5 个标记点，画出所有跨两线的连线段以及两条整直线。假设两线之间每个交点恰由两条线段相交形成。求有界区域数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-II-Q14] | Part C (Q11-Q15)

Nine labeled people each shake hands with exactly two others, and no pair shakes twice. Count the possible undirected handshaking patterns modulo 1000.

九名不同的人各与另外两人握手，每对至多握一次。求不同无向握手关系的数目模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q11] | Part C (Q11-Q15)

Each pair of five labeled towns is joined by one road. Count orientations of all ten roads for which every town can reach every other by a directed route.

五个有标签城镇中，每两镇之间恰有一条道路。将十条路都改为单向，求使任意一镇都能沿有向路线到达任意另一镇的定向数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q13] | Part C (Q11-Q15)

Let $f(n)$ count three-vertex subsets of a regular n-gon forming isosceles triangles, including equilateral ones. Find the sum of all integers $n \geq 3$ for which $f(n+1) - f(n) = 78$.

令 $f(n)$ 为正 n 边形顶点中任选三点组成等腰三角形的个数，等边也计入。求满足 $f(n+1) - f(n) = 78$ 的所有整数 $n \geq 3$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q14] | Part C (Q11-Q15)

In the three-dimensional grid $\{1,\ldots,10\}^3$, count distinct straight lines containing exactly eight grid points.

在三维点阵 $\{1,\ldots,10\}^3$ 中，求恰含八个点阵点的不同直线数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2023-I-Q14] | Part C (Q11-Q15)

Two distinguishable clock hands start at 12. Each move advances exactly one hand clockwise by one hour mark. Count length-144 move sequences that visit all 12^2 ordered hand positions once before returning to the start; report the count modulo 1000.

两根可区分的钟表指针从 12 出发，每步只使其中一根顺时针前进一个刻度。求经过全部 12^2 种有序位置各一次、并在第 144 步返回起点的移动序列数模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2024-II-Q15] | Part C (Q11-Q15)

Inside a fixed regular dodecagon, count all rectangles whose four sides lie along sides or diagonals of the dodecagon. Rectangle vertices may be chord intersections.

在固定正十二边形内部，求四条边均位于其边或对角线上的矩形个数，矩形顶点可以是弦的交点。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q11] | Part C (Q11-Q15)

How many perfect matchings of the vertices of a regular 24-gon use twelve segments all of the same length?

正二十四边形的二十四个顶点两两配对，画十二条等长线段，每个顶点恰使用一次。求画法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2026-II-Q15] | Part C (Q11-Q15)

Count ordered 7-tuples $a_i \in \{1, 2, 3\}$ such that both $\sum_i a_i$ and $\sum_{i=1}^{7} a_i a_{i+1} a_{i+3}$ are divisible by 3, with indices interpreted cyclically modulo seven.

求有序七元组 $a_i \in \{1, 2, 3\}$ 的个数，使 $\sum_i a_i$ 与 $\sum_{i=1}^{7} a_i a_{i+1} a_{i+3}$ 均被三整除，下标按模七循环。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q02] | Part A (Q1-Q5)

A row of ten unit squares has each rising diagonal drawn. A bug travels along edges and diagonals from the bottom-left endpoint to the top-right endpoint, never reusing a segment or moving left along a nonvertical segment. If there are N paths, find $\sqrt{N}$.

十个单位正方形排成一行，各画从左下到右上的对角线。虫子沿边或对角线从整行左下角走到右上角，不重复线段，且非竖直步不得向左。若路径数为 N，求 $\sqrt{N}$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2005-II-Q06] | Part B (Q6-Q10)

Cards numbered 1 through $2n$ are stacked top to bottom, split into top half A and bottom half B, then alternately removed from B, A and added on top of a new stack. A stack is magical when at least one card from each half retains its original top-based position. Find $2n$ for the magical stack in which card 131 stays fixed.

编号 1 至 $2n$ 的牌自上而下排列，分为上半堆 A、下半堆 B，再依次从 B, A 顶端交替取牌并压到新堆顶。若原两堆各至少一张牌位置不变，称此堆神奇。求第 131 张保位的神奇牌堆总张数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2006-II-Q08] | Part B (Q6-Q10)

An equilateral triangle is subdivided by its three midsegments into four small congruent triangles. Color each small triangle uniformly with one of six colors. Up to rotations and reflections of the large triangle, count distinct colorings.

等边三角形用三条中位线分成四个全等小三角形，每个用六种颜色之一染成单色。将整体旋转、翻转视为相同，求不同染色数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-I-Q10] | Part B (Q6-Q10)

Count the quadruples of integers $0 \leq a_0, a_1, a_2, a_3 \leq 99$ satisfying $a_0 + 10a_1 + 100a_2 + 1000a_3 = 2010$.

求满足 $0 \leq a_0, a_1, a_2, a_3 \leq 99$ 且 $a_0 + 10a_1 + 100a_2 + 1000a_3 = 2010$ 的整数四元组数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2010-II-Q10] | Part B (Q6-Q10)

How many integer-coefficient quadratic polynomials have two integer roots, counting multiplicity, and constant term 2010?

求常数项为 2010、两个根均为整数（计重数）的整系数二次多项式个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2011-II-Q06] | Part B (Q6-Q10)

How many integer quadruples $1 \leq a < b < c < d \leq 10$ satisfy $a + d > b + c$?

求满足 $1 \leq a < b < c < d \leq 10$ 且 $a + d > b + c$ 的整数四元组数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2018-I-Q10] | Part B (Q6-Q10)

Two concentric pentagonal rings have corresponding vertices joined by spokes. From an inner vertex a bug may step one vertex counterclockwise on that ring or outward on its spoke; from an outer vertex it may step clockwise or inward. Starting at a fixed inner vertex A, count fifteen-step returns to A, modulo 1000.

两个同心五顶点环由对应辐条相连。虫在内环可逆时针沿环一步或沿辐条向外，在外环可顺时针沿环一步或向内。由固定内顶点 A 出发，求十五步返回 A 的路径数模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2023-II-Q10] | Part B (Q6-Q10)

Fill a labeled 2×6 grid bijectively with $1, \ldots, 12$, requiring neighboring cells that share an edge to have differences not divisible by 3. If there are N fillings, find the number of positive divisors of N.

将 1 至 12 各用一次填入有固定位置的 2×6 网格，要求共边两格的数之差不能被 3 整除。设填法数为 N，求 N 的正约数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2024-II-Q09] | Part B (Q6-Q10)

Using up to 25 identical white chips and 25 identical black chips, fill some cells of a 5×5 grid, at most one per cell. Every occupied row and column must be monochromatic. Count the placements to which no further chip can be added legally.

用最多 25 枚相同白棋子、25 枚相同黑棋子填入 5×5 方格，每格至多一枚。同一行、同一列内已有棋子均须同色。求无法再合法添入任何一枚棋子的布局数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.7 Bijections & structural counting

双射与结构计数

[AIME-2026-II-Q13] | Part C (Q11-Q15)

Two finite integer sets are related if they are disjoint, have the same size, and admit a pairing in which paired integers differ by one. Find the smallest possible size of a set related to exactly 4040 other sets.

若两个有限整数集合不交、元素数相等，且可配成每对整数相差一，则称它们相关。求恰与 4040 个其他集合相关的集合，其最小元素数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-I-Q05] | Part A (Q1-Q5)

Two boxes each contain black and white marbles, with 25 marbles altogether. One marble is selected uniformly from each box, independently. Both are black with probability $27/50$. If both-white probability is m/n reduced, find $m+n$.

两盒中均有黑白两色球，总数为 25。独立从每盒等概率取一球，两球均黑的概率为 $27/50$。若两球均白的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-II-Q03] | Part A (Q1-Q5)

A deck has four copies of each label 1 through 10. Remove two equal-labeled cards, then uniformly choose two of the remaining cards without replacement. If their matching probability is m/n reduced, find $m+n$.

一副牌有 1 至 10 各四张。先移去两张同号牌，再从余牌中不放回等概率选两张。若两张同号的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2002-I-Q01] | Part A (Q1-Q5)

A uniformly random plate consists of three letters from a 26-letter alphabet followed by three decimal digits, with repetitions allowed. If at least one of the two three-symbol blocks is palindromic with probability m/n reduced, find $m+n$.

车牌由三个二十六字母及三个十进制数字依次组成，允许重复，所有排列等可能。若至少一个三符号块为回文的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-II-Q02] | Part A (Q1-Q5)

Three rolls of each of three types are randomly distributed into three labeled bags, three rolls per bag. The probability every bag has all three types is m/n in lowest terms. Find $m+n$.

三种面包各三个，随机装入三个有标号的袋子，每袋三个。每袋恰含三种的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-I-Q01] | Part A (Q1-Q5)

Choose two distinct positive divisors of 2010^2 uniformly. If the probability that exactly one is a square is m/n in lowest terms, find $m+n$.

从 2010^2 的正约数中等可能选出两个不同的数，恰有一个为平方数的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-II-Q04] | Part A (Q1-Q5)

Twelve airport gates lie in a row, with adjacent gates 100 feet apart. Select an old gate uniformly and then a different new gate uniformly. The probability that the walk is at most 400 feet is m/n in lowest terms. Find $m+n$.

十二个机场登机口排成一行，相邻间距一百英尺。先等可能选原登机口，再从其余口等可能选新口。步行不超过四百英尺的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q04] | Part A (Q1-Q5)

A thirteen-cell square array has quarter-turn symmetry about its central cell. The other twelve cells form three four-cell rotation orbits. Color eight cells red and five blue, uniformly among all choices. If the chance of a quarter-turn-invariant coloring is $1/n$, find n.

一个十三方格阵列关于中心方格具有九十度旋转对称，其余十二格分为三个各含四格的旋转轨道。等可能将八格涂红、五格涂蓝。若染色仍具有此旋转对称的概率为 $1/n$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-I-Q02] | Part A (Q1-Q5)

Nine people are split into nationality groups of sizes $2, 3, 4$. A uniformly random three-person subset is chosen. The probability its group-size pattern is exactly $2 + 1$ is reduced m/n. Find $m + n$.

九人按国籍分成 $2, 3, 4$ 人三组，等可能选出三人。所选三人的国籍人数分布恰为 $2 + 1$ 的概率是最简分数 m/n。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-II-Q05] | Part A (Q1-Q5)

Two different cells are chosen uniformly from an $n \times n$ grid, where $n \geq 2$. Find the smallest n such that the probability they share a full side is less than $1/2015$.

从 $n \times n$ 方格中等可能选两个不同小格，$n \geq 2$。求使两格共边概率小于 $1/2015$ 的最小 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q03] | Part A (Q1-Q5)

Shuffle five red and five green cards, then expose the first five in order. Happiness means all exposed reds form one block and all exposed greens form one block, allowing one color to be absent. Reduce this probability m/n and find $m+n$.

将五红五绿牌洗匀，依次翻出前五张。若翻出的红牌各自连续、绿牌也各自连续（允许某色缺席），称为成功。将概率约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q02] | Part A (Q1-Q5)

Choose J uniformly from one through twenty, then B uniformly from the nineteen other integers. If $P(B - J \geq 2) = m/n$ reduced, find $m + n$.

先从一至二十等可能选 J，再从其余十九数选 B。若 $P(B - J \geq 2) = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q05] | Part A (Q1-Q5)

From $(4,4)$ a particle independently takes one of $(-1,0),(0,-1),(-1,-1)$ with probability one third, stopping at its first axis point. If the probability it stops at the origin is $m/3^n$ with $3\nmid m$, find $m+n$.

粒子从 $(4,4)$ 出发，每步独立等可能选左、下、左下一个单位，首次到坐标轴即停止。在原点停下的概率为 $m/3^n$，m 不被三整除，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2023-I-Q01] | Part A (Q1-Q5)

Place five men and nine women uniformly at random in fourteen evenly spaced positions on a circle. Let m/n be the reduced probability that no man faces another man across a diameter. Compute $m+n$.

五名男子与九名女子随机等可能地站在圆周上十四个等距位置。任意男子的正对面均为女子的概率约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2026-II-Q05] | Part A (Q1-Q5)

An urn has r red and b blue marbles, with $r, b \geq 7$. Drawing seven without replacement, the probabilities of exactly four red and exactly five red are equal. Sum the five least possible totals $n = r + b$.

袋中有红球 r 个、蓝球 b 个，均不少于七。无放回抽七个，恰有四红与恰有五红的概率相等。求最小的五个可能总数 $n = r + b$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q06] | Part B (Q6-Q10)

Roll a fair six-sided die four times. The probability that the four results are nondecreasing is m/n reduced. Find $m+n$.

公平六面骰掷四次，四个结果非降的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q10] | Part B (Q6-Q10)

Uniformly choose two distinct points from the lattice box $0 \le x \le 2, 0 \le y \le 3, 0 \le z \le 4$. If their midpoint is also in the box with probability m/n reduced, find $m+n$.

从整点盒 $0 \le x \le 2, 0 \le y \le 3, 0 \le z \le 4$ 中等概率选两个不同点。若中点仍属该整点盒的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-II-Q09] | Part B (Q6-Q10)

Independently color each cell of a 3×3 board red or blue with equal probability. If the probability of no all-red 2×2 block is m/n reduced, find $m+n$.

将 3×3 方格的每格独立等概率染红或蓝。若不存在全红 2×2 方块的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2004-II-Q10] | Part B (Q6-Q10)

Choose uniformly among positive integers at most 2^{40} with exactly two binary ones. If the chance of divisibility by 9 is p/q reduced, find $p+q$.

从不超过 2^{40} 且二进制恰有两个一的正整数中等概率选一个。若被九整除的概率约分为 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-I-Q09] | Part B (Q6-Q10)

Twenty-seven cubes each have two adjacent unpainted faces and four orange faces. Random independent orientations form a $3 \times 3 \times 3$ cube. Its all-orange exterior has probability $p^a/(q^b r^c)$ in prime-factor form. Find $a+b+c+p+q+r$.

二十七个小立方体各有两个相邻面未染色，其余四面为橙色。随机独立定向后拼成 $3 \times 3 \times 3$ 大立方体。外表全橙的概率分解为 $p^a/(q^b r^c)$，求 $a+b+c+p+q+r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2007-II-Q10] | Part B (Q6-Q10)

Independently choose uniform random subsets A, B of a six-element set S. The probability that $B \subseteq A$ or $B \subseteq S \setminus A$ is m/n^r with prime n and reduced numerator. Find $m+n+r$.

在六元素集合 S 中独立等可能选子集 A, B。事件 $B \subseteq A$ 或 $B \subseteq S \setminus A$ 的概率写成最简的 m/n^r，n 为素数，求 $m+n+r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-I-Q10] | Part B (Q6-Q10)

Three distinct vertices of a regular n-gon are chosen uniformly. The probability that they form an obtuse triangle is $93/125$. Find the sum of all possible n.

从正 n 边形顶点中等可能选三个不同顶点，构成钝角三角形的概率为 $93/125$。求所有可能 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q06] | Part B (Q6-Q10)

Twelve distinct books, including three mathematics books, are randomly packed into boxes of capacities three, four, five. The probability that the three math books occupy one box is reduced m/n. Find $m+n$.

十二本不同书中有三本数学书，随机装入容量分别为三、四、五本的三个盒子。三本数学书在同盒的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2020-I-Q09] | Part B (Q6-Q10)

Select three positive divisors of 20^9 independently and uniformly, calling them a_1, a_2, a_3. The chance that $a_1 \mid a_2$ and $a_2 \mid a_3$ is m/n reduced. Find m.

从 20^9 的正约数中独立等概率选取 a_1, a_2, a_3。若 $a_1 \mid a_2 \mid a_3$ 的概率约分为 m/n，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2022-I-Q09] | Part B (Q6-Q10)

Twelve blocks consist of two copies of each of six colors. A uniformly random row is formed. What is $m+n$ if the probability that every same-color pair has an even number of blocks between its copies is m/n in lowest terms?

六种颜色各有两块，共十二块，随机等可能排成一行。若每对同色块之间的块数均为偶数的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2025-I-Q07] | Part B (Q6-Q10)

Randomly pair letters A through L, alphabetize each pair, then sort the six two-letter words. If the probability that the last word contains G is m/n in lowest terms, determine $m+n$.

将字母 A 至 L 随机配成六对，每对按字母顺序排列，再把六个双字母词排序。最后一个词含 G 的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q15] | Part C (Q11-Q15)

Uniformly label the eight faces of a regular octahedron by 1 through 8. Consecutive labels, including 8 and 1, must not lie on edge-sharing faces. If the probability is m/n reduced, find $m+n$.

将 1 至 8 等概率标在正八面体八个面上。要求循环相邻标号（包括 $8,1$）不能位于共边面上。若概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-II-Q12] | Part C (Q11-Q15)

Nine delegates, three from each of three countries, sit uniformly at random in nine labeled chairs around a circle. The probability that everyone has at least one neighbor from another country is reduced m/n. Find $m+n$.

三国各三名代表随机等可能地坐在环形九个固定座位上。每人至少有一名邻座来自其他国家的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-I-Q12] | Part C (Q11-Q15)

Choose f, g independently and uniformly from all functions on a four-element set to itself. The probability that their image sets are disjoint is reduced m/n. Find m.

从四元集合到自身的全部函数中，独立等可能选 f, g。两者像集不相交的概率为最简分数 m/n，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-II-Q13] | Part C (Q11-Q15)

Ten people contribute one left and one right shoe each. Pair all left shoes bijectively with right shoes uniformly. Require that no selection of k resulting pairs contains exactly the shoes of k owners for any $1 \leq k < 5$. Reduce the probability to m/n and find $m+n$.

十人各提供一左一右鞋，将十左鞋与十右鞋随机等可能一一配对。要求任取 k 双配成鞋都不能恰好包含 k 人的全部鞋，$1 \leq k < 5$。将概率约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q12] | Part C (Q11-Q15)

Choose a subset uniformly from $\{1, \ldots, 18\}$. If the probability its element sum is divisible by three is reduced m/n, find m. The empty sum is zero.

从一至十八的所有子集中等可能选一个，空集和为零。若元素和可被三整除的概率为最简分数 m/n，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2004-II-Q02] | Part A (Q1-Q5)

A jar contains ten red and ten blue candies. One person draws two uniformly without replacement, then another draws two from the remainder. The probability of equal unordered color combinations is m/n reduced. Find $m+n$.

罐中十红十蓝糖，第一人等概率不放回取两颗，第二人从余下再取两颗。两人所取无序颜色组合相同的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2006-II-Q05] | Part A (Q1-Q5)

Two independent identical biased dice have opposite face labels summing to seven. Four faces each have probability $1/6$; face F has larger probability and its opposite smaller. The chance of total seven is $47/288$. Express $P(F) = m/n$ in lowest terms and find $m + n$.

两个独立同分布的偏骰，其对面点数和为七。四面概率均为 $1/6$，面 F 概率更大，其对面更小。总点数七的概率为 $47/288$。若 $P(F) = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-I-Q02] | Part A (Q1-Q5)

One urn holds four green and six blue balls; another holds sixteen green and N blue. Independent uniform draws give a same-color probability of 0.58. Find N.

第一罐四绿六蓝，第二罐十六绿及 N 蓝，各独立等可能抽一球，同色概率为 0.58，求 N。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q02] | Part A (Q1-Q5)

Three events A, B, C have each exactly-one region probability 0.1 and each exactly-two region probability 0.14. Given $\Pr(C \mid A \cap B) = 1/3$, reduce $\Pr(\overline{A \cup B \cup C} \mid \bar{A}) = p/q$ and find $p + q$.

三个事件的每个恰一事件区域概率为 0.1，每个恰两事件区域概率为 0.14。已知 $\Pr(C \mid A \cap B) = 1/3$，将无事件发生且已知 A 不发生时的条件概率约为 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-I-Q05] | Part A (Q1-Q5)

Five sock colors have two socks each. On each of three successive days, two socks are sampled uniformly from those remaining, without returning any. If the probability that the third day is the first matching-color pair is reduced m/n, compute $m+n$.

五种颜色的袜子各两只。连续三天，每天从剩余袜子中等可能取两只，不放回。第三天首次取到同色一双的概率为最简分数 m/n。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2015-II-Q02] | Part A (Q1-Q5)

The four grade levels contain $40\%, 30\%, 20\%, 10\%$ of a school. Their Latin-enrollment rates are respectively $100\%, 80\%, 50\%, 20\%$. If the probability that a uniformly chosen Latin student is in the second grade level is reduced m/n, find $m+n$.

学校四个年级人数依次占 $40\%, 30\%, 20\%, 10\%$，各年级选修拉丁语的比例依次为 $100\%, 80\%, 50\%, 20\%$。从拉丁语学生中等可能选一人，其属于第二年级的概率为最简分数 m/n。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-I-Q02] | Part A (Q1-Q5)

Two independent six-sided dice have $\Pr(k)$ proportional to the face value k, for $k = 1, \ldots, 6$. If the probability of a total of 7 is reduced m/n, find $m + n$.

两个独立六面骰的各面概率与点数 k 成正比，$k = 1, \ldots, 6$。若总点数为 7 的概率为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2016-II-Q02] | Part A (Q1-Q5)

The probabilities of rain on Saturday and Sunday are $2/5$ and $3/10$. Sunday rain is twice as likely conditional on a rainy Saturday as conditional on a dry Saturday. If the probability of rain on at least one day is reduced a/b, find $a+b$.

周六、周日降雨概率分别为 $2/5, 3/10$。已知周六下雨时，周日下雨的条件概率是周六不下雨时的两倍。若至少一天降雨的概率为最简分数 a/b，求 $a+b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q02] | Part A (Q1-Q5)

Teams T_1, T_4 contest one semifinal, while T_2, T_3 contest the other; winners meet in a final. In any match, T_i beats T_j with probability $i/(i+j)$, independently of other matches. If T_4 wins the title with reduced probability p/q, find $p+q$.

半决赛由 T_1 对 T_4、T_2 对 T_3，胜者进入决赛。任一场 T_i 对 T_j 的胜率为 $i/(i+j)$，各场独立。若 T_4 夺冠概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2021-I-Q01] | Part A (Q1-Q5)

Two runners contest six races. Runner Z wins the first; subsequently the winner of the previous race wins again with probability $2/3$. If the probability that Z wins exactly five races is m/n reduced, find $m+n$.

两人赛跑六次，Z 赢得第一次。以后每次前一场胜者再次获胜的概率为 $2/3$。若 Z 恰赢五次的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2022-II-Q02] | Part A (Q1-Q5)

Four players A, C, J, S receive uniformly random semifinal pairings; the winners meet in a final. Player A beats C with probability $2/3$. Either A or C beats either J or S with probability $3/4$. Match outcomes are independent. Express the probability that C is champion as reduced p/q and find $p+q$.

四位球员 A, C, J, S 随机等可能配对打半决赛，胜者争夺冠军。A 对 C 的胜率为 $2/3$；A, C 中任一人对 J, S 中任一人的胜率为 $3/4$。各场结果独立。若 C 夺冠概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2024-I-Q04] | Part A (Q1-Q5)

A lottery ticket chooses four distinct numbers from $1, \ldots, 10$. A random four-element winning set gives a prize for at least two matches, and the top prize for four matches. Given that a prize was won, write the probability of the top prize as m/n in lowest terms and find $m+n$.

彩票从 1 至 10 中选四个不同号码，开奖也等可能选出四个。至少匹配两个号码即获奖，四个全中为头奖。已知获奖，头奖的条件概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2014-II-Q06] | Part B (Q6-Q10)

Choose equally between a fair die and a die whose six-probability is $2/3$ and each other face has probability $1/15$. Roll the selected die three times. Given the first two are sixes, reduce the chance the third is also six to p/q and find $p+q$.

等可能选公平骰或一枚六点概率为 $2/3$、其余各点概率 $1/15$ 的骰子，固定该骰投三次。已知前两次均为六，第三次也为六的概率约为 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2017-II-Q09] | Part B (Q6-Q10)

A deck has one card for each of the49 combinations of seven numbers and seven colors. Choose an eight-card set uniformly, conditional on covering all numbers and colors. The probability that one card can be discarded while retaining complete coverage is reduced p/q. Find $p+q$.

七个号码与七种颜色的49种组合各对应一张牌。等可能选择八张牌组成集合，条件是每种号码和颜色均出现。若可丢弃一张仍保持完整覆盖的条件概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2026-I-Q09] | Part B (Q6-Q10)

Roll a fair blank six-sided die six times. After roll j, place sticker j on the top face, covering earlier stickers there. Conditioned on stickers $2, 4, 6$ all being visible at the end, the probability of exactly one blank face is m/n in lowest terms. Find $m + n$.

公平空白六面骰掷六次，第 j 次后在朝上的面贴上编号 j，覆盖该面的旧贴纸。已知最终 $2, 4, 6$ 都可见，恰有一面空白的概率为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2010-II-Q13] | Part C (Q11-Q15)

Four players draw distinct cards numbered 1 through 52. The lower two form one team, the higher two another. Condition on two designated players holding $a, a+9$, and let $p(a)$ be their chance of being teammates. Find $m+n$ for the smallest value $p(a) \geq 1/2$, written as reduced m/n.

四人各抽取一张不同的 1 至 52 号牌，较小两牌为一队，较大两牌为另一队。已知指定两人持牌 $a, a+9$，其同队概率为 $p(a)$。将不小于二分之一的最小可能 $p(a)$ 约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.2 Conditional probability & independence

条件概率与独立性

[AIME-2011-I-Q12] | Part C (Q11-Q15)

Six men and n women line up randomly. Given that each man has a male neighbor, let p be the probability of a run of at least four men. Find the smallest n for which $p \leq 1\%$.

六男与 n 女随机排成一行。已知每名男子旁边至少有一名男子，设至少四男连续的条件概率为 p。求使 $p \leq 1\%$ 的最小 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2010-II-Q02] | Part A (Q1-Q5)

Choose a uniform point in a unit square, and let d be its distance to the nearest side. Reduce $\Pr(1/5 \le d \le 1/3) = m/n$ and find $m+n$.

在单位正方形内均匀选一点，d 为它到最近边的距离。若 $\Pr(1/5 \le d \le 1/3) = m/n$ 为最简分数，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2017-II-Q03] | Part A (Q1-Q5)

Choose a point uniformly by area inside triangle with vertices $(0,0),(12,0),(8,10)$. If its probability of being nearer $(12,0)$ than either other vertex is reduced p/q, find $p+q$.

在顶点为 $(0,0),(12,0),(8,10)$ 的三角形内按面积均匀取点。若该点到 $(12,0)$ 比到其余两个顶点都近的概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2020-II-Q02] | Part A (Q1-Q5)

Choose (x, y) uniformly from the unit square. Let $Q = (5/8, 3/8)$. The probability that the line from Q to (x, y) has slope at least $1/2$ is m/n reduced. Find $m + n$.

在单位正方形内均匀选点 (x, y)。固定 $Q = (5/8, 3/8)$，连接两点的直线斜率至少为 $1/2$ 的概率约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2004-I-Q10] | Part B (Q6-Q10)

A radius-one circle is placed with its center uniformly distributed among positions keeping it inside a 36×15 rectangle. The probability that it does not meet a given diagonal is m/n reduced. Find $m+n$.

半径一的圆完全位于 36×15 矩形内，圆心在全部允许位置中均匀分布。不与给定对角线相交的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q06] | Part B (Q6-Q10)

An isosceles triangle with equal base angles x degrees is inscribed in a circle. Two independent uniformly sampled circumference points determine a chord. Its probability of meeting the triangle is $14/25$. Find the largest possible x minus the smallest.

一个等腰三角形内接于圆，两底角均为 x 度。在圆周上独立均匀取两点形成弦，该弦与三角形相交的概率为 $14/25$。求 x 的最大可能值与最小可能值之差。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q08] | Part B (Q6-Q10)

Independently choose a, b uniformly from $(0, 75)$ degrees. Let $OP = 200$. On the same side of OP, choose Q, R with right angles OQP, ORP and angles $POQ = a, POR = b$. If $\Pr(QR \leq 100) = m/n$ in lowest terms, find $m + n$.

角度 a, b 独立均匀取自 $(0, 75)$ 度，$OP = 200$。在 OP 同侧取 Q, R，使 OQP, ORP 均为直角，且 $POQ = a, POR = b$。若 $\Pr(QR \leq 100) = m/n$ 为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2023-II-Q06] | Part B (Q6-Q10)

Choose two independent uniform points in $L = [0,1]^2 \cup ([0,1] \times [1,2]) \cup ([1,2] \times [0,1])$. The chance that their midpoint also belongs to L is the reduced fraction m/n. Find $m+n$.

在区域 $L = [0,1]^2 \cup ([0,1] \times [1,2]) \cup ([1,2] \times [0,1])$ 内独立均匀选取两点。两点中点仍在 L 内的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.3 Geometric probability

几何概率

[AIME-2003-I-Q11] | Part C (Q11-Q15)

Choose x uniformly in $(0^\circ, 90^\circ)$. Let p be the probability that $\sin^2 x, \cos^2 x, \sin x \cos x$ fail to form a nondegenerate triangle. If $p = d/n$ where d is the degree measure of $\arctan m$ and positive integers m, n satisfy $m + n < 1000$, find $m + n$.

在 $(0^\circ, 90^\circ)$ 均匀选角 x。设 $\sin^2 x, \cos^2 x, \sin x \cos x$ 不能构成非退化三角形的概率为 p。若 $p = d/n$，d 为 $\arctan m$ 的度数，正整数 m, n 满足 $m + n < 1000$，求其和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2019-II-Q02] | Part A (Q1-Q5)

A frog begins on pad one and independently jumps forward one or two pads with equal probability. If its probability of visiting pad seven is reduced p/q, find $p+q$.

青蛙从第一片荷叶开始，每次独立等可能前跳一片或两片。到过第七片的概率为最简分数 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2023-I-Q06] | Part B (Q6-Q10)

A randomly shuffled deck contains three red and three black cards. Before each reveal, a player predicts its color using all previous reveals. Under an optimal strategy, the expected correct predictions equal the reduced fraction m/n. Find $m+n$.

三张红牌与三张黑牌随机洗匀后逐张翻开。每次翻牌前，可根据已翻牌预测颜色。最优策略的猜中次数期望为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2026-II-Q07] | Part B (Q6-Q10)

On each fair-die roll, outcomes $1, 2$ give a coin to Alice, $3, 4$ to Bob, and $5, 6$ to Carol. The chance that Alice and Bob both receive two coins before Carol receives her first is m/n in lowest terms. Find $100m + n$.

公平骰子每掷一次，$1, 2$ 给甲一枚硬币，$3, 4$ 给乙，$5, 6$ 给丙。甲乙均各得至少两枚而丙尚未得第一枚的概率为最简分数 m/n，求 $100m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2015-I-Q12] | Part C (Q11-Q15)

For every 1000-element subset of $\{1, \ldots, 2015\}$, record its smallest element. The mean of all recorded values is reduced p/q. Find $p + q$.

对 $\{1, \ldots, 2015\}$ 的每个 1000 元子集记录其最小元素。所有记录值的平均数为最简分数 p/q。求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-I-Q13] | Part C (Q11-Q15)

A lattice walker starts at height 21 between a reflecting fence $y = 0$ and absorbing river $y = 24$. Away from the fence it independently chooses east, west, north, south with equal probabilities. At the fence it chooses east, west, north equally. Find the expected number of unit jumps until it first reaches the river.

格点随机行走者从高度 21 出发，$y = 0$ 为不可越过的围栏，$y = 24$ 为到达即停止的河流。不在围栏上时，东、西、北、南四方向等可能；在围栏上时，东、西、北三方向等可能。求首次到河流所需单位跳跃次数的期望。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2016-II-Q13] | Part C (Q11-Q15)

Place six identical rooks on a 6×6 board with one in each row and column. An occupied cell in row r, column c has value $r + c$. The score is the minimum occupied value. If the mean score over all configurations is reduced p/q, find $p + q$.

在 6×6 棋盘放六个相同车，每行、每列各一个。第 r 行第 c 列格子的值为 $r + c$，一配置的得分为所占格子的最小值。所有配置平均得分为最简分数 p/q，求 $p + q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2018-II-Q13] | Part C (Q11-Q15)

Roll a fair six-sided die until the first occurrence of consecutive results $1, 2, 3$. If the probability the total number of rolls is odd is reduced m/n, find $m + n$.

反复投公平六面骰，首次连续出现一、二、三时停止。总投掷次数为奇数的概率为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2021-I-Q12] | Part C (Q11-Q15)

Three frogs start at vertices $4, 8, 12$ of a cycle of twelve vertices. Each minute each independently moves to either neighbor with equal probability. They stop at the first simultaneous landing that places two frogs together. If the expected stopping time is m/n reduced, find $m+n$.

十二顶点环上三只蛙初始位于第 $4, 8, 12$ 点。每分钟每只蛙独立等概率跳到左右相邻顶点之一，首次出现两蛙同时落在同一点时停止。若期望分钟数约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.4 Expected values & stopping processes

期望与停止过程

[AIME-2025-I-Q13] | Part C (Q11-Q15)

Two perpendicular diameters divide a disk into four quadrants. Draw 25 independent random chords by choosing uniform perimeter endpoints conditioned on lying in different quadrants. Find the expected number of regions determined by all 27 segments.

两条垂直直径将圆盘分成四个象限。再独立画 25 条随机弦，每条的两端在圆周上均匀选取，条件是位于不同象限。求全部 27 条线段分出的区域数的期望。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-I-Q03] | Part A (Q1-Q5)

A coin has head probability $0 < p < 1$. In eight independent flips, the probability of three heads is one twenty-fifth of that of five heads. Write $p = m/n$ reduced and find $m + n$.

硬币正面概率为 $0 < p < 1$。独立掷八次，恰三正面的概率为恰五正面概率的二十五分之一。若 $p = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2010-I-Q04] | Part A (Q1-Q5)

Two players independently toss the same three coins once each. Two coins are fair; the third has head probability $4/7$. Their head counts agree with reduced probability m/n. Find $m+n$.

两名玩家各独立投掷三枚硬币一次；其中两枚公平，另一枚正面概率为 $4/7$。两人正面数相同的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2006-II-Q10] | Part B (Q6-Q10)

Seven teams play an independent fair round robin without draws. Given that team A beats B in their game, the chance that A ends with more wins is reduced m/n. Find $m+n$.

七队进行无平局、各场独立公平的单循环赛。已知 A 战胜 B，最终 A 胜场数更多的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2008-I-Q09] | Part B (Q6-Q10)

Ten independent rectangular crates have dimensions $3, 4, 6$ feet, with each possible vertical dimension equally likely. The probability their total stacked height is 41 feet is reduced m/n. Find m.

十个长方体箱尺寸均为三、四、六英尺，各自独立等可能选择一种尺寸为高度，叠高恰四十一英尺的概率为最简分数 m/n，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-II-Q08] | Part B (Q6-Q10)

Two players independently roll fair dice until their first six. If the probability their roll counts differ by at most one is reduced m/n, find $m+n$.

两人独立掷公平骰直至第一次出现六。掷次相差至多一次的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2021-II-Q08] | Part B (Q6-Q10)

An ant begins at a bottom vertex of a cube. Its first step chooses uniformly among the three edges. Later it chooses uniformly among the two edges that do not return immediately to the previous vertex. The chance of being on the top face after eight steps is m/n reduced. Find $m+n$.

蚂蚁从立方体底面顶点出发，第一步等概率选三条边之一。以后每步从不立即返回前一顶点的两条边中等概率选择。八步后在顶面的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2001-II-Q11] | Part C (Q11-Q15)

A team plays six independent matches, each equally likely to be a win, loss, or tie. If the probability of more wins than losses is m/n reduced, find $m+n$.

某队独立比赛六场，每场胜、负、平概率均为 $1/3$。若胜场多于负场的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2002-II-Q12] | Part C (Q11-Q15)

Each independent basketball shot succeeds with probability $2/5$. After ten shots, require exactly four successes and no prefix of one through nine shots having success fraction above $2/5$. The probability is $p^a q^b r / s^c$, where p, q, r, s are primes and $a, b, c > 0$. Find $(p + q + r + s)(a + b + c)$.

每次投篮独立命中，概率为 $2/5$。要求十次后恰中四次，且前一至九次任一前缀命中率都不超过 $2/5$。将概率写为 $p^a q^b r / s^c$，各底数为质数、指数为正，求 $(p + q + r + s)(a + b + c)$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2003-II-Q13] | Part C (Q11-Q15)

A bug starts at a triangle vertex. On each of ten steps, it independently chooses either other vertex with equal probability. If its return probability after the tenth step is m/n reduced, find $m+n$.

虫从三角形一顶点出发，每步等概率选另外两顶点之一前往。十步后回到起点的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2014-I-Q11] | Part C (Q11-Q15)

A token makes six independent uniform steps on the integer grid from the origin, each one unit north, south, east, or west. Its chance of finishing on $|x| = |y|$ is reduced m/n. Find $m + n$.

棋子从原点独立等可能地向上下左右各一步，连续六步。终点落在 $|x| = |y|$ 上的概率为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2001-I-Q02] | Part A (Q1-Q5)

A finite set of distinct real numbers has mean M. Adjoining 1 lowers the mean by 13, while adjoining 2001 raises it by 27. Find M.

一个有限且元素互异的实数集合均值为 M。加入 1 后均值降低 13，加入 2001 后提高 27。求 M。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2021-II-Q01] | Part A (Q1-Q5)

What is the mean of all three-digit decimal palindromes?

所有三位十进制回文数的算术平均数是多少?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2024-II-Q02] | Part A (Q1-Q5)

A finite list of positive integers sums to 30, has unique mode 9, and has an integer median absent from the list. Find the sum of its squared entries.

一列正整数的和为 30，唯一众数为 9，中位数为不在该列中的整数。求所有项的平方和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.6 Data & descriptive statistics

数据与描述统计

[AIME-2017-I-Q11] | Part C (Q11-Q15)

Arrange digits1 through9 bijectively in a 3×3 array. Take the median in each row and then the median of those three medians. If Q arrays give final median5, find $Q \bmod 1000$.

将数字1至9各用一次填入 3×3 数组，先取每行中位数，再取三个中位数的中位数。若最终为5的数组有 Q 个，求 Q 模1000的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2024-I-Q03] | Part A (Q1-Q5)

Two players alternately remove either 1 or 4 tokens from a pile, with the last removal winning. The first player moves first. For how many initial sizes $1 \le n \le 2024$ can the second player force a win?

两人轮流从一堆筹码中取走 1 或 4 枚，取走最后一枚者获胜。求 $1 \le n \le 2024$ 中有多少个初始筹码数能保证后手获胜。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2010-I-Q12] | Part C (Q11-Q15)

Find the least $m \geq 3$ such that every two-coloring of $\{3, 4, \ldots, m\}$ contains same-colored numbers a, b, c, not necessarily distinct, with $ab = c$.

求最小整数 $m \geq 3$，使 $\{3, 4, \ldots, m\}$ 的任意二染色中，必有同色的 a, b, c 满足 $ab = c$，三数允许重复。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2012-I-Q11] | Part C (Q11-Q15)

From $(0,0)$, a frog may repeatedly add $(7,2),(2,7),(-5,-10),(-10,-5)$. Let M count the reachable integer points with $|x|+|y|\le 100$. Find $M \bmod 1000$.

青蛙从原点出发，每跳可加向量 $(7,2),(2,7),(-5,-10),(-10,-5)$ 之一。设可达且满足 $|x|+|y|\le 100$ 的整数点有 M 个，求 M 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2001-II-Q05] | Part A (Q1-Q5)

Find the largest n such that every ten-element subset of $\{4, 5, \ldots, n\}$ contains three distinct numbers forming a nondegenerate triangle.

求最大 n，使 $\{4, 5, \ldots, n\}$ 的每个十元素子集均含三个不同数可作为非退化三角形边长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2004-I-Q05] | Part A (Q1-Q5)

Alpha scores $160/300$ on day one and $140/200$ on day two. Beta also attempts 500 points in total, attempts a different first-day amount from 300, earns positive integer scores each day, and has a strictly smaller daily success ratio than Alpha on both days. If Beta's greatest possible total success ratio is m/n reduced, find $m+n$.

Alpha 两天分别得分/作答分值为 $160/300, 140/200$。Beta 两天共作答五百分，首日作答分值不为三百，每天得分为正整数，且两日成功率均严格低于 Alpha 对应日。若 Beta 最大总成功率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2006-I-Q02] | Part A (Q1-Q5)

Among all ninety-element subsets of $\{1, \ldots, 100\}$, how many distinct element sums occur?

在 $\{1, \ldots, 100\}$ 的所有九十元素子集中，元素和有多少种不同值?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2007-II-Q08] | Part B (Q6-Q10)

Full lines parallel to the sides of a 4×5 paper rectangle have total drawn length 2007. A basic rectangle is bounded by adjacent drawn lines in each direction; paper edges do not count unless drawn. Maximize the basic-rectangle count N and give $N \bmod 1000$.

在 4×5 纸上画与边平行的完整横纵线，总长 2007。基本矩形由两方向各相邻的画线围成，未画出的纸边不算。求最大基本矩形数 N 除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-I-Q07] | Part B (Q6-Q10)

Assign the digits $1, \ldots, 9$ bijectively to $a, b, c, d, e, f, g, h, i$. Minimize the positive value $(abc - def)/(ghi)$, where products are meant. If the minimum is m/n in lowest terms, compute $m + n$.

将 1 至 9 九个数字各用一次分配给 $a, b, c, d, e, f, g, h, i$。式 $(abc - def)/(ghi)$ 中相邻字母表示乘积。求最小正值的最简分数 m/n，并计算 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2025-II-Q08] | Part B (Q6-Q10)

Coins of values $1, 10, 25$ are unlimited. For how many target amounts $1 \le N \le 1000$ does repeatedly taking the largest affordable coin use the minimum possible number of coins?

有无限枚面值 $1, 10, 25$ 的硬币。对 $1 \le N \le 1000$，每次取不超过剩余金额的最大面值硬币。求这种贪心算法能得到最少枚数的金额个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2003-II-Q12] | Part C (Q11-Q15)

Each committee member votes for one of 27 candidates. For every candidate, the exact percentage of total votes is at least one less than that candidate's number of votes. Find the smallest possible committee size.

委员会每人给二十七名候选人之一投一票。对每名候选人，其精确得票百分数须比得票数至少小一。求委员会最小可能人数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2009-II-Q12] | Part C (Q11-Q15)

Choose disjoint pairs of distinct elements from $\{1, \ldots, 2009\}$, requiring all pair sums distinct and at most 2009. Maximize the number k of pairs.

从一至 2009 中选互不重用元素的数对，每对两数不同，所有对和互异且不超过 2009。求最多对数 k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-II-Q14] | Part C (Q11-Q15)

For positive integers $a < b < c$, a stamp collection must include each of these three denominations and must have subcollections totaling every integer from 1 through 1000 cents. Let $f(a, b, c)$ be the least number of stamps in such a collection, when possible. Find the sum of the three smallest c for which some a, b satisfy $f(a, b, c) = 97$.

邮票面值为正整数 $a < b < c$ 分，要求三种面值各至少一张，且用邮票子集能凑出 1 至 1000 的每个整数金额。若能实现，令最少张数为 $f(a, b, c)$。求使某些 a, b 满足 $f(a, b, c) = 97$ 的最小三个 c 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2026-I-Q11] | Part C (Q11-Q15)

Place $1, \ldots, 64$ bijectively in an 8×8 grid. Maximize the sum M of absolute differences over all horizontally or vertically adjacent pairs, and give this maximum modulo 1000.

将 1 至 64 各放一次于 8×8 方格，令 M 为所有水平、竖直相邻格数字差绝对值之和。求最大 M 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

[AIME-2021-II-Q11] | Part C (Q11-Q15)

A teacher gives each of four perfectly logical students one distinct member of an unknown set S. Publicly, S consists of four consecutive two-digit positive integers, including a multiple of 6 and a different multiple of 7. All four initially say they cannot determine S from their own number; after hearing the four replies, each can determine it. Sum all possible largest members of S.

老师给四名完全理性的学生各一个不同整数，组成未知集合 S。公开信息是：S 为四个连续两位正整数，含一个 6 的倍数及另一个不同的 7 的倍数。四人起初都表示不能仅由自己的数确定 S，听到四人的回答后各自都能确定。求 S 最大元素所有可能值之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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